OUR ENVIRONMENT

SAFEGUARD
NURTURE
CHERISH



NATIONAL ENVIRONMENT AGENCY

Celebrating 10 years

OUR ENVIRONMENT

SAFEGUARD
NURTURE
CHERISH

National Environment Agency

Safeguard • Nurture • Cherish

Published by
National Environment Agency
Environment Building, 40 Scotts Road, #19-00
Singapore 228231
http://www.nea.gov.sg

Produced by
Ink on Paper Communications Pte Ltd
57 Ubi Ave 1, #06-03 Ubi Centre, Singapore 408936
Tel: (65) 68414028 Fax: (65) 68410943
http://www.iop.sg

Project Manager: Alicia Tan
Researcher Writer: Tan Chui Hua
Editors: Marcus Ng, Jennifer Hollie Chen
Creative Director: Tan Boon Hua
Photographers: Richard Koh - Amaranthine Photos, Lua Hwa Beng

Copyright © 2012 National Environment Agency

All rights reserved. No part of this publication may be reproduced, stored in a retrieval system, or transmitted in any form or by any means, electronic, mechanical, photocopying, recording or otherwise, without permission of the copyright owner. To the best of our knowledge, information presented in this book is accurate at the time of printing.

National Library Board, Singapore Cataloguing-in-Publication Data

Tan, Chui Hua, 1979-
Safeguard, nurture, cherish our environment / [researcher writer, Tan Chui Hua ; editors, Marcus Ng, Jennifer Hollie Chen]. – Singapore : National Environment Agency, c2012.
p. cm.
ISBN : 978-981-07-2528-0

1. Singapore. National Environment Agency – Anniversaries, etc.
2. Environment policy – Singapore. 3. Environmental protection – Singapore. 4. Natural resources – Singapore.
5. Hazardous wastes – Singapore – Management. I. Ng, Marcus. II. Chen, Jennifer Hollie.
III. Singapore. National Environment Agency. IV. Title.

TD171.5 363.7095957 -- dc23 OCN794345758

Printed by NPE Print Communications Pte Ltd

FSC logo

CONTENTS

4 Foreword

6 Chairman’s Message

7 CEO’s Message

8 A Bold Step Forward

18 Safeguard: A Place To Live

70 Working Among Us

120 And The Weather Tomorrow Is...

136 A Novel Approach

164 The Environment Is Ours

190 A Future We Share

208 A 10-Year Journey For Environmental Sustainability

210 Glossary

212 Acknowledgements

214 References

Foreword

Dr Vivian Balakrishnan, Minister for the Environment and Water Resources

In the 1960s, a walk around the city area would sometimes entail close encounters with pungent piles of rubbish on the roadside, choked drains and illegal hawkers operating from makeshift stalls with compromised food hygiene. If you happened to be near the Singapore River, the familiar stench would be an indelible memory.

Such was the Singapore many of us grew up in. Fortunately, our early leaders took a strong stand to protect the environment and safeguard public health, way before being clean and green became fashionable. It formed the Anti-Pollution Unit in 1970, and the Ministry of the Environment (ENV) in 1972. In a few decades, the situation was completely transformed. The clean air, clean water, clean land we enjoy today is a result of the government's and our people's commitment to the vision of a safe, clean and sustainable Singapore.

However, protecting the environment is a continuous effort. At the turn of the 21st century, it was clear that environmental and public health challenges had become more complex and cross-cutting. ENV thus underwent strategic restructuring to become a lean headquarters focusing on policy issues while a new statutory board, the National Environment Agency (NEA), was formed to take care of operations.

Within the last decade, NEA fulfilled its role to safeguard and enhance Singapore's environment. Through the use of innovative technological solutions, it has improved its situational awareness and operational capacity. The agency can tell us how our air quality is faring, whether construction sites are flouting noise control regulations or what the weather is going to be like in the next three hours, at all times of the day.

Not only has it made its operations more effective and comprehensive, the agency has also proven itself to be invaluable during many episodes in Singapore's history. One such instance was the SARS outbreak in 2003. Even though the outbreak took place less than a year after NEA's formation, it mobilised its resources swiftly to support Singapore through the crisis, from contact-tracing in the early stage to raising awareness of public hygiene. Another crucial role of the agency is one which many of us are familiar with – that is dengue control and prevention. As the lead agency in Singapore's battle against dengue, NEA has put together a noteworthy strategy which covers swift operational response to dengue outbreak clusters, rigorous scientific research to support vector and pathogen surveillance and effective mass outreach to involve every citizen to do his or her part.

But the greatest transformation NEA has wrought is in its successful partnership with the community to safeguard and cherish our environment. More and more youths, working professionals, retirees, grassroots leaders and corporations are stepping up to be change-makers for the environment, thanks to NEA's efforts over the years. At the same time, NEA has also undergone metamorphosis

and shed the rigid enforcer image of environmental and public health protection. On top of these achievements, NEA is also playing an important role in positioning Singapore as the region's environmental hub, as it actively identifies channels to share and export our environmental solutions.

2012 marks the 10th anniversary of NEA. As the agency readies itself for an even more challenging environmental future beset with issues such as climate change and diminishing resources worldwide, it is also time for us to reflect upon how far NEA has come, and the difference it has made to Singapore. Much of the work in environmental and public health protection is hard, onerous and often goes unnoticed. But the beautiful, clean, healthy environment we have today would not have come about if not for the relentless drive of the people behind the work – from ENV times to NEA. Every single person has contributed, from the leaders and management to the professionals and the rank and file. These people deserve our tribute and appreciation. This is the reason for this book, to tell the stories of those who have made a difference in our living surroundings.

As we move forward, I am certain NEA will continue to serve the public well, with the strong sense of pride and dedication it has always displayed. I wish the agency a happy 10th anniversary, with many more fulfilling years to come.



Chairman's Message



Chew Gek Khim, Chairman, NEA

In less than 50 years, Singapore has rapidly urbanised from a country with a population of about 1.89 million to over 5 million today. Concurrently, it has achieved a more than 90-fold increase in Gross Domestic Product. We saw a country with *kampongs* (Malay: villages) and many street vendors transform into a bustling metropolis with a reputation for being one of the cleanest cities in Asia with one of the highest rates of wealth per capita. It is not uncommon for cities with such rapid economic growth to lose their greenery and turn into concrete jungles with massive traffic jams and heavy pollution.

However, we were fortunate enough that Singapore's first prime minister, Lee Kuan Yew, took a keen interest in ensuring that Singapore's environment was protected. Today, Singaporeans enjoy a high quality of life, with good air, access to public parks and minimal pollution despite being one of the most densely-populated countries in the world. Even more noteworthy is the fact that Singapore has the reputation of being amongst the top most liveable cities in Asia.

However, it cannot be assumed that this will always remain the case as our country and our neighbours continue to grow economically and urbanise. The physical constraints that we face in terms of land will require us to balance the needs of economic growth with environmental sustainability even more. We will also see more complex environmental challenges such as climate change, potential trans-boundary environmental pollution which affects Singapore's air and shores, changing demographics and rising public expectations. It cannot be business as usual for NEA.

When I took on the role of Chairman in 2008, climate change arising from the global increase in carbon emissions, pollution, new vector-borne diseases as well as rising sea levels were but some of the challenges that NEA had to deal with. While NEA has achieved much since it was formed in 2002, further effort must be made for us to work towards higher standards of environmental sustainability.

To prepare NEA to meet future external and domestic challenges, we need to aim for higher environmental standards and targets, because it is environmental sustainability that will enable us to continue to enjoy our current quality of life. However, higher environmental standards cannot be achieved by NEA alone. The community must take ownership. For it is only when our community and businesses are prepared to change the way we conduct business, manufacture and live can we then build a sustainable, liveable and vibrant Singapore, one we can be proud of to call our home.

On our part, we at NEA have to build up new capabilities and leverage on new technology to prepare for these emerging issues. To do so, we must attract and retain talent in NEA. Over the last two years, NEA has been working on a blueprint that will chart the direction for NEA in the next 5 to 10 years. To achieve this, NEA as an organisation must keep up with the times, and remain responsive, resourceful and relevant. With the blueprint in place and implemented, NEA will be on a more robust footing, and our organisation should be confident that we are ready to take on new challenges. The transformation of NEA will enable the organisation to respond better, faster and with greater agility while working together with the community to address local issues and potential environmental risks. The outcome will be a stronger and more resilient NEA.

On behalf of the NEA Board of Directors, I wish NEA a happy 10th anniversary.

CEO's Message

Andrew Tan, Chief Executive Officer, NEA



NEA is an agency which wears many different hats. It safeguards our environment, ensuring that every Singaporean can enjoy clean air, land and water. It covers a wide range of issues, from pollution control, public cleanliness, waste management and resource efficiency to food hygiene and vector control. In many ways, NEA plays a vital role in the lives of Singaporeans.

The work that NEA does allows us to live, work and play in a beautiful, healthy city. With a more complex and dynamic environment and new challenges such as climate change and other emerging threats confronting us, NEA has to be ever ready to safeguard and protect Singapore's environment. Beyond our own efforts, we must be able to enlist the support of every Singaporean to safeguard, nurture and cherish our environment.

When I joined NEA in 2009, my key priority was to transform and equip the agency to become future-ready. NEA needs to develop better, more effective ways to keep Singapore clean and green even as the population grows and urban development steps up. We need to raise our standards for environmental performance, and involve every citizen in our endeavours. Yet at the same time, we must make sure that Singapore stays competitive economically.

We have developed a strategic road map in line with the Sustainable Development Blueprint to achieve our vision of a clean and green Singapore. In this road map, we have four key thrusts — achieve higher environmental standards, deliver better services to the public, build new capabilities in the agency and develop each and every one of NEA's officers. Some of the steps we are taking include further enhancing our air and water quality, raising cleanliness standards, encouraging greater resource efficiency, building better hawker centres with affordable food for all, increasing our efforts in nurturing Singaporeans to come forth and take ownership of their environment and investing more resources to develop our people to their fullest potential. All these new measures we are putting in place will help position Singapore as the cleanest and greenest city in the region.

NEA's good track record in safeguarding our environmental and public health, operational effectiveness, service standards and engagement of the community puts us in good stead for the future. In the last decade, NEA officers have done their utmost for Singapore's environment. They have made NEA the agency it is today and they will take NEA to greater heights.

This 10th anniversary publication acknowledges the efforts of our officers over the last decade and explains our role. It also captures the spirit of the agency, the remarkable dedication our people have demonstrated in their work, from keeping our air, water and streets clean to preventing lapses in public hygiene. I am confident that this culture of excellence will see us through the many challenges ahead.


KENT


a bold step
Forward

Behind Singapore's clean, green environment, and keeping it going day to day is the National Environment Agency. Its officers work tirelessly to control pollution, ensure hygiene in food establishments, make sure the waste we generate is taken care of, prevent dengue outbreaks and provide weather forecasts and air quality readings.



Take a walk. Take a deep breath. Drop by East Coast Park and have a swim. Round the day off with a feast at one of the many hawker centres dotting the island.

On almost any given day in Singapore, these are typical activities. We seldom pause to think how unusual it is that streets are litter-free, the air is fresh, the water is clean and the food is safe. In fact, the quality of environment we enjoy is nothing short of extraordinary.

We live in one of the most high density, compact cities in the world. Singapore spans just a little over 700 km^2 in size. More than five million residents live side by side with some 10,000 units of commercial developments, over 9,000 industrial establishments and close to a million vehicles. Unlike many other countries, Singapore does not have swathes of rural hinterland to counter the effects of urban development. Inevitably, our activities bring about pollution and generate domestic and industrial waste. And in such a densely-populated country, diseases borne by vectors such as mosquitoes have the potential to spread fast and on a large scale.

Yet we do not inhabit a polluted, unpleasant or even unhealthy city. On the contrary, pollution, litter and public health hazards are rarely daily concerns.

Behind this clean, green environment, and keeping it going day to day is the National Environment Agency (NEA). Its officers work tirelessly to control pollution, ensure hygiene in food establishments, make sure the waste we generate is taken care of, prevent dengue outbreaks and provide weather forecasts and air quality readings. They plan and put in place strategies and procedures to safeguard our surroundings. And, taking it one step further, they engage the community and industry to build a Singapore where everyone has a role to play in cherishing the environment. Without the work of these individuals, the environment we know and enjoy today cannot be sustained.

For all the work it does, the agency is surprisingly young. It was only in 2002 that the statutory board came into existence. Before that, day-to-day operations to protect and manage the environment fell on the Ministry of the Environment (ENV), now known as the Ministry of the Environment and Water Resources (MEWR). Today, the ministry takes charge of policy-making to ensure long-term environmental sustainability while NEA shoulders the operations of environmental management.



"We went through a process of reflection. We had to find a solution for water, we had to find a solution to the disposal of solid waste. We had to find a way to solve our problems with public health. People were still littering, hawker centres and our public toilets were still not clean enough. I would say that provided the sense of long-term crisis, sense of urgency to change things."
— Lim Swee Say, Minister for the Environment (2002 - 2004)



(Source: Ministry of Information, Communications and the Arts, Courtesy of National Archives of Singapore)

AN ACTING MINISTER, A RANKING, AND NEW CHALLENGES

To tell the story of how NEA came into being, one must start with a newly-appointed acting minister, a surprising international ranking and a horizon of increasingly complex environmental issues.

What got things rolling was a report ranking the state of environmental sustainability across 122 countries. The report was released in January 2001 by the World Economic Forum, a non-profit foundation. Singapore was ranked 65th. In several areas, Singapore's performance was rated so poorly that it was placed near the bottom.

The assessment came as a shock to many. Even the authors of the report felt compelled to prepare a defence entitled "Isn't Singapore's score too low?" In a parliamentary sitting following the release, questions were posed to ENV. "... I am quite surprised to come across this study ... which rated Singapore as one of the 10 worst performers in the world," expressed one member of parliament.[1] Another member opined, "... I would like to take the World Economic Forum Report as an early warning signal, a wake-up call, so to say. The question we should be asking ourselves is: are we still doing enough?"[2]

There were those who felt the assessment was unfair, or "simply ludicrous"[3], as a journalist in *The Straits Times* put it. After all, another report released at about the same time by the Greening Industry Network ranked Singapore as one of the top performers among 36 countries surveyed in managing the environment. Moreover, compared to many other major cities, Singapore's quality of environment then was clearly in good stead.

For someone who had just joined ENV, Singapore's low ranking was out of the blue. Lim Swee Say, appointed in 2001 as Acting Minister for the Environment, came across an article reporting the rankings in *Newsweek* magazine. He recalls, "What got me thinking was this article, saying that Singapore was at the brink of environmental disaster. On one hand, we had always felt that we had done very well. Why was it that from an external angle, we were on the brink of environmental disaster? So that got me thinking, there were areas in which we might be doing well in the year 2000, but there were areas that we ought to try to do better for 2010, 2020 and so on. So I took that article in a positive way, more to remind us that no matter how well we did up to a point in time, we must always adopt the mindset that whatever we did was good enough for up to 2000, but might not be good enough for the year 2020 or beyond."

Which was a timely reminder indeed. The environmental challenges Singapore faced at the onset of the 21st century had certainly grown in magnitude and complexity. Global issues, such as climate change, scarcity of potable water and dwindling natural resources, were increasingly pertinent. The transmission of infectious diseases had grown much faster and wider with globalisation. International conventions and standards for environmental health and management were also becoming more stringent.

On home ground, rising population density, more intensive land use, and growing consumption of resources and waste generation demanded longer-term and more innovative solutions. As a small and open country, Singapore was vulnerable to trans-boundary pollution, especially as its neighbouring countries developed.

"Since its formation in 2002, the National Environment Agency has been playing the critical role of safeguarding our environment so that Singaporeans can continue to enjoy a high standard of living. The agency has a strong team of professionals to tackle the wide range of environmental issues such as keeping our streets clean, reducing the number of dengue cases, checking on hygiene standards of food outlets, encouraging the 3Rs (Reduce, Reuse, Recycle) *and promoting energy efficiency. It also has specialists in weather forecasting, environmental health research and radiation protection. From day one ... NEA understands that the challenge of sustaining a high quality living environment will not be fully addressed if it does not have the support of Singaporeans."*

— Dr Yaacob Ibrahim, Minister for the Environment and Water Resources (2004 - 2011) at NEA Community Day, 9 January 2010



As pointed out by a member of parliament, "The ... report was to measure the ability of a country to sustain its environment as it develops further into the future ... So in other words, I take it that, while the quality of our environment is high now, very good, the question remains that: can we sustain it and continue to keep the quality high in the future as more demands, such as population growth and economic development, are made on it? ... There are many disadvantages we have in this area, such as small land mass, lack of forest and rural areas, high population density, or relatively large industrial base, and the like, work against us in sustaining a good environment."[4]

REINVENTION

Clearly, it was time to foment change. At the ministry, Lim Swee Say immediately began a push for transformation. "We went through a process of reflection. We had to find a solution for water, we had to find a solution to the disposal of solid waste. We had to find a way to solve our problems with public health. People were still littering, hawker centres and our public toilets were still not clean enough. I would say that provided the sense of long-term crisis, sense of urgency to change things," Lim remembers. "The article in *Newsweek* was actually very timely and most helpful for me, because it gave me a framework to rally the ministry. That, yes, we have done well, but we must always remind ourselves, that environmental sustainability is a journey with no end."

Rounds of brainstorming, reflection, discussions and meetings later, the leaders in the ministry finally decided to take a bold step forward. A couple of months after the rankings were published, Lim announced to the parliament in March, "We are able to stay away from the threshold of environmental disaster because, all these years, we have been able to harness technology creatively and effectively. At the same time, we have applied our minds innovatively to protect our living environment with a great sense of commitment, responsibility and social discipline ... Looking ahead, our challenge in environmental management is to protect and enhance our living environment ... ENV will reinvent itself ... This will not happen overnight. Our aim is to achieve this over the next 18-24 months."

This reinvention was to take more than a year. Over time, the responsibilities and purview of ENV had expanded considerably. To streamline the functions of the ministry, the first step was to separate policy-making and operations. In short, the ministry would become a lean outfit focusing on policies. New statutory boards, responsible for operations, would be formed under it.

Such a move would have several advantages. Freed from day-to-day operations, the ministry could concentrate on envisioning Singapore's environmental future, identifying challenges on the horizon, and ensuring that the right policies and strategies were in place to bring about long-term sustainability. The statutory boards would focus on managing the environment and improving their operations to become more efficient and effective. Being closer to the ground, they would be faster and more flexible in addressing and responding to environmental needs and issues.

The transformed ministry would have two statutory boards under it. The Public Utilities Board (PUB), formed in 1963, would take charge of the country's water resources, and the new National Environment Agency (NEA) would cover all aspects related to environmental health and protection.

A New Agency

"We had a large workforce strength of about 3,000, including our daily-rated workers. Hence, it was clear that we needed everyone to know what was expected of him or her to be an NEA person. Every NEA person can be the eyes and ears of NEA to gather feedback on our environment. Furthermore, how they should behave and represent NEA is all encapsulated in a handbook for their reference."
— Lam Joon Khoi, CEO, NEA (2002 - 2005)



On 1 July 2002, the National Environment Agency was formally established, bringing under it the Environmental Public Health Division as well as the Environmental Policy and Management Division from ENV. The Meteorological Service Singapore under the Ministry of Transport joined the new agency as well, as its capabilities would enhance the agency's work, particularly in the areas of climate change and trans-boundary haze pollution.

It was an exciting time. The agency was to redefine environmental management in Singapore. Its new leaders, namely Lam Joon Khoi, CEO (2002-2005), and Professor Simon Tay, Chairman (2002-2008), set about the task immediately together with NEA's workforce, many of whom had come from ENV with years of experience. Lam came from previous stints in ministries including the Ministry of Defence, bringing fresh perspectives to the operations, while Professor Tay was noted for his passion for the environment.

"I was happy to think about NEA and its mission and it was also helpful that I teach environment law. For me, passion was important," reminisces Professor Tay. "I would have a weekly meeting with the CEO for debriefing. NEA has many functions under it, from waste management to hygiene. So I made it a point that I saw everything within a year, to get a texture of the job and what the staff did on a daily basis. I took it far beyond turning up once in two months for a meeting. This is a subject one can love. It affects everyone, it affects our children, it affects the quality of our lives, it affects the whole of our cityscape, so how can you not be interested or passionate about it?"

Building on the legacy of ENV, the fledgling agency took progressive steps within its first few years. Divisions were restructured for more synergy and efficiency. New roles were created. Different ways to perform its duties were explored, including implementing technological solutions more extensively. At the same time, as a statutory board, it had to manage its resources more closely than ever.

Above all, a new identity had to be shaped. In his first year, Lam, together with NEA's management, formulated a code to define the identity of the NEA Person. The comprehensive code covered diverse facets, from core values to guidelines for ethical behaviour in different situations. NEA's culture had to undergo a fundamental shift as well. Environmental management had long relied on surveillance and enforcement. But, for long-term environmental sustainability, it was essential for NEA to partner the community in protecting the environment. Steps were thus put in place to nurture a new mindset of partnership. These early years put in place a framework for NEA to carry out its mission.



"The NEA Person does not stop caring about the environment no matter where he/she is. As the eyes, ears and nose of the organisation, their environmental vigilance helps everyone in NEA do their job better."

— The NEA Person Code

"I think the most important transformation in NEA during my stint was in the overall approach and manner in which NEA carried out its work. The agency was then seen as a strict and rigid enforcer of environmental rules and regulations. However, such an approach was not sustainable in the long term. For NEA to succeed in its goal of safeguarding the environment and keeping Singapore clean and green, it was critical to gain the support of the people. We needed to inculcate in the people and community a pro-environment mindset."
— Lee Yuen Hee, CEO, NEA (2005 - 2009)



CONSOLIDATION AND TRANSFORMATION

Consolidation, fine-tuning and furthering the mission of NEA marked the key concerns of the agency after the first few years. In 2005, Lee Yuen Hee, who had filled senior positions in several ministries such as the Ministry of Trade and Industry, joined as the second CEO of NEA. He recalls, "Although NEA was still young as a statutory board, I was fortunate that my predecessor had laid a strong foundation which I was able to build upon. As CEO, my main priorities were to consolidate and build upon the good work that had been laid down to deepen the capabilities of NEA in dealing with existing and new environmental challenges that surfaced."

Under Lee, NEA's bid to become a facilitator and partner also received greater focus. "I think the most important transformation in NEA during my stint was in the overall approach and manner in which NEA carried out its work. NEA was then seen as a strict and rigid enforcer of environmental rules and regulations," shares Lee. "However, such an approach was not sustainable in the long term. For NEA to succeed in its goal of safeguarding the environment and keeping Singapore clean and green, it was critical to gain the support of the people. We needed to inculcate in the people and community a pro-environment mindset." The vision of NEA was revised to become more inclusive, "embracing all in caring for our clean and healthy environment".

Partnering the private sector reached greater heights when Chew Gek Khim, a well-known corporate leader, assumed the position of Chairman in 2008. Her experience in the private sector, network and knowledge of good corporate governance proved to be invaluable to the agency. In her first term, she noted, "Whilst some may quip that NEA stands for 'nearly everything agency', its mission can be summed up by rallying society to embrace the mantra, 'Clean and Green Singapore'... In this respect, I hope to complement NEA's efforts in engaging the corporate sector to espouse the Clean and Green Singapore concept."

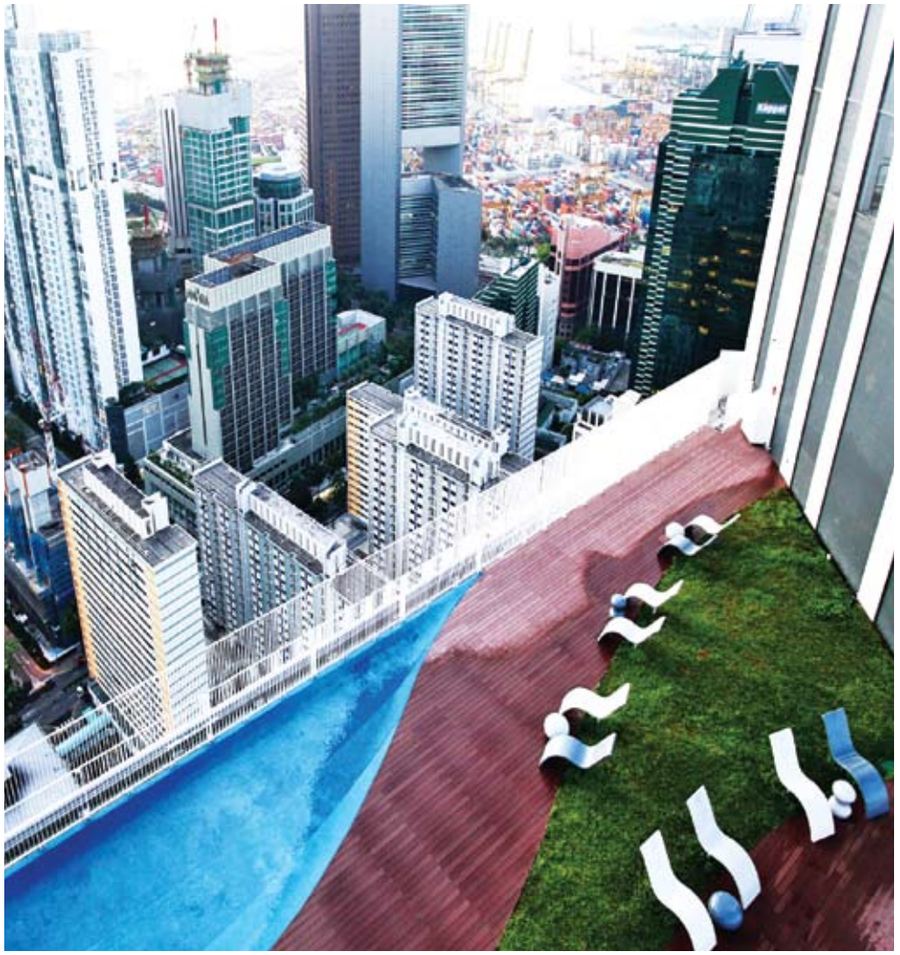
By the late 2000s, the environmental challenges faced by Singapore had evolved to become more complex and cross-cutting. Just as ENV once underwent transformation to meet rising challenges of the time, likewise, it was time for NEA to renew itself. This was top of the agenda when Andrew Tan took over the helm as CEO in 2009. He says, "I would say that my first priority, coming in with fresh perspectives from my previous postings in the Ministry of Foreign Affairs and the Ministry of Defence, is whether this organisation is prepared to respond to the challenges ahead. This may seem like a very simple question, but actually it goes back to the fundamental mission of NEA and also whether we have the capabilities to achieve our mission. To achieve this, I would say that we are tackling a very huge task, because we are talking about organisational transformation. So that was the platform which I came on, to transform the organisation, to get it future-ready for the 21st century."

"I would say that my first priority coming in with fresh perspectives from my previous postings in the Ministry of Foreign Affairs and the Ministry of Defence, is whether this organisation is prepared to respond to the challenges ahead. This may seem like a very simple question, but actually it goes back to the fundamental mission of NEA and also whether we have the capabilities to achieve our mission. To achieve this, I would say that we are tackling a very huge task, because we are talking about organisational transformation. So that was the platform which I came on, to transform the organisation, to get it future-ready for the 21st century."
— Andrew Tan, CEO, NEA



In addition to the various initiatives launched, such as the development of new capabilities, re-allocation of resources and increased focus on research to support operations, Andrew Tan also applied the concept of total defence to the environment. "We reframed our mission to 'safeguard, nurture and cherish'. So it comes out very strongly that there's something important enough, not only for NEA but also the community at large to safeguard. 'Nurture' and 'cherish' have to do with not just looking at the current generation, but at future generations. We want to bring up future Singaporeans with a sense of responsibility towards the environment and whom we are able to hand over the environment to."

Within a short decade of its founding, NEA has come far in its journey. It has faced and overcome diverse challenges, renewing itself constantly to become better and faster. It has played critical roles in national crises time and again. This is the story of NEA, and the people who made a clean, liveable, healthy, sustainable Singapore a daily reality.



1 Kenneth Chen Koon Lap. 2001. *Budget, Ministry of the Environment.* Sitting No.: 10. 15 March.
2 Heng Chiang Meng. 2001. *Budget, Ministry of the Environment.* Sitting No.: 10. 15 March.
3 Nathan, Dominic. 2001. Singapore: Green Paradise or Polluted City? *The Straits Times.* 29 January.
4 Heng Chiang Meng. 2001. *Budget, Ministry of the Environment.* Sitting No.: 10. 15 March.

Safeguard

a place to live





"At the end of the day, if you want the sky of Singapore to be always blue, the land of Singapore to be always green, the water to be always clean, the public health to be always pink, it counts a lot on these many faceless heroes in MEWR, NEA, PUB as well as many faceless heroes in the industry and the workers in the environment sector. We have good people in MEWR and NEA. They know their jobs well, and they know the challenges we face. They are passionate and committed to meeting these challenges. Most of all, they are prepared to adapt to changing circumstances so as to meet new challenges with new mindsets."

— Lim Swee Say, Minister for the Environment (2002 - 2004)



Top: Chinatown in 1970. Scenes such as this were common in the past. (*Source: The Straits Times © Singapore Press Holdings Limited. Reprinted with permission.*)

Opposite: Pagoda Street in 1972. (*Source: The Straits Times © Singapore Press Holdings Limited. Reprinted with permission.)*

Snapshot. Singapore in the 1960s. Streets were more often than not cluttered with all kinds of refuse. One could see discarded tyres, bottles, nets and unidentifiable waste floating and choking up streams, canals and rivers. The water would be black, and stenches were often released. Right next to sawmills, workshops and factories, residences would be found, and people would live amidst the dust, noise and unpleasant smells.

In a matter of decades, a complete transformation had taken place. Singapore had become clean. So much so that it is something we hardly think about today.

Clean air, clean water, clean land. These do not come by accident. When Singapore gained self-governance in 1959, it inherited a heavily polluted environment, especially in the core of the city. In the outskirts of Singapore, there was severe land pollution following decades of dumping untreated refuse in swamps, coupled with liberal use of pesticides such as DDT to prevent mosquito breeding.

At the same time the new government grappled with urgent issues such as housing needs, unemployment and high crime rates, it also began to look into eradicating and controlling pollution through appropriate planning and zoning of land use on the island. With independence in 1965, the government had greater control over the master planning of Singapore's development. Alongside an aggressive agenda for economic development and industrialisation, the government had the foresight to lay down policies ensuring that our environment would not be compromised. This commitment to protect and safeguard our living surroundings is a cornerstone of Singapore's transformation.

Such a commitment is all the more crucial in the light of the challenges we face today. With rapid globalisation and technological advancements in industrial processes, the scale of environmental incidents is potentially further-reaching and more complex than before. Increasing industrialisation also means a growing amount of emissions from our industries. Within Singapore, the environment faces more stress as the population grows and land use becomes more intensive.



The task of bringing about clean air, clean water and clean land falls on NEA, which translates the policies and direction set by MEWR into action. NEA shapes the planning of Singapore's land use, regulates the development of industries and makes sure that there is a balance between economic growth and environmental well-being. It stays on the alert and constantly monitors the quality of our air, water and land. It implements a stringent routine of licensing and enforcement to prevent environmental hazards from taking place. It also goes beyond its enforcer role to reach out to industries to educate, inform and encourage innovation.

This will not be possible without the people in the agency, who work to safeguard and protect the environment. Much of what they do is not glamorous, and for some, it is tedious. Yet talk to any of the workers and they often speak of their work with pride. They are the people who bring industries in line with our strict requirements, who go to the ground to iron out issues. They are planners, inspectors, incineration plant operators, technicians, scientists and engineers, working in concert to keep our home clean and green. A home where we can breathe in clean air, enjoy clean beaches and work in liveable, sustainable surroundings.

"The transition of the operations of environmental protection from the ministry to NEA in 2002 was smooth as the planning had been thorough and comprehensive. Nonetheless, there was a sense of autonomy from the ministry and with that came greater responsibility and accountability. We worked to ensure that Singaporeans could continue to enjoy clean air, clean water, clean land and a high standard of public health, and this was realised with the strong commitment, conviction and the camaraderie of the officers in NEA working to protect the environment."

— Loh Ah Tuan, Director-General, Environmental Protection Division, NEA (2002 - 2007)





Safeguarding Essentials

Sound principles, a strong legal framework, thorough planning. Add to that a rigorous regime of surveillance, licensing and enforcement, constant innovation and the roping in of different stakeholders. Such is the formula employed by NEA to safeguard Singapore's environment, to make sure that our air, water, and land are clean.

THE PRINCIPLES OF IT ALL

Pollution that goes unchecked is one of the greatest ills of industrialisation.

Take, for example, the case of widespread mercury poisoning during the 1950s in Minamata, Japan, after a chemical factory released industrial wastewater with high mercury content into the city's waterways. Or the discovery in the 1950s and 1960s that the use of leaded petrol was linked to lead poisoning in humans and lower intelligence in children.

There are many other similar cases. Singapore, being a relatively newcomer to industrialisation, has the advantage of learning from such experiences in other countries. Which is how it formed three simple principles for environmental protection.

The first principle is to exercise "pollution control at source". Literally, this means controlling pollution right where it happens before it escalates into bigger issues. Key sources of pollution – industrial facilities, power plants and vehicles – are regulated to make sure they do not pollute the environment.

Next, the "polluter pays principle" is an idea rooted in fairness. Those who pollute are held accountable for managing the resultant pollution. For example, industries that produce toxic waste are responsible for treating the waste before they can dispose it.

The third principle, to "pre-empt pollution and take early action", is derived from the fact that our understanding of the environmental impact of technology is far outpaced by its advancement, as the case of leaded petrol demonstrates. Thus, NEA watches closely for potential sources of pollution and takes pre-emptive action where possible. An example is Singapore's control of endosulfan, a pesticide harmful to health, in 2008. Three years later, this pesticide was added to the list of persistent organic pollutants governed by the Stockholm Convention, an international treaty to restrict organic environmental pollutants.

Armed with these three principles, everything else follows, from a legal framework to planning infrastructural developments.

LAWS FOR A BETTER ENVIRONMENT

Right up to the 1970s, laws and regulations in Singapore were still inadequate to protect the environment. For example, even though it was illegal to discharge matter that could be a nuisance or danger to health, no legal quantities were set. The provisions were also ineffective since it was debatable what constituted a nuisance or danger to health.

This began to change in the 1970s when several laws were introduced to improve public health and environmental protection. A key milestone was the Clean Air Act of 1972, which specified industrial emission limits, among other regulations. These laws were precise in their requirements and allowed better control over pollutive activities. Together with the establishment of ENV in 1972, dramatic improvement in Singapore's environment was accomplished in a matter of decades.

"The role of law in Singapore cannot be divorced from the administrative backbone to implement it. The fact that we don't have corruption on a systemic level means that companies can't just pollute and get away with it. Pollution control is partly technological. We have monitoring everywhere. In major factories, we literally can monitor what is coming out of the smokestack the moment it is happening. And I'm glad that we have moved beyond law as enforcement to co-enforcement. Now you can see a lot of major companies tracking their own pollutants. The environmental fact, the law as lived, is clearly strong."

— Professor Simon Tay, Chairman, NEA (2002 - 2008)



Over time, Singapore's environmental laws have undergone several rounds of revisions and consolidation to meet changing needs. Today, NEA administers a number of acts with their accompanying regulations, such as the comprehensive Environmental Protection and Management Act.

Under these laws, the private sector is also required to do its part in controlling pollution. For instance, any development with the potential to cause serious pollution has to conduct a Pollution Control Study. The study identifies sources of pollutants and recommends measures to prevent or reduce the impact of pollution. Industries also have to implement their own monitoring programme and ensure timely response to any incidents.

To make it easier for industry stakeholders to understand Singapore's environmental requirements, a guidebook, called the Code of Practice on Pollution Control (COPPC), was published in 1994. The code, which has been updated several times, lays down the requirements for every type of development, from industrial estates to business parks. It also shares good design practices with architects, engineers, industrialists and consultants.

A CASE FOR FLEXIBILITY

Did you know that for trade effluent to be discharged into sewers, its pH value must be between 6 to 9 and its temperature cannot exceed 45°C? NEA also sets the limits of substances in trade effluent such as sulphates, chlorides and heavy metals. These are just some of the requirements listed in the Code of Practice on Pollution Control to ensure that trade effluent does not cause undue pollution in the waterways. The regulations are necessarily demanding to make sure that our environment is not compromised by industrial activities.

The code does not specify how trade effluent, emissions and such are controlled. Instead, it lists the desired outcomes of pollution control. This approach was taken after consultations with industry stakeholders between 2000 and 2004. Businesses can decide on the best and most cost-efficient methods to control the pollution they create, and exercise creativity in coming up with systems of environmental management.



Making A Difference At The Onset

Have you ever wondered why your home is not placed right next to a noisy factory? Or why, if you are enjoying a meal at a hawker centre, the smells of industries are not part and parcel of your experience? Much of our daily lives go unaffected by industrial activities, thanks to intensive planning of land use by the government, a sound regime of inspection and enforcement, and NEA's close collaboration with the industry.

PLOTTING MATTERS

Land-use planning was adopted in earnest in 1971, when the first Concept Plan was launched. The plan mapped out broad strategies and long-term use of land for a period of 40 to 50 years, and has been updated every decade since. Details for development for the next 10 years, such as specific zones for housing or industrial development, are captured in Singapore's Master Plan.

The drafting and reviews of these plans are driven by the Urban Redevelopment Authority (URA), with inputs from various government agencies including NEA. This inter-agency collaboration allows for optimal use of land resources, ensuring that the environment remains clean, liveable and sustainable even as the economy develops.

Walk around the housing towns in Singapore and you would observe that within the estates, factories that may cause pollution are hardly to be found. Light industries are usually sited some distance away, and general and special industries operate only in purpose-built parks. This deliberate arrangement establishes distance buffers between industrial and residential zones to reduce the potential impact of pollution and nuisance on Singaporeans. Planning is carried out even for every plot within the zones. For instance, care is taken to locate compatible industries near each other. Industries sensitive to cross-contamination, such as food factories, will not be situated near general or special industries.

There are other factors taken into consideration too. For instance, facilities sensitive to noise, such as schools and hospitals, are located further away from expressways. In allocating land use around airports and airbases, the Noise Exposure Forecast (NEF), a measurement of exposure to aircraft noise, is used. Areas experiencing NEF *35* and above are used for industrial developments while residential development is allowed only in areas experiencing NEF below *35*.

Protecting our water catchment areas is particularly critical as water is a scarce resource here. Development in protected catchment areas is strictly controlled. In unprotected water catchment areas, only clean and light industries that do not use large amounts of hazardous chemicals are allowed, so as to minimise the risk of pollution to our water resources. Pollutive industries are located strictly outside catchment areas.

By planning far in advance, a balance is achieved between the demands of development and a high quality of environment.



"For every land use, we have to assess its suitability in terms of its potential environmental impact and dis-amenities that it can pose to the surroundings. Through the assessment, we will then state our environmental requirements to the developers to ensure that their activities can co-exist with the other activities in the area. The challenges that we face include meeting the expectations of industrial developers and the public, and to balance economic growth with our environmental objectives. This will be a greater challenge in the future, because of our limited land area coupled with continuous growth in population and industry. What we are doing now, results may not be imminent. Time will tell if we have done a good job. We often work behind the scenes, and at times we may only be perceived as having increased the operating cost of the industry. Nevertheless, if we can continue to enjoy clean air, clean land, clean water at the end of the day, I will call it an achievement!"

— Chow Chee Kiong, Senior Engineer, Central Building Plan Department, NEA

"Factories operate in different ways, and inspectors need to have a good understanding of the various industrial processes and the pollution control systems. The inspector must understand the factory, the production lines, and if the equipment fails, why it fails and what may go wrong. The inspector must also know the ground, the factory and its surroundings. For example, if a factory is near the water catchment area, the inspector must know how it will impact the water. Every day, every inspector carries out four to six inspections. When a factory is causing pollution, our inspectors also have to become witnesses for prosecution. They play a very key role in making sure that there is compliance."
— Jothieswaran Poobalasingam, Chief Engineer, Pollution Control Department, NEA

CONTROL, RIGHT AT THE SOURCE

Controlling pollution right at the source is a simple idea, but a complex act to follow through.

Before an industry is set up here, NEA first assesses if it is suitable for Singapore. Industries highly detrimental to the environment, and those generating significant amounts of toxic industrial waste, are not permitted even if they are able to add significant economic benefits to Singapore. Some examples would be paper mills and ore smelting plants. Companies bringing in obsolete, pollutive technologies are not welcomed either.

When the Economic Development Board (EDB) brings in potential investors for industrial projects, NEA informs these investors of Singapore's pollution control requirements. By laying out all the regulations in advance, NEA ensures that investors can make an informed choice on the suitability of Singapore for their projects.

As soon as an investor submits a proposal to build an industrial facility in Singapore, NEA will determine if the land parcel is the right fit, taking into account the needs of nearby users. Once the proposal is approved, a process of licensing and inspection will kick into motion. NEA officers will visit the site, ensuring that from design to completion, measures are in place to control pollution. In an average year, NEA receives around 100,000 related applications, from issuance of licences to temporary occupation permits.

A RIGOROUS REGIME

Without a rigorous regime to administer our laws and regulations, there is no way for environmental protection to be truly effective.

First, NEA uses licensing to track and regulate the use of materials which may impact the environment. Licences for factories or usage of hazardous substances, for example, are only issued when applicants show that they are adhering to legal requirements, such as having the necessary pollution control equipment and containment facilities in the premises.

Next, the agency's factory inspectors carry out regular checks on industrial premises to ensure that there is compliance with regulations on the ground. If the premises are found to be flouting requirements, enforcement measures will be taken. These measures range from the issuance of warning letters to prosecution in court, depending on the severity of the violations.

Finally, NEA practises constant vigilance to keep pollution in check. The idea is simple – watch the health of the environment closely and pollution can be nipped in the bud before it escalates. The constant monitoring of our air and water not only provides an up-to-date report card of the environment, it also allows NEA to detect and respond to pollution incidents quickly. More importantly, it helps NEA to track the effectiveness of its programmes and policies in protecting and improving our environment.

"In the early days, you could find sawmills at Lavender Street! Can you imagine, a sawmill at Lavender Street? And people were living next to it. That was how it was, because there was no need for people to get permission to set up a factory. But slowly, this changed. After the Anti-Pollution Unit was formed, any industries that came in, they had to get approval for the piece of land and we would see whether or not it has any impact on the environment. Also at the same time, we would see if this industry is suitable for Singapore. Because Singapore is so small, we cannot have industries that generate a lot of waste. And certain industries use a lot of water, so they are also out, because Singapore is very short of water. And if an industry generates waste that cannot be treated in Singapore, let's say radioactive waste, or some waste so toxic that we don't have the facilities to treat them, we would not permit the industries to set up here. So firstly, whether the industry is suitable. And if the industry is suitable, whether the siting is suitable. For example, we cannot allow sawmills to be in residential areas. Then steel mills cannot be sited next to food factories. So within the factories, we must also plan, otherwise there may be cross-contamination. And we make sure that industries sited close to each other are compatible. After we look at the site, we look at processes. We don't want a company to bring in pollutive, outdated or obsolete processes and technology to Singapore and generate a lot of pollution. So we will make sure that they will bring in the right kind of technology and pollution control equipment."

— Joseph Hui, Deputy CEO, NEA





A STORY OF PLANNING

Think about it. The tiny island state of Singapore, without oil resources of its own, is one of the top petrochemical hubs in the world.

What is even more interesting is how the quality of Singapore's environment remains one of the best in the region, even with over 100 companies including leading petrochemicals companies that are heavily pollutive in nature. Talk about having your cake and eating it too.

The key to the puzzle lies in Jurong Island, located just a few kilometres off the shore of mainland Singapore. The result of land reclamation carried out in the 1990s to form a land mass out of seven smaller islands, Jurong Island is located offshore to minimise the degradation of the quality of environment and impact of potential accidents on mainland Singapore. Reclamation works began under Jurong Town Corporation (present-day JTC Corporation) in 1995. Working closely with ENV (present-day MEWR) and other government agencies, JTC took care to incorporate facilities for pollution control in Jurong Island's infrastructure.

Today, Jurong Island is the premier hub for the chemical industry in the region, with one of the largest oil refining and fuel storage terminals in the world. Even more impressively, its air and water quality, monitored by NEA, is within international standards and consistently comparable to that in mainland Singapore.

The importance placed on planning for the environment is all the more pronounced in the latest initiative rolled out by the government – Jurong Island Version 2.0. More petrochemical companies are expected to set up here, which means that several environmental challenges are posed. New plants and chemical pipelines will have to be located closer to existing facilities. As such, chances of a "knock-on" effect, whereby an explosion from a plant may cause secondary explosions or accidents in neighbouring plants, are higher. The storage, transportation and use of hazardous chemicals will be significantly increased. There will also be more sources of pollution. The consumption of resources such as electricity and water, and generation of waste will also go up.

Driven by EDB, Jurong Island Version 2.0 aims to address these new challenges as well as improve existing environmental infrastructure. Multiple agencies, including NEA, are involved in the project. NEA plays a critical role in the realisation of Jurong Island Version 2.0 by providing technical expertise and advice. It works with other agencies to develop detailed guidelines and safety procedures to address the "knock-on" effect in potential accidents. More stringent control of air emissions from the industries is looked into to make sure that air quality in mainland Singapore and Jurong Island will not deteriorate. Conserving resources being a priority, NEA advises the agencies involved on reducing and recycling waste and maximising use of waste energy, such as the possible conversion of waste heat into electricity. NEA has also commissioned consultancy studies to study waste management, treatment and utilisation in Jurong Island.

By ensuring that such measures are integrated in the planning and development stage, a more competitive, self-sustainable Jurong Island Version 2.0, one of the few petrochemical hubs in the world where the quality of environment remains safe and high, is set to be realised.

INNOVATING ON ONE HAND, ROPING IN THE INDUSTRY ON THE OTHER

With the number of industrial premises, vehicles and developments in Singapore, making sure that each of these complies with environmental regulations is a great challenge. To address this, NEA constantly explores new technology, employing innovation to improve and enhance operations. This approach can be seen in all aspects of NEA's work in environmental protection, from remote monitoring of pollutive emissions to investing in informaton technology systems to manage operations.

Another key strategy to control pollution right at the source lies in roping in the industry to do its part. The industry is closer to the ground and can better assess whether certain solutions are viable. The government can also free up resources for other uses if the private sector takes more initiative at its end. Most importantly, this will foster a culture in the industry whereby environmental management becomes an integral part of business concerns.

One way in which NEA engages the industry is through regular consultations. During such sessions, both parties share insights and put forward their points of view. With the feedback, NEA can gauge if its policies and regulations are feasible, and implements changes where necessary. NEA also actively encourages industries to take on more self-policing and strive to reach higher than stipulated standards. For instance, the agency supports the Responsible Care Programme launched by the Singapore Chemical Industry Council to promote good management and handling of chemicals.



"ExxonMobil enjoys a collaborative working relationship with the National Environment Agency through the years. The agency makes an effort to engage with industry players and to understand the complexity and specific requirements of the refining and petrochemical industry, as well as its associated needs."

— Derk Jan Hartgerink, Manufacturing Director, Singapore Chemical Plant, ExxonMobil Chemical Asia Pacific

The Drive For Clean Air

Breathe in, deep. And enjoy just how fresh the air is. Despite having to share a small island with a large number of factories and vehicles, taking a deep breath is rarely a health hazard for Singapore's residents. NEA plays no small role in bringing this about. It tracks industrial emissions, enforces against polluters, implements higher benchmarks for air quality and monitors the health of our ambient air round the clock.

"NO" TO SMOKE ON THE ROADS

Vehicles, factories and power plants – these are the key sources of air pollution anywhere in the world. What makes vehicular emissions all the more dangerous is the fact that pollutants such as carbon monoxide, nitrogen oxide and particulate matter are released at ground level, mixed in with the air we breathe.

Restrictions on emissions have become increasingly stringent as the years passed. For instance, emission standards for motorcycles and scooters were further upgraded in 2003. And in 2006, the Euro IV emission standards, which demand lower sulphur content in diesel, were imposed for all new diesel-powered vehicles.

Fuel quality is also tackled by the agency. Take sulphur content. Sulphur is naturally found in crude oil, which is refined into diesel fuel. The substance is largely responsible for the release of particulate matter and sulphur dioxide, both of which are pollutants. In 2005, NEA reduced the permissible sulphur content in diesel to 0.005% by weight to pave the way for ultra-low sulphur diesel.

Imposing such requirements involves a lot of work. For the implementation of Euro IV standards, NEA had to consider issues such as the availability of Euro IV compliant vehicles, and social and economic costs. Generally, manufacturers need about two years to roll out new designs of vehicles equipped with more efficient engines and pollution control devices. Another two years are needed for local car dealers to clear existing stock and bring these newly-improved vehicles to Singapore. Altogether, it would take a few years before a more stringent standard can be effectively implemented.

NEA also enforces penalties upon those who drive and who own smoky vehicles. In the past, this would be manually performed by officers armed with smoke meters. Nowadays, officers equipped with video cameras stake out the roads. They capture smoky vehicles on video, scanning at least 5,000 vehicles a day. Drive past the checkpoints at Woodlands and Tuas, and you may very well see officers on the lookout as well. Malaysia-registered vehicles are also subject to the regulations and those emitting smoke measuring more than 50 Hartridge Smoke Units (0 HSU is clear while 100 HSU is opaque) are fined. Those emitting smoke more than 70 HSU are not allowed to enter Singapore.

Drivers who leave their engines idling are taken to task, except under specific conditions such as taxis and buses waiting in a moving queue. This is to minimise the emission of exhaust gases, odour, noises and fuel waste caused by idle engines.

Education also plays a key role in NEA's efforts to combat vehicular emissions. The agency works with large vehicle fleet owners such as taxi and bus companies and the National Association of Travel Agents Singapore to educate drivers to turn off their engines while having meal breaks so as to prevent idling engines. There is also more public feedback on idling engines as Singaporeans become increasingly aware of the harmful effects of pollution caused by inconsiderate motorists.

As a result of NEA's efforts, the number of smoky vehicles on the road has dipped considerably to about 0.6% of the vehicle population in 2011 from about 5.8% in 1990.



PASS THE TEST FIRST!

No person shall use or cause or permit to be used on the road any motor vehicle from which any smoke or visible vapour is emitted.
— Regulation 19 (1), Environmental Protection and Management Act

If it is smoky, it is not going anywhere. Every vehicle in use is required to go for a smoke test during its mandatory periodic inspection at an authorised inspection centre, which checks for the amount of carbon monoxide and smoke released. Depending on the type of vehicle and year of registration, permissible emission limits vary. Any vehicle that fails the periodic inspection is not allowed to renew its vehicle road tax until it passes.

Today, all authorised inspection centres adopt a procedure known as Chassis Dynamometer Smoke Test (CDST) to measure the smoke emissions of diesel vehicles. Piloted in 2000 and made mandatory since 2007, the test simulates actual driving conditions, making the test more accurate than the previous procedures used. This enables NEA to be more effective in its inspection, and further reduces the number of smoky vehicles on the road.

A 24/7 Watch

Fifteen ambient air-monitoring stations around the island, sensors installed in the chimneys of factories, video cameras trained on smoke stacks in industrial parks – NEA keeps an eye on signs of air pollution 24/7.

This continuous monitoring achieves some important goals. It allows NEA to pick up on any deterioration of air quality straightaway and respond immediately. There is ready evidence for these acts of non-compliance when it comes to enforcement against culprits. And NEA can know for certain if its policies and programmes to ensure good air quality are effective.

Up to the mid-2000s, monitoring industrial air emissions was still a time-consuming task. NEA's engineers had to scale structures near the towering smoke stacks to extract air samples. The samples would be put through a series of laboratory tests to check for impurities. Such a system meant that NEA could only assess the quality of air emissions at a particular time, whereas the contents varied through the day.

As technology advanced, better solutions were invented, such as sensors which could analyse air pollutants continuously. Pollutive industries were required to install such devices to track their own emissions. To monitor emissions from these industries, NEA hit upon the idea of linking these sensors up to a central data network. This would give NEA specific, real-time information on industrial emissions. The system piloted in 2007 and officially kicked off in 2010. Major emitters of air pollutants such as power plants are required to install the monitors.

NEA also carries out round-the-clock visual surveillance on the smoke stacks of industries. Video cameras scan key industrial areas for visible smoke. With this, NEA can address public feedback on smoke emissions immediately, as the agency can instantly identify the offending smoke stack using real-time video. Officers can be deployed quickly to ascertain the ambient air quality in the area as well.

Equipped with such an extensive surveillance system, NEA's efforts in ensuring good air quality for Singaporeans are now much more effective and efficient.

READING THE AIR

"Monitoring allows us to find out where we stand in comparison to international benchmarks for the environment. We have formulated air pollution control strategies and programmes to address air pollution sources. But even as we put in place stricter regulations, the sheer numbers of vehicles, our demands for energy, urbanisation and industrialisation pose real challenges. And as international standards become higher, it becomes even harder to reach the new targets. At the same time, pollution control measures come at a cost. So we have to strike a balance. All of us have a part to play."

– Indrani C Rajaram, Chief Scientific Officer, Environmental Monitoring and Assessment Unit, NEA

On the surface, it sounds really simple. Sophisticated hi-tech analysers are fitted in 15 ambient air monitoring stations island-wide. They detect the levels of six main pollutants in ambient air – carbon monoxide, nitrogen dioxide, ozone, particulate matter 10, particulate matter 2.5 and sulphur dioxide. The data, stored in a central network, allows Internet-based access for NEA officers.

Certainly, compared to the 1970s and 1980s, tracking ambient air quality is now more efficient. Formerly, officers had to manually collect samples and conduct laboratory tests to determine the air quality. Automated continuous monitoring of ambient air kicked off in 1994 with the installation of a telemetric system. Upgrades were carried out in 2005 to enhance the system. Today, air quality is tracked at ambient and roadside stations. There is even a mobile air monitoring station to aid on-site investigations.

But even with such a sophisticated system, monitoring the air still requires a strong team of scientific and technical experts. Indrani C Rajaram,







Chief Scientific Officer, leads NEA's team in monitoring and assessing the performance of Singapore's air and water quality. Says Indrani, "The instruments we have are highly specialised equipment that have to be calibrated and checked regularly, and the data they produce must be validated constantly. This is to make sure that what the instruments are giving us are not aberrations, but accurate measurements of the pollutants in the air. There is quite a lot of work to be done."

Indrani's team is behind the daily Pollutant Standards Index (PSI) reading we hear every day. The PSI uses a scale ranging from 0 to 500 to measure air quality. At readings of 50 and below, the quality of air is rated as good, while any reading above 100 indicates that the ambient air is unhealthy. Armed with such information, members of the public can decide if they should undertake certain activities such as outdoor sports. This service becomes even more critical during events such as the periodic haze.

"We had to deploy people round the clock to report the PSI reading every hour during the haze," recalls Indrani. "The PSI reading is based on the worst concentrations of air pollutants, because the reading is for members of the public to take the necessary precautions to protect themselves from the harmful effects of the haze. So we take the conservative approach. We compute the PSI reading based on the highest concentrations of the air pollutants using data from all the ambient stations, even if there are pockets of Singapore where the concentrations may be lower."

As NEA is the national authority on the quality of air, the team also plays a critical role in assuaging public concerns when the haze comes around. "Coping with the sheer number of queries was a big challenge!" says Indrani, remembering the haze in 2006. "The public was very concerned. We had callers wanting to know what NEA was doing about the quality of air, people who had doubts about the PSI readings and such. Many of them wanted to know if they could carry on with their normal routines even with the haze. So in between working round the clock in shifts to report the PSI readings and making sure that our equipment was performing all the time, our team had to answer these queries and allay their concerns."

What is even more challenging is helping to ensure that Singaporeans continue to enjoy a high quality of air even as the country develops. Indrani and her colleagues support NEA's drive for better air quality by contributing to new, higher benchmarks for Singapore's environmental performance and constantly improving their monitoring competencies. The unit is looking at launching more powerful systems to monitor the environment and even real-time sensors for our coastal water quality. To Indrani, who has been in this line since 1978, these challenges are exciting opportunities to find better solutions. She shares, "There has never been a dull moment. Work here has been so exciting! We know that whatever little we can do is going to help. When new standards for air or water quality are announced, we know that we are the guys behind coming up with these standards for Singapore. That's really rewarding."

"October 2006, in the middle of the fasting month. The haze was really very bad. There was a breaking of fast I attended and on the way back, I drove down River Valley Road. I couldn't see in the night! So the next day, I said, we need to do something about it. So we called for a regional meeting."

– Dr Yaacob Ibrahim, Minister for the Environment and Water Resources (2004 - 2011)

ALL THE WAY TO JAMBI

It was one of the worst periods of haze Singapore had ever experienced. In October 2006, the city was shrouded in grey smoke. The smell of burning permeated the air. On 7 October, the PSI reading shot up to a high 128, well in the unhealthy range. NEA's advisory for the day read, "The smoke haze situation has deteriorated overnight ... The latest satellite pictures showed that there were 506 hotspots and thick smoke haze in Sumatra ... persons with existing heart or respiratory ailments to reduce physical exertion and outdoor activities. The general population should reduce vigorous outdoor activity."

This was not the first time Singapore had experienced the haze. The last bad onslaughts were in 1997 and 1998. Periodically, when the southwest monsoon comes around between June and September, it brings the dry season along. In Borneo and Sumatra, peat fires, as well as those started by farmers using the "slash and burn" method to clear the land, spread fast and more intensely during this period. The situation is further exacerbated when it coincides with the El Niño phenomenon, which usually leads to an extended dry spell. The fumes from these fires are carried by prevailing winds to Singapore, Malaysia and parts of Thailand.

In 2006, the situation was critical. There were significant health risks posed to Singaporeans. The quality of life and economy suffered too. Given the seriousness, Prime Minister Lee Hsien Loong wrote to President Susilo Bambang Yudhoyono on 10 October to express concern. Singapore also called for an urgent regional meeting of environment ministers from the countries affected by the haze.

The meeting took place on 13 October in Pekanbaru, Indonesia. Following the meeting, ministers from Singapore, Brunei, Indonesia, Malaysia and Thailand formed a steering committee to oversee initiatives to address the haze. They also decided on meeting once every three months to oversee a plan of action.

Senior officials from ASEAN member countries, international and regional non-governmental organisations (NGOs) and inter-governmental organisations came together in November to brainstorm and develop a comprehensive plan. In the plan, 35 fire-prone areas in eight provinces of Indonesia were identified. Next, Indonesia called for ASEAN member countries and international organisations to aid these provinces in building up capacities to prevent and fight land and forest fires. A technical working group, with members from the affected countries, was formed to support such initiatives.

For Singapore, it was clear that if the initiatives were to be effective, they would have to be implemented at ground zero, right where the fires





Left: Initiatives, such as this ministerial steering committee workshop in 2011, are regularly held to address the haze issue.
Right: An ambient air monitoring station installed by NEA in Jambi, Indonesia.

took place. Hence, the government decided to respond to Indonesia's call and offered its help in Muaro Jambi Regency. The regency was selected by the Jambi Provincial Government as it is fire-prone, and its proximity to the city area means that many residents are adversely affected when the haze happens.

Together with the Jambi Provincial Government and Indonesia's State Ministry of Environment, Singapore developed a master plan to deal with land and forest fires. A team comprising officials from MEWR and NEA was sent to Jambi in January 2007 to understand the ground better. After much research, the master plan was finally drafted and finalised in March 2007.

The plan outlined strategies for sustainable land-clearing practices, fire prevention and suppression. Specific programmes were also designed to realise the plan's objectives. While the Jambi Provincial Government was responsible for implementing the programmes, it also received assistance from Singapore and other ASEAN countries as and when it made such requests. At the invitation of the provincial government, Singapore signed a Letter of Intent on 7 November 2007 to help execute a number of programmes, setting aside $1 million to do so. The key agency representing Singapore in these efforts was NEA.



NEA officers with locals in a village in Muaro Jambi Regency.

GROUND UP EFFORTS

"It impressed upon me very strongly that the haze is not just an environmental problem, and it cannot be addressed from the environmental aspect alone."

– Rohaya Saharom, Deputy Director, Pollution Control Department, NEA

Rohaya Saharom was one of the engineers sent by NEA to Jambi to help tackle the haze issue. A member of the fact-finding mission in 2007, her technical expertise and fluency in the Malay language were critical assets on the ground. She recalls, "Once we were there, we could get a sense of the magnitude of the problem. The district is very big, and deployment of resources is a challenge. Some of the fires took place deep in the forests and gaining access to them was difficult. There was also the question of where to draw the water for fire-fighting from. Besides, the issues were complex, affecting livelihoods. Clearing land using fires was also attributed to certain sectors having lack of access to land-clearing equipment such as bulldozers and tractors."

Together with her team, she made more than 10 trips to the regency between 2007 and 2009 to help launch the programmes chosen for joint implementation. These programmes were carefully planned to develop specific capacities in Jambi officials, increase their situational awareness of fires and hotspots, improve their system of fire prevention and suppression and educate local farmers on more sustainable agricultural practices.

"We organised workshops for the locals, followed up on projects and worked with the authorities and NGOs," says Rohaya. "Forest conservation, peatland management and forest fire prevention and suppression – these are subject areas outside our usual scope of work, an unfamiliar territory for us. So we engaged experts and consultants who were field practitioners in these areas to assist in activities to build capacity and raise awareness on fire prevention and mitigation, and peatland management. The villagers were very enthusiastic. Many were concerned about the state of their land, and the floods they were experiencing. The experts shared how the improper use of land could lead to such consequences. We had a consultant to review the fire-fighting capacities of the locals and present recommendations on bridging the gaps. To me, these efforts were meaningful because we were working from the ground up."

NEA also trained the Jambi officials in reading and interpreting satellite imagery and hotspot information, ambient air quality monitoring and using Geographical Information System to support fire and haze monitoring. Three air and weather monitoring stations were set up so that residents could know the state of their air quality. Land-use maps were developed to help Jambi officials identify and monitor areas affected by fires. "With the ambient air monitoring stations, the locals are able to know the state of the air quality and take the appropriate precautions to safeguard their health during a haze," shares Rohaya.

With strong support from the Indonesian authorities, the programmes were successfully implemented by 2011. However, the regency is only one of the many fire-prone provinces in Indonesia and the scale of the issue remains considerable. Nonetheless, the joint efforts have made a difference in one regency, and in time to come, they may be replicated in other fire-prone provinces in Indonesia.

A Resource To Protect

Water is the essence of life, and a scarce resource in Singapore. This makes NEA's work all the more crucial as it monitors the quality of raw water streams in our catchment areas, ensuring that it is safe for treatment into drinking water. Our coastal waters, especially those around recreational beaches, do not escape the agency's watch either.

BE IT INLAND OR COASTAL WATERS

Every Sunday, it is either kayaking at Sembawang Park Beach or dragon-boating at Kallang River for Eddie Foo and his friends. To Eddie, contaminated waters are the last thing on his mind when he is immersed in these activities. "I grew up in an era when we sometimes swallowed mud and sand while playing! So this is just part and parcel of life; if you want to go onto water, you will sometimes swallow water and that beats mud any day," says Eddie. "I'm not afraid of contamination. It's not that I don't care. If I see pollutants or rubbish in the water, I will try to steer clear of it. But if I take a spill, and I have no choice, I just take it."

But thanks to NEA, if indeed the recreational beaches, inland and coastal waters in Singapore are contaminated, beach goers and water sports enthusiasts will be warned against entering the waters.

Few are aware of this. NEA commissions the weekly sampling of water quality around six recreational beaches (Sentosa, Seletar Island, Sembawang Park, Changi, East Coast Park and Pasir Ris Park) and inland waterways. There are also 9 sampling points set up in the Straits of Johor and 10 in the Straits of Singapore.

Samples taken from coastal areas are tested for the pathogen, *Enterococcus*, and fresh water samples are tested for both *Enterococcus* and chlorophyll-a. If the amount of *Enterococcus* exceeds 200 counts per 100 ml of water in more than 5% of the samples over a one-year period, the site will be declared unsafe for activities such as swimming, where users have primary contact with the water. Chlorophyll-a has to be below 50 µg per litre of water.

This is done for a simple reason. *Enterococcus*, found in the faeces of humans and warm-blooded animals, causes gastrointestinal and respiratory illnesses such as diarrhoea and throat irritation. Chlorophyll-a, on the other hand, is an indicator of algal blooms in a fresh water body. The blue-green alga produces toxins harmful to humans and animals. People may fall ill if water contaminated with these substances gets into their mouths, noses, eyes or open wounds. When they take part in water activities, chances of this happening are higher.

Besides testing for biological contaminants, NEA also analyses the water samples using physical and chemical parameters such as the level of dissolved oxygen and amount of suspended solids. This is done to affirm that our waterways and recreational beaches are healthy and safe.

Should any of the water bodies under its purview fail the guidelines, NEA will investigate the causes of pollution, and take necessary action to remedy the situation.

Using these parameters, NEA has found that the quality of water in Singapore's beaches and inland water bodies has not deteriorated over the years. As of 2011, the Ministry of Health (MOH) has not received any reports on disease outbreaks due to poor recreational water quality. The efforts of NEA and other agencies in monitoring and maintaining our water quality have allowed Singaporeans to use the beaches and waterways with peace of mind.



Guidelines for measuring the quality of water for recreational use were implemented in August 2008, based on a World Health Organization matrix. The decision to do so was taken after an extensive study by an inter-agency committee comprising NEA, PUB, Maritime and Port Authority of Singapore and the Ministry of Health, as well as advisors from the National University of Singapore and Nanyang Technological University.

SAFE TO SPLASH!

If Pasir Ris is one of your favourite hangouts, you would remember the years when signs advising beach goers against swimming in the sea lined the beach. Those signs were put up on 1 August 2008 by NEA after it launched more stringent guidelines for water quality based on WHO standards. Under the new guidelines, the waters at Pasir Ris Beach were no longer deemed healthy for activities where there was contact with water. This was because only 93% of the water samples taken over three years showed readings of less than 200 counts of *Enterococcus* per 100 ml.

Besides warning the public against swimming, NEA also advised beach goers to take precautions, such as avoiding water sports if they had open sores, skin infections or were unwell, not to touch their mouths and eyes with hands soiled with sand, and to wash their hands thoroughly before handling food.

In the same year, NEA also commissioned a study to investigate the origin of pollution affecting our recreational beaches. The results showed that our beaches were mainly affected by local sources. In the case of Pasir Ris Beach, the key culprits were sewage treatment plants and leakages from the sewer network. Besides, water currents in the concave part of the Pasir Ris Beach were weak and pollutants were not as quickly dispersed and diluted as in other beaches.

The aging sewers in the Pasir Ris Beach area were repaired under PUB's sewer rehabilitation programme. PUB also launched a plan to phase out the sewage treatment plants.

In 2010, the water quality review showed that the amount of *Enterococcus* at Pasir Ris Beach was again too high. Seven per cent of the samples showed *Enterococcus* counts of higher than 200. Though the measures had taken effect some time after implementation, NEA decided to take extra precaution and continued monitoring the situation to ensure that the water quality was stable. The beach was finally opened to the public in January 2012.

"I love the sea. I like how immense it is, how elemental and primal. In the case of Singapore, there is nothing quite like being able to run into the water after a day at work. That is what the sea and coast mean to me – it is a taste of freedom and getting away from it the pressures of the rat race and other mundane troubles for a time. I have swallowed water while swimming and have wondered whether the water is safe. Given the amount of shipping traffic Singapore accommodates, it is very important to know if our waters are safe to swim in. I think such sampling done by NEA is absolutely vital!"

– Alexandra May Serrenti, a frequent user of East Coast Park Beach



Samples are collected at a depth of 1 m in the sea. Officers have to take proper precautions to prevent the water from entering their mouths, ears or nose.

Keep The Noise Down!

The sounds of people conversing, vehicles on the roads, television and radio sets, hawkers busy at work in coffee shops and hawker centres – such is the usual medley of noise we have to live with every day. NEA's work is crucial in striking the balance between urban activities and a high quality of living environment. Materials used to pave roads are noise-absorptive to some extent. Industrial buildings, multi-storey car parks and parks are located between major roads and residential areas to act as noise shields. New vehicles must meet noise standards before they can be registered and vehicles-in-use are tested for the noise they emit. These measures help to off-set the amount of noise in our urban environment.

SHHH ... IT'S SUNDAY

A construction site near your home? Thanks to a regulation set by NEA in 2011, you no longer have to put up with the maddening, mechanical noise of piling and construction work on Sundays and public holidays. In fact, by 10 pm the night before, peace would have set in.

Construction sites are never really pleasant. Besides noise, they generate dust, debris and odours. But these are a must if Singapore is to develop and progress. To ease the impact of construction noise on the public, NEA has put in place several measures.

On 1 October 2007, NEA implemented new regulations to lower the noise limits permitted from construction sites for night-time, Sundays and public holidays. The maximum fine for construction noise violations was raised from $20,000 to $40,000. Contractors who repeatedly violate noise limits would also be restricted in their operation hours.

Another round of review was held in 2010 and more stringent measures were imposed on sites located within 150 m of residential premises and noise-sensitive premises such as hospitals. For such sites which commenced work from 1 September 2010 onwards, construction activities were not allowed from 10 pm on Saturdays and eves of public holidays to 10 am on Sundays and public holidays.

The prohibition of work on Sundays and public holidays was further extended to new sites which began work on or after 1 September 2011. Since then, no construction work is allowed on these sites from 10 pm on every Saturday and public holiday till 7 am on the following Monday or day after a public holiday.



A handheld noise meter is used to gauge noise emitted at a construction site during inspections.

A REMOTE WAY OF SURVEILLANCE

Gone are the days when NEA's inspectors had to stand outside construction sites, a noise meter in hand to gauge if noise pollution regulations were being flouted.

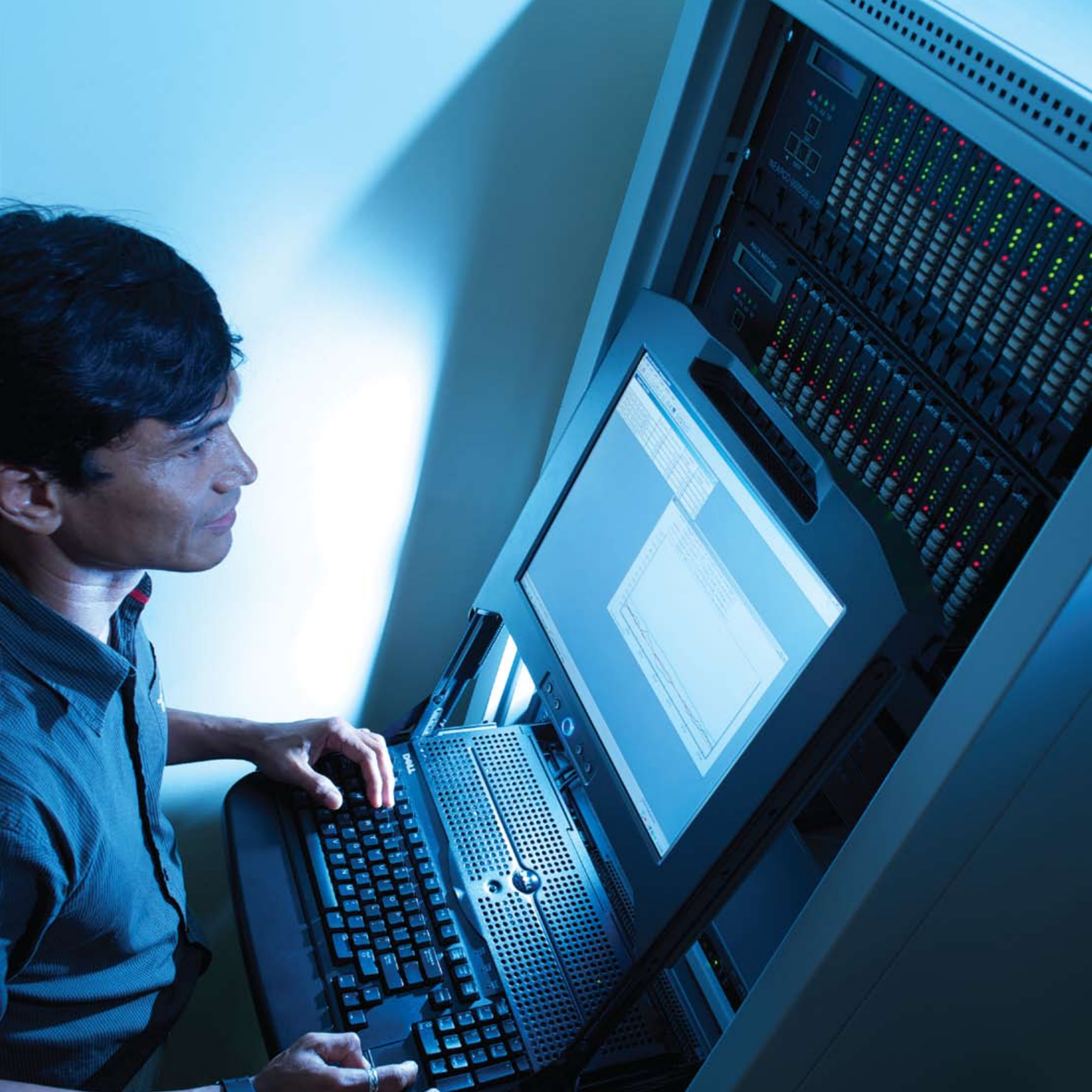
Inspectors can now keep a ear over construction sites all over Singapore, and know who's breaking the rules.

Since 2008, NEA has put in place a smarter, faster and better system. Solar-powered noise meters fixed to a residential building nearest the construction site pick up the noise emitted continuously. The real-time data is relayed via wireless communication to a central database. Both NEA and the construction companies can access the database online any time of the day to check on noise levels. When the levels exceed legal limits, the times at which the offences take place are highlighted in the system. Alerts will also be sent out to NEA and to the company.

The idea for such a system was hatched by NEA and the solution was developed by industry players. In the past, NEA would have to despatch an officer on-site to investigate every public complaint on construction noise. With such remote surveillance in place and more companies adopting the system, such work has been cut down dramatically. As of 2011, any construction project with a value of over $3 million is required to use the system.

"Remote monitoring is good both for NEA and for the contractor. Contractors can monitor and see for themselves the levels of noise being generated. In the past, using stand-alone meters, the data would only be downloaded at the end of the week. So through the week, they would not know whether the levels were high or low. On our end, once the complaints from the public come in, we can check the real-time data from the site in question, so we can update the complainants straightaway. Public complaints do not necessarily translate into violations. It may be higher expectations of the public. For us, it is important to establish for each complaint the noise level generated, whether it is exceeding the permissible limits. If it does, we will take the necessary enforcement action."

— Khairul Sani Samsudin, Senior Engineer, Pollution Control Department, NEA



Containing Hazards

They may be toxic, flammable, poisonous or highly corrosive. But in many ways, they are necessary. Many things in our daily life are linked to chemicals, from toothpaste to fuel and processed foods. Hazardous chemicals are also used on a daily basis in our chemical industry. With so many chemicals around, it is important that they are properly regulated to protect the public. NEA is in charge of the control and regulation of hazardous substances.

CHECKING HAZARDOUS MATERIALS

In such a densely-populated urbanised city, hazardous materials must be well-controlled to curtail the impact of potential accidents and abuse involving the substances. NEA regulates more than 500 chemicals designated as hazardous. One of the agency's key priorities is to ensure that residential and water catchment areas are protected from these substances.

NEA relies on a tight regime of licensing and inspection for regulation. First, any user who intends to import, export, store or use the chemicals must be licensed by the agency. The user must conduct an assessment to confirm that the chemicals do not pose health and safety hazards during its use, and provide mitigation measures during accidents. Those who handle the chemicals must also go through NEA's course on the management of hazardous substances and pass an examination before they are licensed. NEA also requires specific storage conditions and adequate pollution control measures. For instance, nitric acid must be stored separately from alkaline substances to prevent violent reactions. The requirements help to contain the impact of potential accidents within the premises so that they will not affect surrounding areas.

To make sure that these requirements are complied with, regular inspections are carried out. Public feedback and tip-off on lapses in factories will be investigated as well. With over 2,000 companies under its purview, NEA turns to risk-profiling to deploy its resources more efficiently. Companies are classified into one of three categories, depending on the amount and type of chemicals kept on-site. Category 1 companies pose the highest risk and are audited and inspected more frequently than the companies in the other two categories.

Companies which flout rules may be warned, given a show cause letter or even prosecuted if they are consistently recalcitrant. Should an accident occur, NEA will work alongside other government agencies such as Singapore Civil Defence Force (SCDF) and Ministry of Manpower in a coordinated response. NEA monitors the environment to assess the accident's impact. It also makes sure that the company cleans up the place to ensure the safety of its neighbours and workers.

Additionally, companies generating toxic industrial waste have to register with NEA and report the volume of such waste produced and where it is treated. The collection and treatment of toxic industrial waste can only be undertaken by NEA licensed toxic waste collectors.

Certain off-the-shelf products that contain chemicals exceeding stipulated limits come under scrutiny too. NEA conducts checks regularly. If products are found to contain controlled substances, or



if the levels exceed exemption limits, action will be taken against the importers.

Exceptional care is taken when it comes to transport of hazardous substances. Unlike fixed installations, where safety features are in place and workers are trained to handle hazardous substances, there are very few measures that can be undertaken to mitigate any incident that occurs during transportation.

To reduce the risk of an accident, companies have to seek NEA's approval to transport hazardous substances that exceed a stipulated quantity. Drivers must undergo a relevant course run by SCDF before they are permitted to transport hazardous chemicals. The routes and timing are restricted as well. Sundays, public holidays and peak traffic hours are to be avoided. This is so that if accidents do take place, their impact will be manageable.

In addition, tracking devices and remote-activated brakes are installed in the vehicles used for transporting highly toxic substances. This allows the location of the materials to be tracked at all times and prevents the vehicles from being used for sabotage.

"When we conduct inspections, we need to know which chemicals are stored at the companies, how they are used and whether the storage areas are secure. Proper documentation and pollution control equipment must be in place. We also need to educate businesses that it is essential to install pollution control measures. Sometimes, there are companies which do not understand why our requirements are so stringent. Because the chemicals they use may be cheap, but the pollution control measures are very expensive. For example, any company that stores more than 100 tonnes of hazardous substances must install an underground retention basin. The basin is meant to contain the water used for fire-fighting if fires do break out, so that the contaminated water does not pollute our land or water catchment areas. This basin may have to be as big as a swimming pool. So here they have to build a pool which must be kept empty and it does not seem to make sense. So we need to educate them, because we want to protect lives and the environment."

— Sulaiman Bin Abdul Rahman, Technical Support Officer, Pollution Control Department, NEA


TOXIC
6

BRIDGING THE SAFETY GAPS

It was just another Thursday morning. But if you happened to be one of the motorists using the Tuas Second Crossing on 28 July 2011, you would have seen industrial drums scattered on the crossing and emergency response personnel from SCDF and BOMBA Malaysia containing a mock chemical spill in earnest. Dummies were rescued from what appeared to be a multiple collision of an isotanker, a lorry, a passenger car and a motorcycle.

What took place was a joint field exercise between Singapore and Malaysia, during which both countries ran through emergency response operations in a simulated accident involving hazardous chemicals on the crossing.

Such exercises are critical as the transport of all hazardous chemicals between Singapore and Malaysia takes place on the Tuas Second Crossing. Every year, about 111,000 tonnes of hazardous chemicals, such as sulphuric acid, sodium hydroxide, phenol and formaldehyde, are transported between the two countries.

If you take a look at the areas surrounding the Causeway at Woodlands, it is clear why the transport of hazardous substances is confined to the Tuas Second Crossing. Woodlands, Bukit Timah and Johor Bahru are all congested and densely-populated zones, while Tuas and the western part of Johor are relatively less populated. By using the Tuas Second Crossing, the potential impact of accidents on residential areas is reduced.

The emergency response plan is the result of planning and discussions among the Malaysia-Singapore Joint Committee on the Environment, the Department of Environment, Johor and NEA. A procedure is set up to alert both countries as soon as possible when an accident takes place so that action can be taken early on. To make sure that agencies on both sides can coordinate their operations and get the situation swiftly under control, a detailed response procedure is laid out as well. Joint field exercises are held regularly so that all the agencies involved are familiar with the procedures, with Singapore and Malaysia taking turns as organisers.

Between 2000 and 2011, seven exercises were conducted. A number of agencies from Singapore and Malaysia are involved in every exercise, including emergency response forces such as SCDF and BOMBA Malaysia. Chemical companies were also invited to take part by providing emergency response personnel, vehicles, drums and other props.

During every exercise, different scenarios involving vehicles, motorists and chemicals are enacted to test the responses of both countries in various aspects. For example, in the 2011 exercise, the accident was simulated 500 m away from the Singapore Tuas Checkpoint. The site was under Malaysian territory even though it was nearer to Singapore, and the Singaporean team arrived at the scene before the Malaysian forces. The exercise thus put to test Singapore's ability to effectively hand over command of the operations to Malaysia in a time-critical scenario. SCDF and BOMBA Malaysia also had to race against time to rescue "victims" from vehicles and attend to bystanders affected by the chemicals.

These joint exercises build closer working ties between the response forces in both countries. Existing gaps in operations are also looked into and bridged as a result of the collaboration. Together with the emergency response plan, the exercises ensure that forces on both sides of the Tuas Second Crossing are always on the ready for any incidents involving hazardous chemicals.







Participants in a display of solidarity after the exercise in 2011.

From X-Rays To Laser Pointers

Radiation protection in Singapore began when the Radiation Protection Inspectorate (RPI) was formed in 1972, under MOH. The following year, the Radiation Protection Act was passed. The responsibilities for radiation protection and nuclear science were consolidated under NEA in July 2007. Today, NEA discharges its duties to ensure safe and secure use of radioactive materials as well as ionising and non-ionising irradiating devices through a system of licensing, inspection and enforcement.



Thermoluminescent Dosimeter

RADIATION UNDER SCRUTINY

The next time you visit an x-ray room, see if you can spy the Thermoluminescent Dosimeter (TLD) worn by the operators of the machine. This gadget, distributed by NEA to all radiation workers, clocks the amount of radiation they are exposed to. The TLDs are analysed periodically by NEA to ascertain that the annual radiation dosage of each worker does not exceed 20 millisieverts (mSv) per year, the safety limit.

This is just part of the work NEA does to regulate radiation in Singapore. From x-ray machines and laser pointers to microwave ovens and smoke detectors, the agency controls just about any radiation-emitting equipment and substances that can potentially harm humans. Radiation is commonly classified into two forms – ionising and non-ionising. Both can cause harm to life and the environment, depending on the dosage. Ionising radiation also has the energy to split atoms and molecules in its path into ions, and can damage DNA. Its effects on living organisms are cumulative. The uses of radiation, such as in nuclear medicine, radiotherapy, industrial applications and household appliances, however, mean that irradiating equipment and substances are very much a part of our surroundings.

The fact that such controls were established early on gave Singapore an advantage, as there have been relatively few cases of irradiating equipment and substances found without documentation of their origins. This is particularly important as Singapore does not have the capacity to treat radioactive waste, and such equipment and substances have to be re-exported to their countries of origin after their useful lifespan ends.

Part of the challenge of NEA's work is to keep up with advancements in nuclear science and technology, and the growing applications of radiation. In 2008, the agency issued a total of 35,481 licences. By 2010, the number had increased to 37,170. As the medical and industrial sectors expand, so do the scale and complexity of NEA's duty to protect Singapore from the potential misuse and abuse of radiation.

Altogether, NEA administers 17 types of licences to manage the import, export, use, ownership and transport of radioactive materials or irradiating apparatus. Radiation workers have to sit for a test to make sure that their knowledge and skills are sound before they are licensed. Inspections are conducted before and after a licence is granted to ensure that the right conditions are in place. For example, the radiation level outside rooms housing irradiating equipment should not exceed 10 µSv/hr. If the premises or activities are found to be unsafe, the licensee will be warned and required to take mitigatory measures, or even face prosecution.

Opposite: Inspecting to see if radiation levels emitted by the angiography equipment are within permissible limits.



Part of the challenge of NEA's work is to keep up with advancements in nuclear science and technology, and the growing applications of radiation. As the medical and industrial sectors expand, so do the scale and complexity of NEA's duty to protect Singapore from the potential misuse and abuse of radiation.



SHOULD A "DIRTY BOMB" FALL

It is a thought that hardly occurs to the man on the street – what if Singapore were to be hit by a "dirty bomb", or radiological dispersal device, one day?

At the Centre for Radiation Protection and Nuclear Science (CRPNS) in NEA, imagining such scenarios and planning for them are part of the job. As with any other industrialised country, there are facilities in Singapore with equipment and materials that emit radiation. Nuclear-powered naval ships visit the island from time to time. Moreover, a number of neighbouring countries have announced plans to build nuclear power plants in the near future.

Planning for radiological and nuclear crises is not the work of just one agency. Rather, it is a coordinated effort which involves government agencies such as SCDF, MOH, Agri-Food and Veterinary Authority (AVA) and the Ministry of Foreign Affairs (MFA). CRPNS, with its experience and knowledge, brings to the table technical and scientific expertise. In the event of a crisis, its scientists and technical officers will undertake critical tasks to ensure the public's safety, such as monitoring radiation levels, deciding on the extent of the warm zone (zone within which the threat still exists but is not immediate) and testing soil samples for ground contamination.

Readiness goes beyond planning on paper. Joint emergency exercises are conducted together with SCDF. Scenarios involving nuclear-powered warships or radiological dispersal devices are simulated. The centre also works with MFA to participate in regional nuclear safety co-operation.

Having procedures in place does not mean that the work is done. At the end of the day, it is access to relevant information and timely analysis which hold the key to an adequate emergency response. As Ang Kok Kiat, Director of CRPNS puts it, "To respond to such threats, we need to know what nuclear technologies are out there and their inherent safety problems, and the consequences should incidents happen. We must be able to assess the impact from the release of radioactive materials – what will be released, how much will be released and for how long. And meteorological data to tell us whether the materials will come to Singapore, how long it would take, and how long it would last. These are things that are critical in planning for an emergency response."

AN EXERCISE IN VIGILANCE

In March 2011, the tsunami triggered by an earthquake off Japan's coast led to equipment failure and release of radioactive materials at Fukushima Daiichi Nuclear Power Plant, resulting in a nuclear crisis.

Even though the incident took place 5,500 km away, the Singapore government decided not to take a chance. It mobilised relevant agencies to prepare contingency plans and launched responses appropriate to the level of danger presented by the situation.

NEA played a crucial role in assessing the threats faced by Singapore. The Meteorological Service Singapore provided weather data to project if the radioactive plume released would reach Singapore. Using the data, CRPNS calculated that the radiation levels would be negligible by the time it arrived.

Despite the findings, the government adopted a vigilant approach. Four ambient radiation monitors were set up by CRPNS in different parts of Singapore to monitor radiation levels. Water sampling was conducted to check for radiation contamination in our coastal areas. The AVA stepped up its screening of food imported from Japan and CRPNS pitched in to help.

To assure the public of Singapore's safety, readings of radiation levels were published daily on NEA's website. The government released press statements and set up a dedicated website for Singaporeans to keep track of the situation. To allay concerns that radioactive materials could be released when contaminated produce from Japan was incinerated, some four tonnes of food were double-bagged and secured in durable plastic bags before they were carefully disposed of in Semakau Landfill, even though contamination was low and incineration was acceptable under international guidelines.

At the height of the crisis in March and April, NEA's Ops Room, where emergency response operations are coordinated during a crisis, was activated to function 24/7 and daily situation reports were prepared. However, as the crisis began to stabilise in Japan, and readings in Singapore showed constant low background radiation levels of 0.08 – 0.1 μSv/hr, the operation was eventually scaled down by early May 2011.





WHAT TEXTBOOKS DON'T TEACH

"Radiation, what does that have to do with the environment?" That is the most common response when Dr Phua Tan Tee tells people what he does.

Dr Phua, one of the few radiation scientists in Singapore, has been in the line of radiation protection and safety since 1983, when he joined the former Radiation Protection Inspectorate. Today, he heads the Ionising Radiation Control Unit in CRPNS, where much of his work revolves around leading his team to license, inspect and monitor irradiating equipment and materials, and to provide scientific expertise and advice where needed.

Having the right qualifications, however, is not enough for a scientist working here. Effective communication and knowing how to handle human relationships are highly prized skills as well. "The textbooks and teachers can only give you this much. Our knowledge comes from on-the-job training," shares Dr Phua. "Experience on the ground, going out to do inspections, meeting the licensees face-to-face, understanding what they are doing, their environment, the materials they are using, their difficulties and what you are licensing for; that is important. Anyone who joins us, if he doesn't like site work or cannot handle human relationships, he may not be able to stay on."

One of the most striking things about Dr Phua is his ability to sidestep scientific jargon and talk about radiation science in layman's terms. Which is necessary in his job, where he has to work with people from all walks of life, from technicians to doctors, answering their queries and convincing licensees and users of irradiating equipment and substances that NEA's regulations are reasonable. During the Fukushima crisis, the number of public queries relating to radiation spiked. "If the car is imported from Japan, is it ok? Can I mix my daughter's laundry with the rest of our laundry if she just came back from Japan? Are appliances and electronic parts from Japan safe for use?" Dr Phua elaborates, "These are just some of the queries we received and we need to assure the callers. Because there is a lot of misconception about radiation safety and people are very influenced by the movies they watch! So in this line, it is important to have good techniques in communication, to use the right words, the right tone. This can't be taught. Sometimes we even have to go down to the site to assure and counsel those who were excessively worried about radiation in their vicinity."

Like many of his colleagues in NEA, Dr Phua finds his work meaningful. "If there is no control over radioactive materials or equipment, the issue of safety will arise – safety for the user, and for the public," he says. "And on the medical side, safety for the patients as well. And it's quite interesting work! If you love your job and you love to share knowledge, it is enjoyable, sharing information and to being able to assure others that they are safe. That is the most satisfying part."

Trashing Waste For Clean Land

Dealing with waste is a serious affair here. Besides having to dispose of more and more waste every year, this waste has also got to be treated quickly before it becomes a public health and environmental hazard in our hot and humid climate. Waste management in Singapore relies on a few core pillars. Enforcement, investment in infrastructure, innovation and promoting the 3Rs – Reduce, Reuse, Recycle.

CRACKING DOWN ON DUMPING!

Dumping of untreated waste, especially from industrial sources, leads to land pollution and may even cause water pollution if it takes place near waterways. The way the authorities, first ENV, then NEA, cracked down on illegal dumping is an example of how legislation, monitoring, enforcement and education come together in environmental protection.

The usual culprits are earth-moving contractors and general waste collectors. Generally, they do it to avoid paying fees at incineration plants. To escape detection, they often dump at out-of-the-way spots, such as Lim Chu Kang, Tampines, Punggol and Tuas.

Under the Environmental Public Health Act, trashing waste in public and unauthorised places is strictly prohibited. Offenders face fines, jail terms and confiscation of vehicles used to carry out dumping.

The efforts to clamp down on illegal dumping started with toughening the penalties for the act. In 1999, the fine for the first offence was raised from $10,000 to $50,000. This alone, however, could not deter would-be offenders as long as they believed they could get away with it. Effective enforcement was necessary. Surveillance at hotspots was carried out to catch culprits in action. Earth bunds, fencing and bollards were put up to prevent easy access to a convenient dumping ground.

To make the effort even more effective, the public was encouraged to report on illegal dumping. Offenders who were caught also received attention in the press to highlight the stepped-up enforcement. With all these measures, ENV, and later NEA, successfully brought about a sharp decline in such activities. In 1999, there were 190 cases of illegal dumping recorded in Singapore. By 2010, the number had dropped significantly to 39.

ENERGY FROM GARBAGE

The story of how waste-to-energy (WTE) incineration plants became an integral part of our solid waste management strategy illustrates Singapore's willingness to invest in technology for the environment. WTE incineration may seem like an easy choice to make. However, in the early 1970s, it was anything but that. Dumping rubbish directly in landfills was viewed as the most natural course. There were relatively few countries using incineration to dispose of waste, save for some parts of Europe and land-scarce countries such as Japan. Fears of air pollution caused by incineration were widespread, and some developed nations even balked at commissioning incineration plants on a large scale. On top of that, WTE incineration plants were expensive, usually costing more than $100 million each. So when Singapore decided in 1973 to invest in the first WTE incineration plant in Asia outside of Japan, it was a bold move.

Opposite: Refuse bunker at Tuas South Incineration Plant.



In 1970, some 1,200 tonnes of rubbish were disposed of per day. By 2011, the figure had risen to 7,830 tonnes every day.

There were obvious advantages, of course. The volume of waste could be reduced by an amazing 90%, thus saving landfill space. Heat, the by-product of incineration, could be harnessed to produce electricity. An incineration plant took up much less space than a landfill and could process more waste. Scrap iron could also be recovered from the waste. To overcome the concern of air pollution, flue gas cleansing treatment and filter treatment systems were installed to remove acidic gases, dust, and other contaminants.

In 1979, Singapore's first WTE incineration plant, located at Ulu Pandan, was completed at a cost of $130 million. The second WTE incineration plant was commissioned in 1986 in Tuas. Senoko Incineration Plant came about in 1992.

Singapore's fourth and largest WTE incineration plant was commissioned in 2000 at a cost of $890 million. Sited on a 10.5 ha plot of reclaimed land, Tuas South Incineration Plant is one of the largest of its kind in the region in terms of capacity. The plant can burn 3000 tonnes of garbage every day. Industrial and domestic waste, contraband goods, ballot papers and decommissioned currency notes – these are just some of what pass through the plant's furnaces.

An average of 800 refuse trucks roll into the plant per day. They tip their rubbish into a bunker, and are weighed before and after to calculate disposal fees. In the bunkers, air pressure is kept below atmospheric pressure to contain the stench. Cranes mix the rubbish to make sure that the composition is even before feeding it to six incinerators. Advanced combustion control systems regulate the combustion rate so that the refuse can be completely burnt.

The gas produced passes through a precipitator where dust particles, even those smaller than 1 micron, are removed. Next, it goes through a dry scrubber system and a catalytic fabric filter system to neutralise its acid contents and remove dust and other contaminants before exiting through two 150 m tall chimneys. The quality of the flue gas is monitored for pollution control. The resultant ash and slag are transported to pits and electro-magnetic separators to pick out ferrous scrap metal for recycling. The remainder will be sent to Semakau Landfill.

The heat produced is used to generate steam in boilers, which drives generators to produce 600kWh of electricity per tonne of refuse. In total, 20% of the power goes to operating the plant and 80% is sold to the National Electricity Market of Singapore. Exhaust steam is cooled and pumped back into the boilers for re-use.

If there is anything amiss with the equipment, alarms will go off. These alarms work on parameters that are customised and indicate different levels of urgency. Operators can then prioritise their tasks and take pre-emptive action.

Today, Singapore is one of the few countries in Asia to use incineration of refuse to manage waste. Incineration of waste continues to be a controversial topic for reasons such as air pollution and toxin levels in incineration ash. As technology progresses, however, pollution control methods have improved. Emissions by incineration plants are monitored to make sure they are below safety limits. In addition, Singapore's system of separating hazardous waste from incinerable waste and sanitary disposal of ash at Semakau Landfill ensure that the fears of pollution are not realised here.





To keep the plant going, five operations teams work round the clock in three shifts. There are checklists for fabric filters, boilers, generators, turbines – every equipment and process in the plant.

KEEP IT BURNING

"We are the last line of defence. The other incineration plants are smaller. And once their bunkers are full, they will divert the refuse vehicles to us, and we have nowhere else to divert to. So we have got to take in whatever that comes in. We are like their big brother," explains Mohamad Ghazali Bin Ismail, Senior Operations Manager at Tuas South Incineration Plant (TSIP).

Meet the people behind TSIP. These engineers and technicians are a rare breed in Singapore. As Joseph Boey, Senior Manager, says, "Put it this way, there are not too many of us around because there are only four such plants in Singapore." These people keep Singapore's largest incineration plant going, 24/7, 365 days a year. They manage fires that reach temperatures of 1,000°C, so as to combust the tonnes of rubbish delivered to the plant every day. Altogether, over 140 of them work to manage the plant.

NO DOWNTIME!

There are two main groups of people who keep the plant going – the operations team and the maintenance force. One manages the day-to-day running and the other looks after the equipment.

Downtime is the bane of workers here – to be avoided at all costs. If any of the incinerators is down, refuse will start piling, and public waste collectors may not have anywhere else to turn to. Sunny Tan, Higher Technical Officer, has this to say, "We cannot afford downtime. We are always fighting for time."

To keep the plant going, five operations teams work round the clock in three shifts. Then there are checklists for fabric filters, boilers, generators, turbines – every equipment and process in the plant. Selvam, Senior Technician, explains, "The checklists ensure that there will always be men on the ground. A pipe may have a pinhole leak, but the system will show no change in the water pressure. So water will be wasted, and the small leak may lead to something more serious. Or the equipment may appear to be working on the system, but locally, a part may be out of alignment. If we don't correct it, the part may give way! So we do the checks two or three times every shift."

The operations team keep an eye on the refuse coming in as well. Bulky waste, for example, can bring operations to a halt by jamming the incinerator or even damaging the machinery. Four high capacity rotary bulky waste crushers are installed to deal with such garbage. Smouldering refuse and hazardous chemicals are to be watched out for too. Sometimes, waste collectors bring in refuse that may cause fires in the bunkers, which in turn may lead to a plant shutdown. And despite strict regulations banning the disposal of hazardous materials at incineration plants, such substances still end up here now and then.

"We do random checking. Every day, we will check about 30 to 40 lorries. One lorry takes about 10 minutes. If there is rubbish that is not supposed to be there, we will scoop it up and put it back into the vehicle. And we will warn the waste collector," says Mohamad Ghazali. Operations Shift Supervisor, Low Eng Hai, quipped, "Especially during the seventh month or Qing Ming, when people burn offerings. When we see vehicles coming in with smoke coming out, we will tell them to unload their refuse. And if it is on fire, we will spray water. Bunkers catching fire is very rare and we have zonal sprays to put them out. If there is a fire, it's an emergency! Prevention is better than cure!"

The bunkers at TSIP are able to hold up to four days of trash, but forward planning is still essential. During festive seasons and mass events such as Christmas and the Formula One night race, the volume of waste may spike by 30% or more. The operations team has to make sure that bunker levels are low during such times.

Opposite: Inspection and maintenance works inside the furnace during an overhaul.



Every year, each of the six incinerators at TSIP is shut down for a month for an overhaul. The overhaul has to be timed during periods when waste volumes are low. The incinerator and all its parts will be cleaned, thoroughly checked, maintained, repaired and tested before it is allowed to start again.

To keep the equipment at the plant going smoothly, maintenance and incinerator overhauls have to be carried out. The equipment is maintained regularly. When issues crop up, the maintenance crew, which is on call 24/7, will look into them before they escalate. Dennis Lee, Technical Officer, says, "Our work involves maintaining an incineration plant and a power generator. The work here is very complex, from the equipment to the treatment process."

Every year, each of the six incinerators at TSIP is shut down for a month for an overhaul. The incinerator will be cleaned, checked, maintained and tested before it is allowed to start again. Technicians such as Muhammad Khairozi Bin Hassan will inspect each part methodically. He says, "For example, we use ultrasonic equipment to test if the boiler tubes are too thin. Too thin, they will burst. We have to check the inside of the furnace, which is 40 m high. We take all the readings, and the technical officers will analyse to see if repairs are necessary." All things electrical are covered by an electrical maintenance team.

Before the incinerator can start up again, the pressure test, which lasts a day, will be conducted in the presence of an officer from the Ministry of Manpower. The incinerator's boiler is filled with water and the pressure is increased steadily until it reaches 40 barG. Once it is demonstrated that there are no leaks, the incinerator is ready to burn.

AT THE END OF THE DAY

The work at TSIP is demanding, challenging and unpleasant even. Hardship is a constant. The people have to work in adverse conditions of dust, odours, extreme heat and noise. Workplace safety is deeply ingrained, because, as Low points out, "It's a matter of life and death."

Safety procedures, meetings at the start of the day to reiterate safety concerns, personal protection equipment, a website and courses dedicated to safety, drills on chemical leaks and fires – all these go to creating a strong safety culture. This can be seen in operations such as overhauls, when workers have to operate in confined spaces, facing potential threats such as carbon monoxide. Trained assessors will check at least every eight hours if the area is safe to work in. Workers are also equipped with gas meters to alert them to any dangers. Incompatible tasks, such as painting and hot work, are not allowed to take place simultaneously.

Despite all these, TSIP's people are proud of their work. Tan Kee Pin, Principal Manager, highlights, "It's more complicated to maintain an incineration plant, because there is a power plant too. People who join us have to start from zero to understand the process. Some people can't take this kind of environment and can't stay on. But those who stay are those who love the work and challenges despite the hazards."

What about the smell then? "Immune!" laughs Low, "Definitely I'm very proud of my work. It's a unique job. Those who love working with machines will love working here."

And contrary to expectations, people around them are fascinated rather than put off by their work. Tan Kee Pin divulges, "Because there are only four plants in Singapore, a lot of people don't know what we do. When I tell them that we burn rubbish and from the heat, we produce electricity, they are actually very fascinated."

For Joseph Boey, "I burn garbage" is a handy line to explain his work. Joseph goes, "My seven year old son asked me what my job is. And I said, 'I burn garbage'. He replied, 'You burn garbage, how come I don't smell the garbage when I sit beside you?' And this is the amazing part of my job. I'm in this industry and I face tonnes and tonnes of garbage every day. But technology has progressed to such an extent that we don't have to be associated with rubbish in a negative way anymore. This job has got no lack of challenges. The plant takes care of the majority of the waste island-wide. I have to do my part to make sure that this plant serves Singapore. That in itself is a meaningful responsibility."



Top: Checking electrical parts in the plant.
Opposite from top: Part and parcel of work - inspecting the turbine-generator; Making sure that the stokers are in tip-top condition.





THE MATHEMATICS OF TUAS SOUTH INCINERATION PLANT

Incineration capacity
3,000 tonnes a day

Catalytic bag filters comprise
2,240 fabric bags

Temperature at which refuse is burned
900 - 1,000°C

Steam generation per boiler
370°C (35 barG)

105 tonnes per hour

Power generation capacity
80 MW
10.5 kV generator voltage

24 refuse discharge bays

8 Weighbridges

6
Ten-compartment catalytic bag filters
Two-zone electrostatic precipitators
Stoker/furnace units with integral boilers

4
Bulky waste rotary shears
Refuse cranes of 10 m^3 each

3 Overhead wire rope operated ash cranes of 3.5 m^3 each

2
Air-cooled condenser systems with 12 fans each
Bulky waste cranes
Concrete chimneys 150 m in height
Condensing steam turbines coupled to generators
Lime silos and reaction product silos
Magnetic separators to each incinerator unit
Pusher-type hydraulic slag extractors to each incinerator unit
Rain water buffer basins
Separate refuse storage bunkers
Vibrating conveyors to each incinerator unit



The heat produced by the burning refuse does not go to waste. It is used to generate steam in boilers, which drives two steam turbines coupled to generators to produce electricity. In total, 20% of the power goes to operating the plant, and 80% is sold to the National Electricity Market of Singapore. The exhaust steam is then cooled and pumped back into the boilers to be used again.

INNOVATION FOR WASTE

Waste management is complex. Collecting it in a way that does not create nuisances for the public, treating it, making sure that there is a good disposal system – such operations are all the more challenging in this high-density, compact city. By using innovative engineering and technology, NEA manages to improve many aspects of solid waste management.

Public waste collection generally relies on a system of garbage trucks making the rounds to every block and building and emptying their chutes. During these collection times, noise and smells cannot be avoided. Progressively, NEA has implemented a system whereby these issues no longer plague waste collection.

The Pneumatic Refuse Conveyance System (PRCS) uses air suction and a network of pipes to suck rubbish from buildings to a central location for collection. Such a system has several advantages. The odour of garbage during collection, traffic noise, scavenging by rodents and leaking chutes — all these issues become irrelevant as collection is done in a closed system. The time and manpower required for collection are cut down significantly too. Today, PRCS is used in some 60 residential and commercial developments in Singapore, including hospitals, shopping malls, residences and hotels.

Another innovation explored by NEA is the use of incineration bottom ash (IBA) to lay roads, a method used in countries such as Netherlands and Taiwan. IBA is the matter left behind after waste is burned. In 2010 alone, some 405,000 tonnes of IBA were sent to the only landfill in Singapore, Semakau Landfill.

To reduce this volume, NEA embarked on an experiment between 2001 and 2003. Working with the Land Transport Authority (LTA), PUB and a private company, NEA conducted a trial to pave 150 m of road along Jalan Buroh with 1,200 tonnes of treated IBA. The process was tricky as IBA contains traces of heavy metals and incompletely burnt materials. However, the experiment was a resounding success. The quality of road was comparable to those built using conventional materials. Groundwater and storm water run-off from the section did not show signs of contamination. Following that, another two sections along Tampines Road were laid with IBA by LTA. The sections were tested for levels of hazardous substances such as carbon, dioxins, phosphate etc. The conclusion? IBA-built roads are good to go.



Former Minister for the Environment Lim Swee Say laying the road with IBA.

Such application of IBA, however, is yet to be commercially viable in Singapore, for reasons such as the high cost and uncertainty over market demand for the material. Nonetheless, NEA remains open to the idea that this method may be used one day, while continuing to look out for more solutions to tackle the waste issue.






Wildlife that thrive on and around Semakau Landfill include (*from left*) the grey heron, jumping spider, nudibranch, feather star.

HAVEN ON A DUMP

Keep still, and you may very well catch a glimpse of endangered birds such as the Great-Billed Heron and Malaysian Plover. The waters teem with barracuda, trevally and milkfish. During low tide, you may encounter the Giant Carpet Anemone, sea urchins, sickle seagrass, nudibranchs and the impressive Knobbly Sea Star. Mangroves line the edge of the mudflats, offering an oasis for intertidal wildlife.

Sounds like a nature park, but it is actually a garbage dump. Pulau Semakau, 350 ha in size and capable of holding 63 million m^3 of garbage, is Singapore's only landfill. What is unusual about Pulau Semakau is that it is the first-of-its-kind landfill in the world that demonstrates how natural habitats can co-exist with waste.

Managed by NEA today, Semakau Landfill is where all of Singapore's incinerated trash and non-incinerable waste go. Barges regularly ship refuse from Tuas to Semakau Landfill. The rubbish is then transported to a series of cells that serve as landfill. These cells, located in an artificial lagoon, are filled with seawater until they are to be used. To get a cell ready for waste-tipping, water is first pumped out of the cell, which is lined with impermeable geomembrane to prevent waste from seeping into the sea. When a cell is used up, it will be covered with earth. NEA regularly samples the water surrounding the landfill for contamination.

This feat of engineering began in the 1990s, when it was clear that Singapore's last landfill at Lorong Halus could not keep up with the island's waste management requirements. After considering future development needs on the main island, the government decided to build the landfill offshore. In a stroke of creativity, engineers at ENV (present-day MEWR) proposed to turn the sea space between Pulau Sakeng and Pulau Semakau into a landfill.

However, the ministry had to first assess the feasibility of the proposal and the landfill's potential impact on the marine environment. As this would be the first landfill of its kind in the world, it sent engineers to Japan to learn about near-shore landfills and to the United States to look at barging systems for offshore facilities. A consultant was engaged to study the viability of the idea and whether the impact on the marine environment could be mitigated. The conclusion was that the idea could be realised. Approval was given by the parliament in 1994 and construction began in 1995.

That was not all. Biologists were consulted and measures taken to minimise the impact on the ecosystems of Pulau Semakau and Pulau Sakeng. About 400,000 mangrove saplings were planted to replace those destroyed in the construction process. New plots of mudflats were created on the fringes of Pulau Semakau before the replanting could take place. Artificial creeks were built to ensure that seawater could circulate to the saplings. To protect the corals around the island, the engineers used fine mesh silt screens near the work areas to reduce the impact of sediment. Four years after the project began, Semakau Landfill officially opened on 1 April 1999. Completed at a cost of $610 million, it is expected to meet Singapore's needs for landfill space beyond 2045.

Today, the first thing that is noticeably absent when you step on Pulau Semakau is the smell of garbage, usually the signature of a landfill. Instead, you can sight over 80 species of birds, 8 species of seagrasses, more than 40 species of butterflies and a host of corals, anemones and marine life on this unlikely nature haven, found, of all places, on a rubbish dump.

"Some 10 years ago, we were running out of space on the mainland to build any more landfill for our garbage. The engineers in NEA then came up with an innovative solution to build an offshore landfill. It was never tried out in such a big scale anywhere else in the world before, but our engineers persevered for four years with the hard engineering work involving more than 1000 workers to build the rock bund from the bottom of the seabed. They also took care to put in measures to protect the marine ecosystem. That Semakau today is not just a landfill but an island with opportunities for recreation and nature lovers is truly a great testimony to the forward thinking and excellent planning of our engineers."

— Dr Yaacob Ibrahim, Minister for the Environment and Water Resources (2004 - 2011) at the National Engineering Week, 24 October 2005





A PLACE TO BE ENJOYED

The man who pushed for Pulau Semakau to be opened to the public, Dr Yaacob Ibrahim, had the brainwave after his first visit to the island in 2004. "We took the boat from Pasir Panjang," says Dr Yaacob. "If you remember that road, you move away from the landfill station, and then you see the mangroves, you see the greenery. I said, 'This cannot be a landfill, right?'"

The visit was a revelation of sorts. Pulau Semakau was not just any landfill. In the words of the former minister, it had the potential to be a place where "we show Singaporeans that we can balance diversity and environmental management in a very nice way." Dr Yaacob thus took the plunge to open the island to the public. In 2005, NEA approached organisations such as Raffles Museum of Biodiversity Research to look into whether the island could be used for recreational and educational activities. After some studies, it was found that Pulau Semakau offered a rich environment for guided nature walks, bird watching and sport fishing outings.

Nature lovers, anglers, school groups, researchers and just about anyone who was curious started flocking to the island. Its rich biodiversity won converts easily and the story began spreading. International media soon caught onto it. Word of a nature haven on a rubbish dump appeared in the papers, news channels, documentaries, scientific and environmental journals, and along with it, accolades and praise for the innovative solution. Officials from various nations such as Malaysia, the Philippines, New Zealand, Japan and Brunei came to study the possibility of replicating the landfill in their own countries.

Meanwhile, the island has grown to become a favourite recreational spot for Singaporeans. Guided walks are conducted regularly to introduce participants to the fascinating intertidal ecosystem. Bird watchers, anglers and people interested in astronomy find Pulau Semakau one of the best places for their activities. NEA's first environmental charity run, benefitting local environmental NGOs, was also held on Pulau Semakau in 2007, with more than 40 corporations taking part.

To Dr Yaacob, Pulau Semakau remains a special place. He recounts a trip that he once took with his family. "As we were walking out, it was already 7 pm, getting quite dark, and we saw these lights in the trees. Fireflies! We saw fireflies in Semakau! It showed that the water was clean enough for the fireflies to come back. At the end of the day, we managed to bring back the biodiversity. The birds came back, the crabs came back and I thought it was fantastic. It is one of the little areas that reminds you that it is possible for us to create in a sense, a nirvana out of Singapore."



Aerial view of Semakau Landfill, 2011.

WASTE AS RESOURCE

It's time to give waste a new lease of life because it can be a rich resource too.

Resource efficiency is actively promoted by NEA as a strategy to cope with waste. For instance, the agency works closely with recycling companies, the Building and Construction Authority (BCA) and the construction industry to promote the recycling of construction and demolition waste. Today, about 99% of such waste finds new uses. Used copper slag is adapted for grit blasting; steel slag, for the production of road-building materials and used concrete, for foot paths and road curbs. Another exciting project focused on the recycling of ink and toner cartridges. This project, which commenced in 2011, was initiated by manufacturers. NEA supported them by providing advice and facilitating meetings between the manufacturers, recycling companies and the National Library Board, whose libraries housed the drop-off bins for the cartridges.

Recycling went nation-wide in 2001 under ENV. In public areas with high human traffic, recycling bins were placed. Schools became part of the national recycling campaign when NEA started a programme in 2002 to inculcate such habits among students. By 2008, all condominiums and private apartments in Singapore were required to have recycling receptacles within their premises. To drive home the message of "reduce, reuse, recycle" among Singaporeans, NEA kicked off Recycling Day in 2004. This was extended to become Recycling Week in 2010.

The various measures have brought about a gradual improvement in our waste situation over time. In 2001, the national recycling rate was 44%. As of 2011, the recycling rate has reached 59%.

SINGAPORE PACKAGING AGREEMENT

Less of waste means more going for environment sustainability. Today, more than 100 companies and industry associations in Singapore have banded together to reduce packaging waste, thanks to an agreement launched in 2007 by NEA.

The agency began exploring the idea of reducing packaging waste right at source in 2005. After consultations with the industry, it concluded that the best way forward was to facilitate a voluntary agreement among stakeholders. In 2006, NEA formed a task force comprising members from the industry, NGOs and the government to develop the Singapore Packaging Agreement (SPA).

Albert Lim, Chairman of the Packaging Council of Singapore, was invited to co-chair the task force along with NEA and Singapore Environment Council. His expertise and experience were crucial to the development of the SPA.

In 2007, the five-year SPA was finally signed by 32 organisations, including five food and beverage industry associations representing more than 500 companies. The NEA also formed a Governing Board to oversee the implementation of SPA, with representatives from different stakeholders.

The board set to task immediately. A Code of Practice was developed to guide designers and users in addressing environmental concerns in their packaging decisions. Activities were organised. "Every few months, we invite signatories, non-signatories and relevant government agencies for a networking session. We present case studies. Companies can share the problems they face with their waste and others can help, or we can match the businesses for solutions. We also circulate information on what other countries do with regards to packaging waste and invite local and overseas speakers to share their views," says Lim. At these networking sessions, non-signatories were also encouraged to participate in SPA.

Membership began to grow steadily. At first, only food and beverage packaging was covered under SPA. In October 2009, the agreement was extended to include all types of product packaging including detergents and electronics. Commercial building owners, shopping malls and supermarkets also joined the agreement to support the waste reduction drive.

This programme has yielded encouraging results. Within the first year, these organisations had managed to save 850 tonnes of waste collectively. Four years into the agreement, the signatories cumulatively saved more than 7,000 tonnes of waste. In 2008, the 3R Packaging Awards were inaugurated to recognise signatories who had made commendable progress in achieving the agreement's objectives.

A second SPA is to be launched in July 2012. Says Mr Lim, the man who played no small role in making SPA possible, "I have been in the packaging industry for the past 36 years. This is one of the things I want to be involved in purely because packaging has been part of my life. The amount of packaging in Singapore is quite high, so somehow it is also a personal achievement to see the waste being reduced, and the change taking place in the packaging industries now."





3R Packaging Awards Ceremony, 2011.



"It took us about a year to develop SPA. We visited New Zealand to study their packaging agreement and adapted it for Singapore. There were consultations with the industry. One of the biggest challenges was getting signatories on board. We had to talk to the companies and convince them that they could benefit by reducing packaging waste. In the end, they realised that raw material costs were increasing, and the only way to keep packaging costs down was to cut packaging usage or reduce packaging wastage."

— Albert Lim, Chairman, Packaging Council of Singapore and Chairman, SPA Governing Board

"At Tetra Pak, we continuously refine our processes in the manufacturing of beverage cartons. For example, we identified the possibility of reusing trimmed polyethylene from our production line within a closed loop. The material was fed back into the lamination process, thereby reducing waste. This required modifications to our equipment and capability development to ensure that product quality was not compromised. Through the Singapore Packaging Agreement, Tetra Pak was able to share with other signatories our innovations and relevant contacts to facilitate the recycling of composite material containing polyethylene and aluminium, which are also used in other food and beverage packaging. We also work with different stakeholders to make sure beverage cartons are recycled as broadly and effectively as possible. Since 2008, we have been running the nation's largest school recycling competition, to encourage students to recycle used beverage cartons. Today, the Tetra Pak School Recycling Competition has expanded to include 100 schools islandwide."

— Jacques Deschamps, Cluster Leader for Supply Chain Operations in South and South East Asia, Tetra Pak

"Nestle has always embraced efforts towards a sustainable environment. To affirm this commitment to our stakeholders, we readily agreed to be a pioneer signatory to the Singapore Packaging Agreement in 2007. This platform has given us the opportunity to showcase our initiatives towards a sustainable environment, and by doing so, we hope to encourage others to follow suit. We are also kept abreast of what other manufacturing and packaging companies are doing and learning from them as well. By continuously reviewing the design of our packaging and improving our production processes, we have been able to make better and more efficient use of packaging resources. We have saved about 25 tonnes of tin, 67 tonnes of plastics and 27 tonnes of paper every year. Besides walking the talk, we would like to encourage all others to do their part towards a sustainable environment by recycling packaging waste. Hence, we developed a smartphone application, 123Recycle, with Nanyang Polytechnic to help educate consumers on how and where to recycle packaging waste."

— Valerio Nannini, Managing Director, Nestlé Singapore

A Liveable Place

"For me, I think it is important that NEA excels as an effective and efficient regulator. There are three things we want to achieve. Firstly, we want Singapore to have the best environment possible — clean air, clean water and clean land. And also a quieter environment. Secondly, we want to build a technically strong and professional team to ensure operational excellence. And thirdly, we aim to provide excellent service quality to the public and industry."
— Koh Kim Hock, Director-General, Environmental Protection Division, NEA



Of all the days in a year, more than 90% of them are days where Singaporeans enjoy good air quality. The raw water from our catchment areas is safe to be treated into drinking water. The beaches are healthy, and our children may play in the waters without fear. All our inland waters support aquatic life.

In a short space of 10 years, NEA has built on the legacy of the former ENV to improve environmental management and catalyse changes in the way things were done. It has also wrought a closer and more meaningful working relationship between the government and the private sector, as well as raised the quality of the environment in Singapore.

Our environmental performance has also garnered international attention. One such example is Singapore's ranking in the Siemens Asian Green City Index, launched in February 2011. The index was put together by the Economist Intelligence Unit, a business entity within The Economist Group. It measured the environmental performance of major cities in Asia. Among the 22 cities surveyed, Singapore was the only city to receive a ranking in the top band of performance, labelled "Well Above Average". In eight of the categories used to assess the cities' environmental performance, Singapore scored "Well Above Average" for water and waste management and "Above Average" for all the other six categories including energy, land use and transport.

The index report noted, "... it is comprehensive and effective policies that elevate the city to rank well above average overall. A rich city-state, Singapore has access to resources, but unlike other cities in the Index, the government is not split between competing levels of administration. And it has a highly trained civil service, along with a reputation for transparency, which is underlined by Singapore's fourth place in Transparency International's Corruption Perception Index."

The importance of political commitment in bringing about a sustainable urban environment was underscored in the speech given by Mayor of South West District, Dr Amy Khor, at the launch of the index in Singapore. She said, "As you are aware, Singapore is a city with limited natural resources and land space. Since the early days of our nation-building, careful thought has been put into how we manage our scarce resources and achieve economic growth, while maintaining a good quality living environment for our people. Today, Singapore enjoys a reputation as a clean and green garden city, through long-term planning and an integrated approach to urban management. Where we encountered constraints, such as our shortage of water, we turned this into an opportunity and developed NEWater. Likewise, we treat waste as a resource and convert waste into energy. This culture of continuous innovation underpins much of Singapore's efforts to overcome its constraints. Our reputation as a clean and green city is the result of continuous effort and cooperation from all stakeholders from the government, industry and community."

The efforts and hours NEA officers put into safeguarding the environment are well worth it. The reward? A clean, liveable, sustainable place for today, and for Singapore's future generations.



In a short space of 10 years since NEA was formed, it has built on the legacy of the former ENV to improve environmental management and catalysed significant changes in the way things were done. It has also wrought a closer and more meaningful working relationship between the government and the private sector, as well as raised the quality of environment in Singapore.





Working among us

A Framework For A Healthy Environment

"The work that NEA does is important. It affects our lives on a daily basis. NEA as an agency is fantastic. They have people who know how to clean up, they know how to make sure that you can control dengue, how to manage hawker centres, they are very good. These are the silent workers."

— Dr Yaacob Ibrahim, Minister for the Environment and Water Resources (2004 - 2011)



NEA's work touches us in almost every aspect – the clean streets we walk on, the hygienic food establishments in which we dine and the healthy environment we live in.

The agency's officers walk and work among us, covering the ground to inspect, survey, enforce and even educate. Its team of researchers, analysts and officers from various fields support, improve and strategise NEA's operations, for the cause of better environmental and public health.

A sound legislative framework provides the foundation for the agency's operations. A number of relevant laws are administered by NEA, such as the Control of Vectors and Pesticides Act, Sale of Food Act, Infectious Diseases Act and Smoking (Prohibition in Certain Places) Act.

The backbone of NEA's environmental health work is the Environmental Public Health Act. Drafted in 1968 and implemented in 1969, the Environmental Public Health Act was a milestone in the history of Singapore's environmental health. Based on similar health acts in other countries such as Britain, the act encompassed comprehensive regulations on issues ranging from the management of food establishments and public cleaning services to vector control and hygiene in funeral parlours. The act brought Singapore up-to-date in addressing public health issues caused by rapid urbanisation and a burgeoning population.

Over time, various acts have been updated to ensure that they stay relevant and adequate to ensure a liveable, healthy environment. For example, regulations pertaining to food hygiene have been updated as recently as 2011 to incorporate higher standards for catered food.

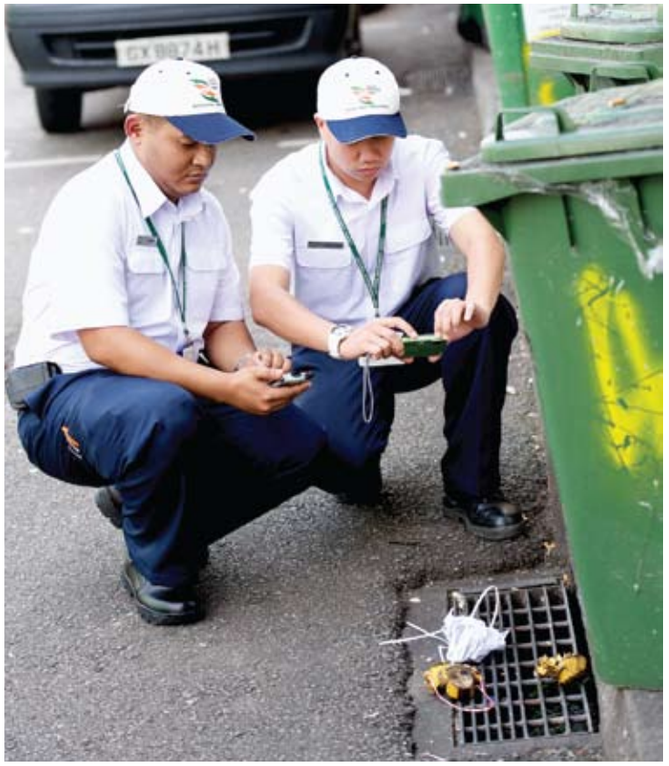
Accompanying these laws are Codes of Practice on various aspects of environmental health. Compiled for industry stakeholders, these codes comprise requirements, guidelines and best practices. For example, the Code of Practice on Environmental Health spells out legal requirements to address environmental health concerns in the design of buildings. Before the premises are allowed to operate, NEA will ensure that guidelines are complied with.

These codes of practice, together with the environmental laws, form the bedrock for NEA's work in regulating environmental public health and hygiene. They give the agency the authority to execute its work, and the mandate and mission to keep us in the pink of health.

Opposite: Mosquitoes can breed in stagnant water trapped in a space as small as this keyhole. Here, NEA officer extracts a sample using a pipette to analyse if breeding is taking place.



A Clean Act



Despite Singapore's clean image, litter still afflicts our environment. At its best, litter is unsightly. At its worst, litter can provide hospitable habitats for the breeding of mosquitoes and food waste for vermin, as well as pollute and choke our waterways. NEA manages Singapore's public cleaning regime, educates the public on anti-littering and enforces against litterbugs to ensure cleanliness in our environment.

A THOROUGH REGIME

Singapore's reputation as a clean city is in part a result of its thorough public cleaning regime, which is undertaken by private companies and NEA's daily-rated workers.

In the past, public cleaning duties were divided among various government agencies. For instance, public parks were taken care of by the National Parks Board (NParks) while vacant state lands were managed by the Singapore Land Authority (SLA). In April 2012, NEA began to progressively take over all the duties for public cleaning except for housing estates, which would continue to be managed by town councils. The move to integrate public cleaning services under NEA brought greater operational efficiency while providing a single feedback channel for the public on cleanliness-related issues.

GETTING TO THE ROOT

A reliable cleaning service alone is insufficient to keep our public areas clean. To get to the root of the problem, NEA uses public outreach campaigns to change mindsets and enforcement to deter and punish offenders.

The first national programme against littering was the month-long "Keep Singapore Clean" campaign launched in 1968. Annual campaigns to drive home the message became a regular feature from then on. After NEA's formation, it increased efforts to battle littering. In 2002, the "Singapore, Litter-free" campaign was launched, with an emphasis on ownership by the community to keep the country clean. The campaign went high-profile when mass events including National Day were targeted to be litter-free. In 2002, schools joined the campaign and in 2006, the campaign was extended to hawker centres. By 2008, coffee shops, bus interchanges and reservoir parks were roped in to extend the message's reach.

To deter would-be litterbugs, uniformed NEA officers patrol public areas regularly. The officers, together with auxiliary police, stake out littering hotspots to catch offenders. When a litterbug is caught, a ticket will be issued on the spot, requiring the offender to either pay a fine or attend court. To reform and deter littering behaviour, penalties such as fines and Corrective Work Orders are meted out depending on the severity of the act. These penalties apply to foreigners too.

"Environmental health is a highly challenging, dynamic field, particularly in a high-density compact city-state such as Singapore. In every aspect of our work, from ensuring high standards of food hygiene and vector control to keeping our public areas clean, from managing hawker centres and markets to running the country's only active cemetery, we seek to strike a balance between individual concerns and the greater welfare of the community, whilst keeping an eye on delivering services in the most cost-effective manner so as to meet public expectations without compromising on operational needs. Ultimately, we need the active co-ownership and participation of every citizen to help make Singapore a clean, healthy and liveable home for all of us."
— Derek Ho, Director-General, Environmental Public Health Division, NEA



REFRESHING THE ANTI-LITTERING CAMPAIGNS

Bins placed in more accessible and strategic locations, mothers playing a key role in spreading the anti-littering message and litter-free ambassadors shaping a new social norm of denouncing littering behaviour. These measures are part of NEA's renewed efforts to stamp out littering following two key studies on the issue.

Despite years of campaigns and strict enforcement, the problem of littering still persists. To understand what motivates such behaviour and how it can be addressed more effectively, NEA commissioned two important studies in 2006 and 2009 respectively.

The 2006 littering behaviour study saw researchers stationed in 203 public locations, from beaches, bus terminals and void decks to foreign worker congregation sites. They observed litterbugs and people who binned their rubbish, and conducted 3,018 face-to-face interviews with both groups of people. The key findings? Two out of 10 Singaporeans responded that they would litter and 83% of respondents felt that imposing fines would deter littering. The study led NEA to place more emphasis on engaging the community to change their mindset on littering.

In 2009, three sociologists from the National University of Singapore and a World Health Organization (WHO) consultant were brought in to conduct a more in-depth study of this issue. The panel researched and reviewed different aspects of littering, such as the profile of typical offenders, public perception of littering, communication strategies, enforcement procedures and the various social, cultural and environmental factors linked to the act. Focus group discussions and detailed interviews were held, and researchers surveyed 1,500 secondary school students and 3,000 households.

The findings were highly insightful. For example, major factors found to motivate littering behaviour include the lack of visible enforcement and bins nearby. Older people are more apologetic for littering than youths, women are more concerned with being caught littering than men and mothers feel that they should be positive role models for their children by not littering. NEA's anti-littering campaign was reviewed and found to be lacking specific messages to different target groups such as youths and smokers. The punitive measure against littering was also under-emphasised.

Armed with these insights, NEA refreshed its campaign again in June 2010. It improved the litter bin infrastructure, increased the frequency and visibility of uniformed enforcement and came up with more targeted public outreach and education efforts. More focus was placed on reaching out to mothers and youths as well.

Four months after the new measures were rolled out, NEA reviewed the effectiveness of the campaign. Visual checks on 20 littering hotspots revealed that the amount of litter had decreased. The number of litterbugs caught dropped from 41,392 in 2009 to 23,898 in 2010. By 2011, it had dipped further to 11,131. Most likely, this is due to the new anti-littering campaign in June 2010 and the deterrent effect of uniformed enforcement, which was carried out daily in 2011.



1967 - more than 1,000 people took up brooms and went on a "Sweep-Clean" campaign in Jalan Besar initiated by the Ministry of Health. *(Source: Ministry of Information, Communications and the Arts, Courtesy of National Archives of Singapore)*

SWEEPING THROUGH TIME

Alleys cluttered with discarded objects, sidewalks littered with piles of sweepings and drains clogged with garbage. Such was the normal state of affairs in Singapore even in the 1960s. Cleanliness in the city was a tall order when public cleaning operations were encumbered by the shortage of manpower and resources. Good social habits were generally lacking and many premises did not even have a dustbin. Dumping the sweepings from one's home or shop onto the streets was quite a standard practice then.

Shortly after Singapore became independent in 1965, the new government began to tackle the problem of public cleanliness. Harsher penalties for littering were imposed, dustbins were provided to households without one, public campaigns were launched and street patrols were organised to catch those who dumped rubbish illegally. In 1968, a programme was put in place for streets to be swept and refuse to be removed every day. The work was carried out manually by daily-rated employees. However, manpower shortage being an issue, the government introduced mechanical sweepers in 1972 for more efficiency. By the 1980s, much headway had been achieved as streets became cleaner and the resettlement of the majority of Singaporeans into Housing and Development Board (HDB) flats made domestic waste collection simpler.

To make the business of keeping the streets clear of rubbish more efficient, the government began outsourcing public cleaning services. By 2004, street cleaning services for the western, eastern and central parts of Singapore had been outsourced. Today, NEA carries out daily audits of the cleaning companies' services to ensure their reliability.

REFORM IN ACTION

You may have seen these people in bright, luminous green vests bearing the words "Corrective Work Order" sweeping away. Corrective Work Order (CWO) was introduced in 1992 to make litterbugs participate in public cleaning for a period of three hours, in order that they would understand the impact of their acts and the difficulties faced by cleaners. The clearly marked vests also identify them as litterbugs to distinguish them from cleaners, so that the shame of being punished in such a public manner would serve as a deterrent to potential litterers.



NABBING HIGH-RISE LITTERBUGS

It seems almost unbelievable that there are people who would throw sanitary pads, food waste and even soiled tissue out of their flats in high-rise buildings, but it does happen. To tackle such behaviour, NEA distributes educational materials and displays banners and posters regularly in the public to highlight the dangers of high-rise littering and to encourage greater social responsibility. When these acts take place, enforcement action requires evidence or statements by witnesses. NEA first serves letters to residents in the suspected unit and those living in the same column of flats to advise against high-rise littering. Warning letters come next when such acts do not stop. And if the behaviour persists, the agency will proceed with stake-outs to gather evidence against the culprits, followed by prosecution. NEA will also be rolling out surveillance cameras for high-rise littering hotspots to aid in nabbing such offenders after a successful pilot in 2011.





"In any policy review, we need to first understand the needs and concerns of all affected parties and find a balance between meeting public expectations and ensuring that the policies and measures introduced are practical and implementable in addressing those needs. We will focus on areas where the public will be impacted by second-hand smoke in their daily lives and at community spaces where human traffic is high. We also take into consideration spaces used by vulnerable groups; hospital compounds, nursing homes, children's playground, etc. Our long-term aim is to allow smoking only at designated areas. Hence, transiting from a prohibitive list to a positive list which states places where smoking is allowed. Think about it, as a citizen, what would you expect? You would expect that the moment you step out of your home whether to work or play, the authorities would ensure that you do not have to pass through second-hand smoke if you do not wish to. When you go to bus stops or train stations or visit any food places, you wish you have a choice to say, 'I don't want to go through a smokescreen'. Together with the support from the public and stakeholders, we will continue to increase the protection against second-hand smoke by making more public places smoke-free. We hope that the community too plays its part, for example by politely advising smokers to refrain from smoking at prohibited places and encouraging them to avoid lighting up where second-hand smoke could impact non-smokers. The environment is for all to enjoy and each of us has an important role to play."

– *Chua Yew Peng, Director, Policy and Planning Division, NEA*

OUT WITH SMOKE!

Now you can enjoy an al-fresco dinner, groove the night away at one of Singapore's many nightlife spots, and work out at designated exercise areas, without having to suffer second-hand cigarette smoke. These are just some of the areas which became smoke-free after 2006 when the Smoking (Prohibition in Certain Places) Act was extended.

The government has been progressively increasing the number of smoke-free public places since 1970 so that there are more spaces where the public may use or enjoy without having to put up with the hazards of cigarette smoke. The prohibition kicked off with omnibuses, theatres and cinemas and was gradually extended over the years to virtually all indoor places and many outdoor community spaces.



The prohibition of smoking in public places is part of the government's National Tobacco Control Programme, led by the Health Promotion Board and supported by agencies including the Health Sciences Authority, Singapore Customs and NEA. The programme aims to reduce smoking prevalence in Singapore by adopting a holistic approach including public education, legislation, taxation and provision of smoking cessation services.

The smoking prohibition policy adopts an integrated approach comprising policy or legislation review, enforcement and public communications. Consultations with the public and stakeholders are carried out to gather public sentiments and address operational concerns. For example, for the extension of the ban to hawker centres and food establishments on 1 July 2006, and entertainment outlets on 1 July 2007, NEA had begun the public consultation process in 2005 through an e-poll. In addition, focus group discussions were held with smokers and non-smokers, and dialogue sessions were carried out with owners of food establishments and entertainment outlets. This enabled NEA to fine-tune the policy to meet public needs while balancing it with operational considerations of stakeholders, thereby ensuring a smooth roll-out of the extension.

The result? While hawker centres, food establishments and entertainment outlets were made smoke-free, provisions were given in the law for operators to set up designated smoking areas so as to cater to patrons who were smokers. The early announcements and engagement efforts gave the operators ample time to prepare for the ban.

In addition, public communications and education efforts played an important role in preparing the ground for the implementation of the policy. Media publicity efforts, guidebooks for premises operators, advertisements and an educational blitz were launched to ensure that the public and stakeholders were fully aware of the upcoming ban. Enforcement was also carried out to ensure compliance and deter smokers from flouting the laws.

The same approach was again adopted for the next smoking ban extension announced in March 2012. Over 8,000 responses were received through the e-poll survey conducted between November and December 2011, with 89% supporting the call for more smoke-free places. Feedback was also garnered on the places where the public would like to be smoke-free. Beyond the e-poll, NEA also collected feedback through social media platforms, discussion forums and feedback channels. Consultations with stakeholders including HDB, LTA, town councils and other premises owners and managers were also conducted. Such exercises allowed NEA to work out the details of implementing the ban's extension with the different stakeholders.



SLICK OPERATIONS

When it comes to cleaning the environment, it's not just about streets and pavements. On 25 May 2010, an oil tanker and a bulk carrier collided in the Singapore Straits just 13 km off Changi East. Some 2,500 tonnes of crude oil from the tanker spilled into the sea.

The Maritime and Port Authority of Singapore (MPA) formed the first line of defence. It moved in immediately to contain and treat the spill. NEA, on the other hand, began mobilising resources to clean up the shores should the oil patches spread. Its air and water quality monitoring teams were despatched on the very same day to assess the impact of the spill on the environment. The teams found the levels of toxic substances in the air and water to be within safe limits.

The next morning, oil patches were detected off Changi Naval Base and a nearby rock bund. By the afternoon, the patches had spread to East Coast Park beach. Within the next few days, Tanah Merah and Chek Jawa at Pulau Ubin were hit too.

NEA straightaway closed East Coast Park and Changi beaches to the public and launched an intensive clean-up operation. Some 30 NEA officers were deployed to conduct surveillance at different stretches along the eastern coast. About 500 personnel from NEA, a cleaning company and employees of the tanker owner took part in the operations.

From 7 am to 7 pm every day, these personnel worked to clean up the patches. Dispersants were used to treat the oil, and contaminated sand was removed and disposed of. Operations sped up after NEA successfully tested a more effective biodegradable agent comprising nanoparticles to clean thick patches of oil at East Coast and Pulau Ubin.

At the same time, NEA led other government agencies and volunteers to play different roles in the operation. Daily meetings were held to coordinate ground operations and bring everyone involved up to speed. At Chek Jawa, oil absorbent booms and padding were put up to keep the oil out of the area. Patches that seeped through were cleaned by NEA, NParks and volunteers. Thanks to the swift response, no major damage was done to the eco-system.

PUB, the national water agency, installed oil absorbent float booms to prevent oil patches from entering drains and canals along East Coast Park beach. It also brought in a vacuum tanker to suck up oil scum from the water surface.

The rapid response and collaborative efforts facilitated a faster, smoother clean-up act. Ten days after the operation took off, the beaches were declared open to the public on 4 June 2010.



"Even as Singapore develops and progresses, and our population increases, we have to make sure that our environment stays clean, liveable and sustainable. It's not enough to just keep it at status quo, though that in itself will be a challenge since the stress placed on the environment will grow as the country develops further. We have to find ways to make our environment even better, cleaner and healthier. The responsibility for this falls upon MEWR, and its statutory board, NEA. Which is why over the years, NEA has consistently introduced measures and initiatives to improve our surroundings. The extension of the smoking ban is but one example. For many members of the public, the environment has become more pleasant as a result. Another important initiative is the consolidation of public cleaning services under NEA for more effective and efficient operations. As we move forward, NEA's work in maintaining food hygiene standards, vector control and research and public outreach will become all the more challenging and critical."

— Grace Fu, Senior Minister of State, Ministry of Information, Communications and the Arts, Ministry of the Environment and Water Resources

A Safe And Clean Food Paradise

NEA relies on stringent systems of licensing, enforcement and ground inspections to perform its duty. However, given the number of food operators and handlers in Singapore, continuous monitoring is impractical. At the end of the day, those in the food and beverage industry need to do their part too in taking care of hygiene and cleanliness.

Talk about Singapore being a food paradise. In 2011, there were a total of 5,855 cooked food stalls in government hawker centres, 13,829 food stalls in private eating establishments, 14,363 private food shops (including restaurants, fast food outlets, eating houses, food courts and caterers), and 367 supermarkets. Over 100,000 food handlers work in the industry, providing sustenance for Singaporeans and tourists alike.

These were also the numbers of food establishments NEA had to inspect and regulate in 2011 alone to ensure a high level of hygiene and cleanliness.

Food safety in Singapore comes under two agencies – NEA and the Agri-Food and Veterinary Authority (AVA). AVA regulates food safety at the stages of import, manufacturing, distribution and wholesale while NEA regulates retail food service establishments to ensure that food is prepared in a hygienic manner.

Maintaining high standards of food hygiene in Singapore is a task of herculean proportions. Every year, more and more food and beverage outlets open for business. With more locals dining out and increased tourist arrivals, the number of patrons at food establishments grows by the day as well. In 2004, a survey showed that 48.4% of Singaporeans ate out, a figure which rose to 60.1% in 2010. On top of that, there is also a great disparity in the standard of hygiene and food safety procedures across the food and beverage industry.

To make sure that this veritable food paradise offers safe and clean food at all times, NEA relies on stringent systems of licensing, enforcement and ground inspections. However, with the number of food operators and handlers in Singapore, continuous monitoring is impractical given our finite resources. At the end of the day, those in the food and beverage industry need to assume the primary responsibility in ensuring that food for sale is prepared under hygienic conditions. Consumers can also play their part by not patronising food outlets with poor hygiene practices.

A SYSTEM FOR TRACKING

It is just a piece of paper, but this licence makes all the difference in many ways. It allows food operators to operate legally and NEA to track their performance and records more efficiently. The system of licensing also aids NEA in working out inspection schedules and taking swift action in the case of any non-compliance.

Every retail food establishment, be it a stall in a hawker centre, a restaurant, or a fast food outlet, is licensed by NEA, subject to compliance with licensing requirements and the Code of Practice on Environmental Health. In the case of hawker centres and markets, NEA also manages the tenancies.

WATCH FOR THE DECAL

Grading is not just for students. At every hawker stall and food shop, you can clearly see a decal displaying the grade of the premises. These grades are given every year by NEA, whose officers assess the cleanliness and hygiene of the premises before giving the food establishments a grade.



"We take cleanliness seriously not just for the 'A' grade, but because people are eating our food, so we must take care of hygiene."

— Kwan Wai Kern, proprietor of Sin Tong Nam, a bakery with an 'A' grading at Tiong Bahru Market

Implemented in 1997, the system's highest grade is "A", when the premises score 85% or more on NEA's checklist. "D" is the lowest grade, when the score is between 40% - 49%. These grades are given based on criteria such as cleanliness of food preparation areas, implements and equipment and the hygiene practices of food handlers.

The grade must be displayed prominently so that the public can gauge the hygiene standards of the outlet. With pressure from both the authorities and rising public expectations, food hygiene standards have gradually improved over time. Back in 1997, less than 50% of food establishments scored "A" or "B", compared to more than 90% in 2011.

Moving forward, NEA will adopt a risk-based assessment framework in its food hygiene surveillance. Key factors such as the types of food, operating hours, clients and past records of operators would be taken into consideration in risk-profiling of food premises for inspections. For example, a restaurant selling raw food such as sushi runs a higher risk of food contamination while caterers serving vulnerable groups including the young, sick or elderly have to be monitored more closely. This system, to be used internally, will allow NEA to deploy its resources more efficiently.

INSPECTIONS, UNANNOUNCED

It's done on a regular basis, it's unannounced, and sometimes, officers in plain clothes make an appearance.

Inspections are part of NEA's efforts to ensure food hygiene. These surprise spot checks allow NEA to know what is happening on the ground and take action where necessary. On average, more than 12,000 inspections were conducted every month in 2011.

Should officers find any flouting of regulations, offenders will be penalised. The Points Demerit System, introduced in 1997, is used to assess the seriousness of the offences. Demerit points are given for lapses in hygiene, such as failing to keep the establishment free of vermin or insects, sweeping food scraps onto the ground and preparing food on the floor. Depending on the number of demerit points accumulated, the business may face suspension or may even have its licence revoked.

NEA officers also go around the island in plain clothes to purchase food from establishments randomly. After the purchase, the officers will reveal their identity to the sellers. The food is then tested for pathogens in accredited laboratories. Besides facilitating the collection of data on food contamination, these spot checks also help motivate sellers to uphold better practices in food hygiene and safety.

PUSHING FOR HIGHER STANDARDS

To push for higher standards of food hygiene in the industry, NEA requires all food handlers and operators to attend and pass relevant courses developed jointly by NEA and the Workforce Development Agency (WDA). Topics covered range from micro-organisms and food contamination to hygienic ways of preparing, handling and storing food. Every three years, food handlers must undergo a refresher course.

NEA also organises regular briefings and dialogue sessions to inform and educate stakeholders in the industry. At the end of the day, food establishments depend on the public for business, which means that public pressure is more likely to be effective. Therefore, NEA holds regular public health education workshops and talks to create more awareness among consumers on hygiene issues. It also briefs grassroots organisations and government agencies on safety guidelines on catering when festive seasons come round. Track records of licensed food caterers are published on NEA's website so that the public can avoid using unlicensed or unreliable caterers.

Opposite: Measuring temperature of cooked food to ascertain that the food has not been left standing for long. NEA officers also take the opportunity during inspections to educate operators on control of food temperature and serving time.


LAU WEI HAR
TAN AIK KEAT

"NEA has made critical contributions to Singapore's liveability over the years through its relentless efforts to uphold high standards of public health and hygiene. This is despite more economic activities and a larger population. It is well-positioned not only to continue in this excellent work, but also to lead in new areas such as climate science, environmental research, and industry development. Much of NEA's success can be attributed to its dedicated and professional staff who continue to seek new solutions to emerging challenges, and to respond swiftly and effectively when faced with new demands. Over the last 10 years, NEA has successfully achieved the operational effectiveness and service responsiveness which were the objectives of its establishment as a statutory board in 2002. It is a cohesive and resourceful organisation that works closely with MEWR and many other stakeholders in the public sector, industry and community."

— Rosa Daniel, Deputy Secretary, Policy, MEWR



THE PROBLEM OF FOOD CONTAMINATION

Have you ever left food in your home only to find it going bad or mould growing on it some time later? Singapore is warm and humid – ideal conditions for bacteria and mould to multiply and spread quickly as compared to countries with temperate or drier climates. These bacteria and mould can produce toxins that are harmful to health.

This is only one of the many challenges of preventing food poisoning in Singapore. Contamination can be caused by many different factors – such as poor personal hygiene, dirty premises, food left standing for too long, unclean cooking implements, contaminated ingredients, undercooked food or cross-contamination with other sources.

When food poisoning incidents do happen, the investigation falls primarily upon three agencies – AVA, the Ministry of Health (MOH) and NEA. AVA is involved if the food source is suspected to be contaminated, such as the food manufacturer or industrial kitchen. MOH investigates which bacteria or toxins are responsible. NEA checks the retail outlet to verify what has gone wrong.

Such investigations are highly challenging processes. As the food and beverage industry is fast-moving, inventory is usually exhausted by the end of the day, sometimes even before afflicted consumers report falling sick. Generally, food businesses also tidy up and clean their premises when the day is over, which makes it hard for investigators to find the source of contamination. For example, if a chopping board cross-contaminated with bacteria is responsible for a outbreak, it will usually be cleaned by the time food poisoning was reported. Over at MOH, ascertaining the cause of poisoning takes time as well. Patients generally consume a wide range of foods from different sources every day, and pinpointing the responsible source is a challenge if the outbreak is not massive. Investigation thus takes considerable time and it is often hard to identify the exact cause of poisoning in many cases.

Despite all of the government's efforts to regulate food safety and hygiene, food poisoning cannot be eradicated completely. Just one hygiene lapse, such as handling cooked food with unwashed bare hands, is enough to cause an outbreak. Ultimately, food handlers and operators are responsible for the safety and health of the food they sell.

"NEA's strength lies in its team of dedicated and professional staff. They are a cohesive team and are very knowledgeable and experienced in their work. It was indeed an honor and privilege for me to have worked with them. They are very committed to their jobs, the unsung heroes of our clean, green and healthy environment. They carry out their duties and responsibilities untiringly, without fear or favour, be it in enforcing against littering, enforcing the smoking ban, sweeping the streets, checking on hawkers, fighting the dengue scourge, operating the incineration plants efficiently, checking on construction noise pollution and so on. They work behind the scenes and are seldom in the limelight. When things are all right, their efforts go unnoticed and often unappreciated, but when things go wrong, they are cast into the limelight. It is important that NEA continues to attract and retain a fair share of the nation's talents and develop a core team of professional officers with the passion for the environment."

— Lee Yuen Hee, CEO, NEA (2005 - 2009)

STAMP THE TIME!

Before you head for the catered buffet line or dig into a packet of catered food, check the time-stamp of the food. If it has been more than four hours since the time-stamp, avoid the food for your health's sake.

Since 15 February 2012, NEA has required all food caterers to time-stamp all buffet and packed meals. A time-stamp sign must be displayed prominently on buffet tables set up by caterers. Catered packed food should also have labels indicating the time-stamp. The agency will take action against those who do not time-stamp their food, or who provide false or misleading times.

This is because when cooked food is stored between 5°C and 60°C, bacteria may multiply quickly to unsafe levels or produce toxins which can cause food poisoning. To prevent this from happening, cooked food should be eaten within four hours if it is left standing within this temperature range. However, consuming the item within four hours is not a guarantee that food poisoning will not occur. Undercooked food and contamination caused by unhygienic preparation, dirty food service equipment and poor handling may still cause food poisoning.

Catered food poses higher risks to consumers than dining in restaurants or hawker centres, because a large amount of food is prepared in advance and stored before it is transported to the venue where it may stand for a while before being consumed. Between 2009 and 2011, caterers accounted for 21 of 75 major food poisoning incidents and 35% of all persons affected in major food poisoning incidents, even though caterers comprise only 1% of the food establishments in Singapore. This led NEA to impose more stringent requirements on food caterers and conduct inspections more frequently. NEA also requires all food caterers to engage at least one certified Food Hygiene Officer to oversee and audit the hygiene standards and operations of the caterer.

The time-stamp requirement came about after consultation with the food catering industry. Educational materials on time-stamping are also made available to inform caterers of the requirement and rationale. Says James Wong, President of Association of Catering Professionals Singapore, "The new regulation can serve to protect both the industry and the consumers from food poisoning ... the association will continue to work with its members to improve on industry best practices to ensure safe food for all."



SAFE WATER TO DRINK

The quality of piped drinking water in Singapore is one of the safest in the world, and well within guidelines set by WHO. As an added measure to guarantee that our piped water is safe for drinking, NEA was appointed in 2007 as an independent auditor to regulate the quality of water supplied by PUB. Using guidelines developed by a technical committee comprising experts from fields such as toxicology, water treatment, medicine and microbiology, NEA conducts audits and visits the suppliers on-site to test the water quality for compliance with these guidelines.

WHEN HYGIENE IS UNDERMINED

4.30 *am*, 4 April 2009. MOH informed NEA of a food poisoning outbreak. 7.30 *am*, Kandang Kerbau Women's and Children's Hospital contacted NEA, reporting 57 cases of food poisoning cases. All the cases were linked to the Rojak Geylang Serai stall in the Geylang Serai temporary market. 8.00 *am*, NEA officers arrived at the suspected stall to conduct a joint investigation with MOH.

Samples and swabs were taken. The stall holder and four assistants were sent for screening for food-borne pathogens at Tan Tock Seng Hospital. Business operations of the stall were ordered to stop immediately. While NEA took charge of the scene, MOH focused on finding out the cause of the food poisoning. As an added precaution, NEA also conducted an inspection of all stalls at the temporary market.

It was the worst case of food poisoning in recent times, with two fatalities and 152 ill, 48 of whom were hospitalised. One victim suffered a miscarriage.

THE "RAT" HERRING

While investigations were ongoing, public speculation swirled. Just two days before the outbreak, a local paper had run a story on rodent infestation at Geylang Serai temporary market, accompanied by vivid images. Despite the lack of evidence, the media and online community leapt to the conclusion that rats were responsible for the outbreak. This was followed by fervent public debate on the government's role in curbing rodent infestation and maintaining hygiene standards.

THE CAUSE: *VIBRIO PARAHAEMOLYTICUS*

Over at MOH, laboratory investigations finally identified the bacterium behind the outbreak. It was *Vibrio parahaemolyticus*, a pathogen found in raw seafood and known to cause gastro-intestinal diseases. However, no one was able to ascertain how the bacteria got into the rojak. Before NEA and MOH arrived on the scene on 4 April, the food handlers had discarded all the remaining food and cleaned up their premises. Most likely, cross-contamination between raw seafood and the cooked food had taken place.

A REALLY UNHYGIENIC STALL

Although the authorities could not establish how the contamination had taken place, two things were clear – that the stall was the source of food poisoning, and that the food handlers practised poor hygiene. The chopping board used in the stall was found to be carrying faecal coliform, which comes from faeces, and *Staphylococcus aureus*, a bacterium is found on human skin. Swabs taken from various areas in the stall showed traces of *Salmonella*, a bacterium which causes food poisoning. The stall holder was subsequently fined and debarred from applying for a hawker or food shop licence and registering as a food handler. His assistants had to re-take and pass the Basic Food Hygiene Course again as a precaution.

ADDRESSING PUBLIC CONCERNS

Meanwhile, NEA had to address public speculation and debate on rats and hygiene. On 18 April 2009, an open letter from the CEO of NEA, supplemented by press interviews with Khoo Seow Poh, then Director-General of Public Health, was published in full in newspapers. In the letter, the CEO expressed his sympathy for the victims and provided assurance that NEA would continue to uphold high standards of food hygiene.

Even though rats did not cause the outbreak, the CEO acknowledged that NEA should have moved in earlier to help address the rodent infestation. During the crisis, NEA, together with the Geylang Serai Temporary Market Management Committee, had stepped in to cut down rodent population at the market.

A STRICTER HYGIENE REGIME

Despite the fact that the outbreak was caused by personal hygiene lapses, NEA decided to use this episode to further strengthen its regime on the control of food hygiene.

NEA stepped up its inspection of markets, food centres and food stalls, and increased fines for lapses. The mandatory Basic Food Hygiene Course for food handlers was revised to emphasise proper hygiene practices to prevent food contamination.

THE QUESTION OF TEMPORARY MARKETS

Though it was established later that the food poisoning outbreak was not caused by rodents, the question of cleanliness in temporary markets became a sticking point during the crisis.

When hawker centres are upgraded under NEA's Hawker Centres Upgrading Programme, temporary markets are built and maintained by the local Citizens' Consultative Committees, who usually form specific management committees to oversee the markets' operations.

After the Geylang Serai food poisoning incident, NEA now includes its officers on the management committees of temporary markets and works more closely with the committees to improve housekeeping practices. In addition, from 1 June 2009 onwards, all operators of temporary markets or food centres have to be licensed. They will be held liable for hygiene lapses so as to ensure accountability.

Off The Streets, Under New Roofs



A typical hawker stall in the 1970s. *(Source: The Straits Times © Singapore Press Holdings Limited. Reprinted with permission)*

Gone were the days when hawkers would file past your home, calling out their wares and ringing bells and such to attract customers on the streets. Instead of sitting out on rickety stools under the hot sun and eating your food off a plate of dubious cleanliness, you can now have your pick of meal in the comfort of well-ventilated, sheltered hawker centres.

Many of us remember the days of street hawking with nostalgia, often forgetting the poor hygiene that went along with it. Food waste was often dumped indiscriminately on the streets and in drains by hawkers and patrons, causing pollution and increasing the vector and vermin population. Hawkers would go around with two buckets of water to be used through the day for washing a small stack of plates for serving food. Such poor hygiene practices led to outbreaks of cholera, typhoid and dysentery. Street cleaning was made difficult as roads and drains were obstructed by hawkers and their props. In parts of the city where hawkers congregated, the areas would look haphazard as hawkers built makeshift structures to act as stalls. Those who sold goods other than cooked food spread their wares out to attract customers, congesting streets, alleys and even roads. It was hard to eradicate street hawking as many of the hawkers were hardship cases.

By the late 1960s, however, the rate of employment in Singapore grew and people became more affluent. Street hawking was no longer the main means of livelihood available. As such, the government began to control the upsurge of street hawking to eradicate the accompanying public health and environmental issues. An island-wide census of street hawkers was conducted between 1968 and 1969 and some 24,000 street hawkers were registered.

To solve the root of the problem, the government began building proper centres to house street hawkers. These centres had potable water supply, proper sanitation and sewage systems, refuse disposal facilities, electricity and ceiling fans installed for ventilation. HDB also started constructing hawker centres and markets as part of its facilities for residents in new public housing estates. By 1986, all street hawkers were finally resettled into the newly-built markets and hawker centres. It was a 15-year effort which cost $100 million.

The time and resources spent are well worth it. Today, hawkers are able to practise their livelihoods in well-equipped facilities. Public health problems arising from poor hygiene practices declined significantly with the resettlement of hawkers. Pollution of waterways and dirtying of streets due to street hawking is curbed. The roads are free from congestion by make-shift stalls and unregulated goods display. And Singaporeans get to have good, affordable, hygienic food in clean hawker centres, which also double as community spaces where friends and families go to socialise. Over time, hawker centres have evolved to become a well-loved national icon, and are even listed as one of the top places to see in the landmark book “1000 Places to See Before You Die”!

"We are exploring new not-for-profit management models, creative architecture and innovative designs that make the centres more pleasant without making it more expensive, which encourage social interaction for Singaporeans from all walks of life. It is formal recognition that hawker centres are an integral component of our social infrastructure."

— Dr Vivian Balakrishnan, Minister for the Environment and Water Resources



CATCHING UP WITH TIMES

If you could recall, just a decade ago, our hawker centres and markets were becoming run-down. Most of them were built 20 to 30 years ago, and the facilities could no longer keep up with changing needs and times. This becomes particularly noticeable as surrounding HDB blocks were gradually upgraded.

Regular maintenance of hawker centres and markets owned by MEWR is undertaken by NEA, while respective town councils oversee those owned by HDB. However, maintenance alone is not enough to ensure that these facilities remain serviceable.

Recognising the central place of hawker centres and markets in the lifestyle of Singaporeans, the government embarked on upgrading these facilities in 2001 with the launch of the Hawker Centres Upgrading Programme (HUP). Under the programme, $420 million was set aside to upgrade hawker centres over ten years. The first batch of centres to be upgraded was announced in July. The upgrading allowed hawkers to better compete with supermarkets and food courts and also increased the appeal of the centres.

The objective of the upgrading was to provide a more pleasant dining and marketing environment for stall holders and the public. Before every HUP project, a working committee would be formed, comprising different stakeholders such as grassroots members and hawker representatives, so that matters such as the upgrading schedule, layout of stalls and design of the centre could be discussed and decided upon. Feedback was also gathered from stall holders so that their concerns could be addressed.

The programme was a success. Dining and marketing in the newly-upgraded centres became more pleasant. Hygiene and cleanliness standards were easier to upkeep in the new buildings. Care was taken to preserve the charm and ambience of old hawker centres in the new structures while giving them a distinctive identity. Under the programme, a total of 107 hawker centres and markets would be upgraded by 2014.

As Singapore's population grows, it became clear that new hawker centres were needed in the future as more Singaporeans were eating out. In October 2011, Dr Vivian Balakrishnan announced plans to build new hawker centres, followed by the confirmation in 2012 that seven new centres would be constructed over the next five years.

The government set up a public consultation panel on the issue in December 2011 and invited members from social enterprises and different fields such as food and beverage, environmental sustainability and architecture to come up with ideas on how the new hawker centres should be. These moves ensured that Singaporeans have a stake and say in the making of our future hawker centres. More importantly, hawker centres would continue to be part of what makes Singapore unique.



THE SPECIAL SQUAD HITS

"There were a lot of complaints where people buy canned abalone at a special price. But when they went back, they open up, it's not what they get ... And then you have these people selling talcum powder ... adulterated the talcum powder with flour. They get this empty container from those people who collect them for sale. So they open up the back part of the container, it's easy to remove it. Take out the talcum powder, say about 80 percent, leave 20 percent on the top and then the rest you will (top) up with flour."

– Phillip Chew Peng Kia, who worked as a public health inspector between 1957 and 1974, on the goods sold by illegal hawkers (Source: Oral History Centre, National Archives of Singapore)

There is a long list of reasons why illegal hawking had to be eradicated. Street hawkers are sources of unwholesome food, fake and contraband goods, pollution and street obstruction. Singapore's efforts to end illegal hawking was carried out in fits and starts till 1973, when ENV began intensifying efforts to stem such activities. An operations room for the Hawkers Department was set up at Princess House. It served as a nerve centre to monitor via radio the movements of ENV's fleet of 21 radio-equipped vans, and to co-ordinate and direct the activities of field staff. Information, such as tip-offs on sites of illegal hawking, could be relayed quickly. Requests for back-ups could also be sent immediately.

Four inspectors from the Special Services Squad, accompanied by CISCO guards, were assigned to each van to patrol the streets. Each day, four or five teams would be sent on raids. During these raids, goods were seized, cooked food thrown away, and the offending lorries and cars impounded. Perishables and fresh produce were given to charities.

It was not an easy job. Most hawkers depended on hawking for their livelihoods and public sympathy was usually with them. There were cases where illegal hawkers turned aggressive and put up resistance during the raids. But by 1975, the situation came under control as over 18,000 street hawkers were relocated into new centres and markets.

Today, street hawkers are a rare sight in Singapore. Many of such hawkers today are part of syndicates and are involved in selling pirated, contraband or illegal goods. The Special Squad, as it is now known, continues under NEA with its mission to crack down on these activities.

A NEW LEASE OF LIFE

Remember the old, dank hawker centres and markets before HUP? Thanks to the programme, these old buildings were given a new lease of life.

Brighter surroundings, better ventilation, cleaner stalls and floors – these are the new features that stand out at first glance. But there's more. Depending on the state of condition the centres were in, different improvements were carried out. The standard list of upgrading features included re-tiling, replacement of tables and stools, re-wiring and replacement of exhaust systems, sewers and water pipes and improvements to toilets and bin centres.

If a hawker centre had an estimated lifespan of another 10 years after upgrading, it would then be reconfigured or rebuilt. This means bigger stalls, better facilities, fire safety features and even a new architectural design and layout. In some cases, hawker centres and markets located near each other were consolidated and rebuilt as a single larger centre, for example Commonwealth Crescent Market.

To facilitate smoother operations in the upgraded centres, central freezer and washing areas were built. Barrier-free access was incorporated to facilitate wheelchair-bound patrons.

As HUP was not run on a cost-recovery model, rental increments were moderate. Stall holders were also given several choices to decide if they want to continue with their livelihoods at the same centre or other non-upgraded centres, or to opt out with an ex-gratia payment package.

These cleaner, brighter, well-ventilated centres and markets were warmly received by the public. A survey commissioned by NEA and the Ministry of Community Development, Youth and Sports in 2005 showed that 92% of respondents felt that upgraded centres were cleaner and 68% said that they would visit the facilities more often.

What's more, hawker centres are also turning into vibrant spaces for the community as NEA works with grassroots organisations, residents and the hawkers to organise food fairs, live music and performances at these venues. So that the public can catch up on the latest buzz at these icons of Singapore, NEA has also launched a website, myhawkers.sg, in 2010 to serve as an information portal.

Newton food centre through time











Top left: Newton Food Centre in 1977 *(Source: The Straits Times © Singapore Press Holdings Limited. Reprinted with permission.)*
Middle, left: The food centre during the 1980s. *(Source: Ministry of Information, Communications and the Arts, Courtesy of National Archives of Singapore)*



A CULTURE OF SAFETY

Devin Otto Kimble, Managing Director of MENU Pte Ltd Food & Drinks Group, the company behind Brewerkz Restaurants and Microbreweries, set up shop in Singapore after stints in the food and beverage industry in Cambodia and Hong Kong. He shares with us his thoughts on food hygiene and NEA.

Of all places, why did you choose to set up business in Singapore?
Singapore is really at the crossroads of civilisations, you can breathe the air here, and the legal system is transparent. When we set up in Singapore in 1997, the main considerations were that the laws were very clear. We could find out what the regulations were for a food business easily.

As someone who is in the food and beverage industry, how would you characterise your relationship with NEA, who is a regulator?
We are all trying to get to the same place, to put safe food on the table. We get 1,000 or more customers a day, and if I start getting people sick, I can hurt a lot of people. So we are on board with NEA on this.

Sometimes, there are things we overlook, and we'd say, 'Argh, we should have done that.' Once, we wanted to move the location of our bar and the NEA officer rejected the plan. So we had to spend another $10,000 to get the architect to draw the plan again, and this time, making sure that sinks for hand-washing and such were included.

But it makes sense and it's incredibly important to have people who are looking after these issues. Because I'm an operator and I'm more concerned with having enough people to work on the floor and that they get their pay checks. So we can depend on NEA as the steward. Because if Singapore is known for making people sick, then people won't feel as comfortable here.

What about the implementation of designated smoking areas and the smoking ban? Did it affect your business in any way?
It didn't affect our business at all! I've always thought it was the right thing to do. I noticed when I first came down to Singapore from Hong Kong that I didn't have to send my jackets and things out for dry cleaning as much. Back in Hong Kong, the cigarette smoke in restaurants was awful so it was a joy to be able to work in a smoke-free environment.

Can you share with us some of the changes you feel NEA has made to this industry?
We used to laugh about the footprints on toilet seats. But I haven't seen them for years, so that's a change NEA has made!

NEA has created a culture of safety, and that's really important. We went through this with the whole SARS incident in 2003, and NEA helped create the awareness for clean restrooms and personal hygiene, like not sneezing on other people.

Here's an example of how the culture has changed. When I first came to Singapore, I remember talking to people from one of these steakhouses. They were famous for their all-you-can-eat salad bar. They said, "Look, it's not working in Singapore because no one wants to eat at the salad bar." You didn't eat raw vegetables in Asia 15, 20 years ago, because you didn't know if it was safe to do so.

People can say having these requirements increase the cost, but they also lower the cost because there is more trust across the system. I don't have to go out and seek suppliers and make sure independently that they are certified to be safe. Having this system means that when you bring somebody on board the food industry, you know they have been through a course on food hygiene. It sets a culture where if somebody gets off a boat and tries to sell you a bunch of oysters, without any licences or certifications, you say, 'I don't know where these came from!'



WE HAVE GROWN TO LOVE CLEANLINESS

Veronica Choo runs the popular Hwa Yuen Porridge at the upgraded Tiong Bahru Food Market. She took over the stall from her parents-in-law, who started out as street hawkers peddling congee along Kim Pong Road many years ago, before they were resettled here. She tells us about food hygiene in the past and today.



What would you say is the biggest difference between the old Tiong Bahru Market and the upgraded version?

Before upgrading, people who came here to eat had to put up with rain and shine. Now, they don't have to worry about rain anymore! It is also much cleaner, better now. Customers can tell at one glance if our stalls are clean. Especially tourists or locals who are here for the first time, they would tell us, it is 'very clean!' And they don't have to worry that if they take hawker food, they may end up with food poisoning. Even though some customers say they miss the old hawker centre and the old atmosphere, in truth, they won't accept it if the surroundings are not clean. It's just talk. In the past, before the upgrading, they would say, 'Why is it that the hawker centre hasn't been upgraded yet, why is it still so old?'

As someone who is a veteran hawker, do you feel that food hygiene practices have changed over the years?

In the past, our expectations of hygiene were not so high! But as we learn more about hygiene and health, our expectations have grown and now if we see poor handling of food, we won't dare to eat. We also train our stall helpers to be hygienic and send them for courses. In the past, it was so hard to make them understand the need for hygiene! We would tell them, 'Don't use your bare hands!' And they would still do it once our backs were turned. But nowadays they won't – they will use gloves or tongs. So practices have improved.

The stall holders in the past were of the older generation, so they didn't have the opportunity to be educated. Nowadays, the stall helpers and holders are more educated, so their mindsets have changed, because there is more knowledge and awareness. Our lifestyle has changed, we have grown to love cleanliness and hygiene more.

Has the smoking ban affected your business in any way?

Now, after the smoking ban, things are much better. The centre used to have so much smoke, and as stall holders, we took in so much second-hand smoke! It's a good move, because those who are sensitive to smoke can now eat in peace at the hawker centre, whereas in the past, they would have to avoid the smokers.

Do you feel that NEA's grading and inspection of food stalls is important?

Whether it is a good grading or not, at the end of the day, what really matters is that we practise cleanliness. Today, most customers expect stalls to have at least 'B' or 'A'. In the past, few stall holders would think this way, they would say, 'It doesn't matter, we grew up on unhygienic food anyway!' But now, nobody talks like that.

I feel quite relaxed when the NEA officers come by to check. Because I know all our things are in order. Such inspection has a role – it serves to remind us that we need to be clean and hygienic at all times. It's a good thing, otherwise, some stall holders may just do a superficial job. In the past, some stall helpers even smoked and handled ingredients at the same time!

Controlling Outbreaks

The duty to protect Singapore from outbreaks of vector and environment-related diseases is shouldered by NEA. The agency sees to the first line of defence for Singapore, through its comprehensive programmes of preventive surveillance and control, inter-agency collaboration, public education, community engagement and scientific research.

Vector: a living organism which carries a pathogen that is capable of causing an infection in humans.

Pathogen: a microbe or micro-organism which can cause an infection to an animal or plant.

Cockroaches, rodents, flies and mosquitoes – these are some of the common vectors found in Singapore today. Of these vectors, the mosquito stands out as the most dreaded. Unlike other vectors, the female mosquito actively seeks human prey. Through its bite, it transmits viruses or parasites it has picked up from an infected person to the victim. The mosquito is responsible for spreading pathogens causing diseases such as malaria, dengue fever, chikungunya and Japanese encephalitis. Today, the *Aedes* mosquito is Singapore's biggest concern as it spreads dengue, an endemic disease in tropical and sub-tropical areas. Effective vaccines or anti-viral drugs for the disease have yet to be developed, and those who contract dengue may face death.

NEA is Singapore's first line of defence against vectors and the pathogens they carry. The island's status as a travel hub makes it vulnerable to imported pathogens from other regions while its high population density means that multiple persons can be infected within a short time. Through its comprehensive programmes of preventive surveillance and control, inter-agency collaboration, public education, community engagement and scientific research, NEA strives to prevent outbreaks of vector and environment-related diseases in Singapore.

COMBATING DENGUE

Look around, are you breeding danger? It's your life, it's your fight. Change water on alternate days. What if unwanted guests move into your house while you are away? Stop breeding danger. These constant warnings on the radio, television, papers and posters are part of NEA's strategies to battle dengue.

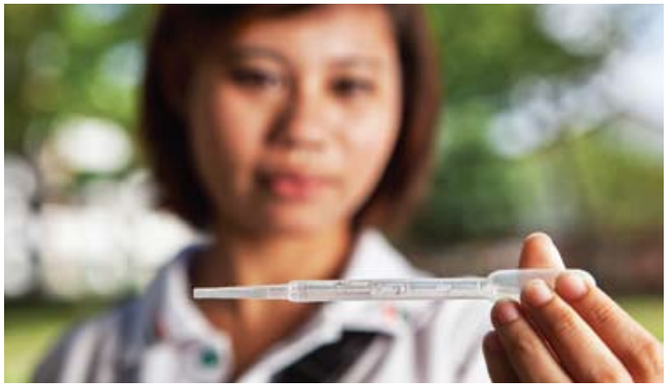
Fighting dengue is a formidable task. The *Aedes* mosquito, which has a lifespan of about two weeks, can breed in a spot of water as small as a twenty-cent coin. The female mosquito can lay about 300 eggs in its lifetime. These can hatch into larvae in less than a day, and grow into mosquitoes in a week. When a mosquito bites a person bearing the dengue virus, it becomes infective in about seven days. In short, they breed fast, infect fast.

To fight them, NEA employs two primary operational methods. One, look for signs of breeding and make the environment inhospitable. Two, break transmission of the disease as soon as dengue cases are detected.

Search and destroy. This is the mission of NEA officers. They scour public areas and residences to hunt for breeding grounds, remove stagnant water and destroy the sites. Property owners found to have mosquitoes breeding on their premises may be penalised for creating public health risks. In an average year, NEA personnel carry out at least one million inspections and ground surveys. When warmer months come around from April to September, dengue cases often spike as *Aedes* mosquitoes breed rapidly. Hence, NEA strikes in advance, launching operations aptly termed "Intensive



The Mosquito Larval Inspection Pole allows NEA officers to check roof gutters for mosquito breeding without having to resort to using ladders.



Source Reduction Exercises" in public and residential areas. These operations seek and annihilate not just breeding sites, but also potential habitats.

The war against dengue is a national issue. NEA leads the government's task force comprising some 25 agencies and institutions in anti-dengue efforts. It also coordinates and works with groups including town councils, construction sites and schools to eradicate mosquito breeding on their premises, and help them develop stringent regimes to prevent breeding. Such collaboration leads to visible results. For example, in 1999, about 30% of construction sites were found to be harbouring mosquitoes. By the mid-2000s, the figure had dropped to about 6%.

But even with all the resources pumped into this fight against the *Aedes* mosquito, it will always be an uphill battle as long as individuals do not exercise vigilance at their end. Research statistics show that mosquito breeding takes place mostly within homes. The public thus holds the all-important key to the success of these efforts. On NEA's part, it does its best to encourage every person to play his or her role. Mass media is exploited to spread anti-dengue messages via every communication channel possible. Programmes are developed to target specific groups of people, such as outbound travellers, domestic helpers and homemakers. Grassroots organisations, community volunteers, government agencies and educational institutions are also involved in dengue prevention efforts on the ground. In 2011, NEA launched a new engagement plan. This plan incorporates more in-depth education and outreach efforts with routine inspection visits by environmental health officers. During these visits, which take place before the warmer months, no enforcement action will be taken against any mosquito breeding detected. Instead, the focus is on public education.

TACKLING VERMIN AND NUISANCES

Besides mosquitoes, NEA also tackles vermin and nuisances in Singapore. It conducts regular checks for breeding and harbourage sites of cockroaches and rodents and educates stakeholders such as town councils, landlords and food operators on good housekeeping practices, refuse management and pest control. It enforces against those who repeatedly exhibit poor housekeeping. NEA works with pest control companies as well to reduce the number of rodent burrows, exterminate vermin and keep the crow population low.

MANNING THE GATES

Singapore is one of the busiest ports in the world. In 2011, more than 120,000 shipping vessels arrived at our ports while 46.5 million passengers passed through Changi Airport. Such high traffic also means that the risk of infectious diseases from other regions entering Singapore is higher. NEA holds the fort against potential outbreaks through its port health duties.

At Changi Airport, NEA officers, together with Immigration & Checkpoints Authority personnel, screen passenger arrivals. They watch out in particular for those who may be infected with yellow fever. Yellow fever is caused by a virus spread by mosquitoes and may lead to internal bleeding and death. To prevent the virus from gaining a foothold in Singapore, passengers from countries where the disease is endemic must carry valid health certificates. Those who are not certified will be examined medically and issued with a health alert card.

At the sea port, NEA officers keep a look out to prevent plague from entering Singapore. Plague is a deadly and fast-spreading disease caused by bacteria borne and spread by rats. In this case, NEA inspects vessels for signs of rat infestation. Ships which have death or a sick person on board, or which are coming in from plague-infected countries are quarantined before they are cleared to enter the port.

FROM HOUSE TO HOUSE



It was just a small note of appreciation, but it made Shaik Amina Bee's day. A family had written in to NEA to commend her for her professionalism and helpfulness. That would not have been so surprising if not for the fact that Shaik Amina had actually issued a summons to this family for breeding mosquitoes on their premises.

Shaik Amina Bee has been working as an Environmental Health Officer since 2005. She is one of the many NEA officers who go on foot, house to house, to check for mosquito breeding as part of the agency's war against dengue outbreaks. "I entered the flat and saw four huge porcelain vases on the family's altar," Shaik Amina recounts the incident. "So I informed the owners that I needed to check those vases for mosquito larvae. I helped them carry the vases and pour the water out and found larvae in one of the vases." What impressed the residents despite having been penalised was Shaik Amina's helpfulness. Not only did she help clean the vases and water spillage, she also shared information and tips on preventing mosquito breeding.

To Shaik Amina, this is all in a day's work. From 7.30 am to 5 pm every day, Shaik Amina visits residences in the blocks assigned to her. When she enters a flat, she checks water receptacles, dish trays, overturned pails, flower pots – anything that may be a breeding spot. If the resident is not at home, she has to return on another day. During dengue cluster outbreaks, gaining access to every unit is all the more important. Shaik Amina, like other environmental health officers, has to work nights to break the source of transmission. When she finds mosquito larvae on the premises, she will inform the resident, point out the larvae and collect them in front of the resident. With her educational kit in hand, Shaik Amina also educates the resident on how to prevent mosquito breeding. The larvae are then sent back to NEA for analysis.

Running into disgruntled or even angry residents is a common occurrence in Shaik Amina's work. She shares, "Sometimes, when I find breeding in the homes, the residents get angry, and they will ask, 'How come there are mosquitoes? I change the water regularly!' I hear all kinds of accusations. Some bang their doors, or complain that I'm rude. Some even snatch the dish tray and throw the water away before I can collect the larvae!"

Such unpleasant encounters only strengthen Shaik Amina's resolve to do her job well. She puts it this way, "I am trying to do my job and prevent dengue outbreaks, but some residents only think about fines. So I will hear them out, and try to explain to them the importance. Sometimes I joke and banter with them, and they will become calmer after a while. Some would say, 'Give chance *la*, officer.' And I have to tell them it's not my decision to make." She learns from these encounters too. Now, when she finds larvae, she takes a picture of them in front of the residents, so as to deter residents from tampering with the evidence.

What really fuels her is her personal conviction in the work she does. Recalls Shaik Amina, "Last time, there were mosquitoes in my flat, but I didn't even know they were mosquito larvae. I thought they were worms. So after joining NEA, I realised, this means a lot of other residents are also unaware. When I point out larvae to the residents, sometimes they will make sounds of disgust, and say 'Aren't those worms?' 'No', I say, 'Those are mosquito larvae and they are very dangerous. You can die from it.' See, a lot of people do not know what larvae are, or what dengue is. So my job is important, to educate them, and explain to them. And this is my job and I represent NEA, so I must do it well."

"When I point larvae out to the residents, sometimes they will make sounds of disgust, and say 'Aren't those worms?' 'No', I say, 'Those are mosquito larvae and they are very dangerous. You can die from it.' See, a lot of people do not know what larvae are, or what dengue is. So my job is important, to educate them, and explain to them. And this is my job and I represent NEA, so I must do it well."

– Shaik Amina Bee, Environmental Health Officer

CRUNCHING DATA TO DETECT CLUSTERS

You may have noticed this in your last encounter with an NEA environmental health officer – that he or she is tapping away at a palm device.

This small device, a Personal Digital Assistant (PDA), has greatly increased productivity for NEA's ground officers. No longer do they need to shuttle to and fro the regional offices as much as in the past. Feedback or complaints sent to NEA's headquarters are transmitted directly to ground staff, who can then respond straightaway. Enforcement can be carried out on the spot as well. Ground officers can also retrieve relevant data on the cases they are working on while out in the field, such as the location of mosquito breeding sites. But the real beauty of this gadget lies in the fact that the data keyed in by these officers is directly transmitted to a powerful system at the agency's headquarters, and used for in-depth analysis.

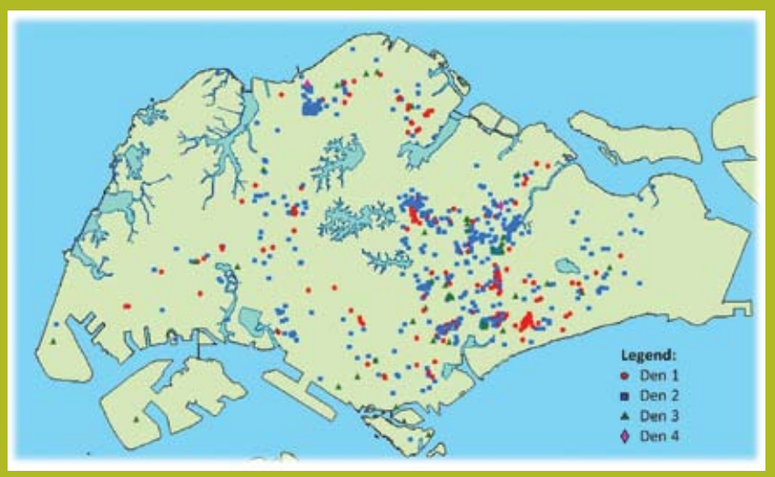
Every day, large amounts of data are created as NEA officers go on routine surveillance – the latest mosquito breeding site found, cases of non-compliance at hawker stalls etc. This data is updated real-time into a system which uses Geographic Information System (GIS) technology and internet mapping to plot the information. Relevant data from other agencies, such as SLA and MOH, is also imported. Analysis is carried out and ground operations are then planned and strategised accordingly.

Since the mid-2000s, this system has played a critical role in the fight against dengue outbreaks in Singapore. When data on breeding sites is keyed into the system, these locations show up on a virtual map of Singapore. Reported cases of dengue from MOH are plotted as well. When two confirmed dengue cases with onset dates within 14 days of each other occur in a radius of 150 metres, the system will flag it out as a dengue cluster. Other parameters are also used to pinpoint areas where there is localised transmission of dengue. Analysts at NEA can also see how dengue cases are spreading over time.

The data is updated in real-time. This allows officers at both the headquarters and regional offices to be in tune with what is happening on the ground. Using the data, they can plan operations and deploy resources more effectively. For example, the mapping system also provides information such as the locations of nearby construction sites and vacant plots of land, housing type and whether there are vulnerable groups such as nurseries or homes for the elderly in the vicinity. With the information, officers can identify which areas may be potential breeding sites and which sites should be inspected first. If the housing in the area is predominantly landed, ground staff can also prepare the necessary equipment to check roof gutters, a common breeding spot in landed property.

On top of this, NEA also takes pre-emptive measures once potential clusters are identified. Clinics located near these clusters are highlighted in the system, and NEA will inform them to watch out for dengue cases and notify MOH immediately when there are suspected cases. Early detection can help NEA to zoom in on problematic areas on the ground more quickly. When the cluster of transmission is broken at an early stage, the outbreak can then be controlled more successfully.





THE SITUATION ROOM

The Situation Room is where NEA's strategians convene, get the lowdown on the ground situation, plan and coordinate NEA's operations to tackle environmental and public health incidents such as dengue outbreaks.

The forerunner of this idea was a dedicated room set up in 1994 in the Vector Control and Research Department under ENV. The room was installed with the Intergraph CAD/CAM system to track the migration of the *Aedes* mosquitoes. GIS mapping technology was later introduced in 1997 to enhance the tracking system.

In 2003, shortly after NEA's formation, the entire GIS system was moved to a room situated on the 21st floor, next to the officer of the Director-General of Public Health. The room was named the "Situation Room". Not long after that, all five regional offices of NEA were connected to the main Situation Room at the headquarters and each regional office had a similar Situation Room. The ground data enabled the regional offices to monitor closely the state of the environment within their respective area of operation.

Over the years, the information captured in the Situation Room has played no small role in NEA's swift responses to incidents such as dengue outbreaks. As environmental issues become increasingly complex, such a function is all the more pivotal. To enhance NEA's situational awareness further, plans are now in place to construct a more powerful and technologically advanced command centre in the near future.

"NEA was formed in 2002, during a period that saw new and emerging diseases, some of them impacting Singapore. In 1998, there was an outbreak of Nipah Virus which took the lives of a number of our abattoir workers. Then, in 2000, Singapore had its first major outbreak of Hand, Foot and Mouth Disease, a normally mild childhood disease, where we saw the Enterovirus 71 turned aggressive. A year after NEA was formed, we had SARS. One of the key concerns then was to ensure and maintain a high standard of hygiene and sanitation to prevent outbreaks of infectious diseases that could endanger lives and create an adverse impact on our economy.

I read an article about how wild elephants were tracked in Africa using GIS. That was sometime in 1994 and GIS was something new then. So, I thought we could use the same idea to track the movement of the Aedes mosquito. We started using the GIS on a modest scale, using the Intergraph CAD/CAM system and upgraded it into a Global Positioning System later. I named the room where the information was centralised and displayed 'Situation Room', after a similar room in the White House in the United States. Each morning, I would go to the Situation Room to feel the pulse of what was happening on the ground. We would also hold meetings in the Situation Room, especially when we saw a rising trend in dengue or some other communicable disease, and those at the meeting could visualise on the screen what the ground situation was at that time."

— Daniel Wang, who was appointed Commissioner of Public Health in 1979 and went on to become the Director-General of Public Health in NEA (2002 - 2003)



SWIMMING POOLS AND COOLING TOWERS

Public swimming pools and buildings with cooling towers or central air-conditioning are monitored by NEA to prevent outbreaks of environment-related diseases. In the case of buildings with cooling towers, NEA requires owners to carry out regular maintenance. This is to prevent the growth of *Legionella*, a bacterium which can cause fatal respiratory infection. Public swimming pools are monitored as they are used by many people, which means that water-borne diseases may spread easily. Every month, pool operators must send samples of pool water for testing by accredited laboratories and submit the results to NEA. The agency also carries out random sampling of the water quality in public pools and inspections are carried out annually.

Scientists For Public Health

"We have been able to keep the overall incidence of vector-borne diseases low, comparable to that seen in developed countries. Take malaria for example, we knew that Anopheles *mosquitoes would normally breed in areas with stagnant brackish water. So we built anti-malaria drains in these areas over the years. This strategy has been a success ... Of course, we are fully aware that this does not mean that Singapore will not have any more cases. Located in a tropical region, malaria will be a perennial challenge. However, the public can take comfort that should a case occur in Singapore again, we have in place a system to detect the source and to swiftly stop the spread of the disease ... Dengue fever is another major challenge. Although the mosquito population has been kept low since the 1970s, dengue remains a concern in Singapore and globally ... So far, we have been able to control dengue in Singapore quite effectively. Even though we are located in a region that is highly endemic, we still manage to keep the incidence to the lowest in the tropic ... mosquitoes are living things. Like all living things, they are able to adapt and overcome our control measures over a period of time. So what we know about their behaviour 10 years ago may not be valid today. Likewise, what we know about them today may not be so 10 years down the road. It is an on-going challenge to keep up with their adaptability. This is why ENV* (present-day MEWR) *is stepping up our investment in applied research, so as to tackle these challenges posed by viruses and their vectors in our effort to stay on top of the changing situation. Today, I am pleased to officiate at the launch of the Environmental Health Institute, EHI in short."*

— Lim Swee Say, Minister for the Environment (2002 - 2004), at the launch of the Environment Health Institute, 25 April 2002



Over at the laboratories in the Environmental Health Institute (EHI), researchers race against time in an invisible war against micro-organisms.

Never in human history have we travelled as often, or as far, as in the 21st century. As we travel, we take along pathogens which have the potential to spread via person-to-person contact, water or vectors in whichever place we stop over.

On top of that, these pathogens evolve and adapt as they enter new territories, giving rise to new strains and reducing the effectiveness of tried and tested control measures. In the field of public health, scientists have to close ranks with ground surveillance forces to combat these pathogens.

In Singapore, scientific research on vectors and communicable diseases has been in place since colonial times, but never on the same scale as today. Recognising that the playing field has evolved significantly, ENV established EHI in 2002 to build capabilities in public health research so that Singapore is better prepared to address existing and emerging vector-borne outbreaks, particularly dengue.

Today, the institute is an integral part of NEA. Within EHI, molecular biologists, virologists, immunologists, veterinarians, epidemiologists and entomologists work together to safeguard Singapore's public and environmental health, assessing potential public health risks and applying scientific findings to formulate evidence-based measures and programmes to address these risks.



SURVEILLANCE IN THE LABORATORIES

To effectively combat potential vector-borne epidemics, we have to know how they spread and what can stop them. That requires firstly, identifying the pathogen responsible for causing the disease and the host or vector that spreads it and secondly, understanding the ecology of disease transmission.

EHI runs a laboratory virus-surveillance programme to provide early warning of outbreaks of vector-borne diseases including dengue, malaria and chikungunya. A network of some 300 general practitioners located all over Singapore supports the programme. When there is a suspected case of infection, the doctor will take a blood sample and send it for testing at EHI. If it is a case of infection, EHI proceeds to determine the genetic fingerprint of the virus.

With such data, the scientists in EHI can accomplish a number of things. Firstly, for diseases that are already endemic in Singapore, such as dengue, the data can provide early warning of outbreaks, and point to locations with higher risk. Secondly, the system enables the detection of new diseases, such as chikungunya. Thirdly, the capability enables thorough investigation of disease clusters through molecular epidemiology. For example, when malaria broke out in 2009, EHI assisted MOH in confirming locations of disease transmission. As a result, the intensity and scale of operations were more efficient and effective. To build and enable its capabilities, EHI continuously develops diagnostic capabilities through in-house research and evaluation.

Recognising that greater cross-border cooperation is needed to tackle tropical mosquito-borne diseases. EHI has established a strong network of international collaborators. There is regular exchange of information on the prevailing viruses and disease situation among researchers and authorities in different countries. These efforts contribute to greater regional preparedness for outbreaks and provide Singapore with an important line of defence against vector-borne diseases.



THE DENGUE VIRUS

The dengue virus has four serotypes, all four of which are circulating in Singapore. Serotypes are variants of the virus classified by the antigens found on the surface of the cell. An antigen is a substance that causes the human body to produce antibodies against it. Certain serotypes may be more virulent or deadly than others. Infection with one serotype confers immunity to the same type but not to the others.

Usually, there is one predominant serotype which is circulating. Over the years, EHI has established that a change in the predominant serotype is associated with the occurrence of outbreaks. Thus, monitoring when a switch takes place provides valuable early warning of impending outbreaks, allowing prompt vector control intervention.

EHI has also found that the dengue virus is genetically diverse, and there is a constant introduction of new strains. What this can mean is that the pressure exerted by our intensive vector control actually selects the 'fitter' viruses that can be more persistent and have a higher potential to cause outbreaks. As with serotypes, monitoring the genetic signature of viruses can also provide an additional early warning for imminent outbreaks.



BUILDING DOSSIERS ON VECTORS

Besides pinning down the pathogens causing outbreaks, EHI also builds up information on the vectors which spread them. Knowing the vectors inside out – their distribution, diversity, behaviour, life cycle and breeding habits – can help NEA decide on the best methods to eliminate specific vectors and assess potential risks of emerging or new vector-borne diseases. For instance, the breeding grounds of the *Anopheles* mosquito, carrier of malarial parasites, are usually found in parks, woods and forests instead of residences. *Aedes aegypti* which spreads dengue is more commonly found in urbanised areas. Hence, an operation targeting malaria will include rural parts of Singapore in its search and kill mission, while an anti-dengue operation will take place mainly in residential and built-up areas.

EHI's continual research has shed much light on the understanding of vectors. For example, it has developed molecular identification tools to characterise the genetic diversity of *Aedes aegypti* population in Singapore. Using them, scientists discovered that the local population of *Aedes aegypti* is genetically diverse. More research can then be conducted on the growth and distribution of the mosquitoes and how competent they are as vectors.

Another example is the institute's research on the flight range of dengue vectors – *Aedes aegypti* and *Aedes albopictus*. Prior to this research, it was commonly held among scientists that *Aedes aegypti* seldom flies more than 50 m in its lifetime. A field study by EHI debunked that belief, after it demonstrated that both *Aedes aegypti* and *Aedes albopictus* can easily and quickly disperse within a radius of 320 m to search for breeding spots. EHI also showed that mosquitoes can effectively disperse in disused high-rise residential buildings. This study led to a revolution in the understanding of Aedes mosquitoes' flight ability and won a Royal Entomological Society Award from the United Kingdom for the year 2004 - 2005. Subsequent anti-dengue operations were revised to cover a wider area.

FINDING APPLICATIONS

On top of its role in supporting ground surveillance and vector control programmes, EHI also applies its research to find practical solutions for public health issues.

Take insecticides, for example. EHI checks the susceptibility of mosquitoes to various types of insecticides so that it can identify if any of the mosquitoes has developed resistance, and which insecticides work best against them. With this understanding, NEA can ensure that insecticides used in mosquito control remain effective.

The institute has also developed a number of interesting applications. A cost-effective protocol for early diagnosis of dengue and serotyping using polymerase chain reaction (PCR) technology, for example, was developed by EHI. This protocol is now used in several laboratories in local hospitals to shorten the time taken to diagnose dengue. It also facilitates more real-time monitoring of serotypes.

Another important innovation of EHI is a non-invasive method to diagnose dengue. Traditionally, patients need to undergo a blood test to confirm an infection. Now, the institute has developed a method which can detect the infection with the saliva of the patient. The test, which is currently undergoing field trials in a few countries, is much easier to perform, more cost-effective and can be used in laboratories without the need for sophisticated equipment.

Mosquito identification tools are also developed by EHI's scientists for education, reference and training purposes. These tools include pictorial booklets and charts featuring common mosquito species and online interactive keys.

NOT JUST VECTORS

Besides vectors, the institute also researches environmental health concerns and food hygiene, such as the mould problem in indoor environments, risk of carbon monoxide accumulation in restaurants with individual cooking stoves and *Legionella* bacterial growth in spas, water storage tanks and mist fans. EHI is also developing the capability to assess the microbiological quality of local retail food, identify the risks associated in the preparation and handling of food and devise ways to mitigate these risks. All these help to ensure that our food will always be safe for consumption.





CURBING CHIKUNGUNYA

The story of how Singapore curbed the spread of chikungunya in 2008 is one which shows how inter-agency vector control operations and laboratory surveillance came together to overcome the outbreak.

Chikungunya is a debilitating viral disease transmitted by mosquitoes. As no vaccine is currently available to prevent infection, the control of vectors is critical in the fight against the disease.

Even before chikungunya reached our shores, EHI had been collaborating with relevant institutions in Sri Lanka on chikungunya research. This offered EHI first-hand experience in handling and analysing the virus, as well as a good understanding of the epidemiology of the disease.

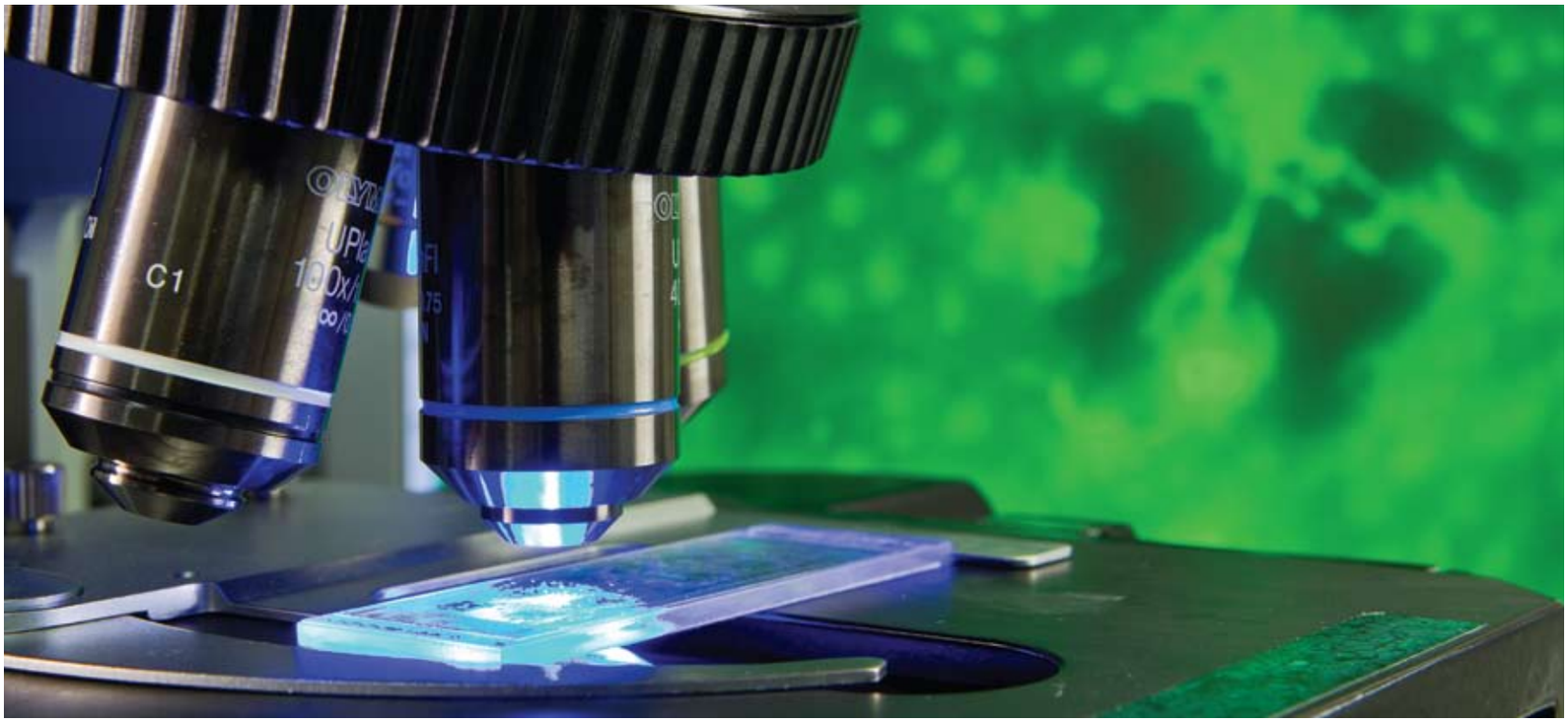
Through EHI's surveillance network, doctors in participating clinics had been alerted on the possibility of chikungunya hitting Singapore. When the disease broke out in January 2008, it was detected swiftly by EHI. Active case detection was initiated by MOH to uncover secondary cases among residents and workers in Little India and Serangoon, the vicinity of the index case. On its end, NEA promptly mobilised its ground forces to rein in the spread by stepping up vector control.

Over at EHI, researchers processed 400 to 500 blood samples daily, given the surge in number of suspected cases. NEA also involved other agencies such as URA, the Ministry of Manpower and Singapore Contractors Association Limited to clean up the area. Community outreach programmes in Serangoon were intensified to urge residents, shop owners and foreign workers to prevent mosquito breeding and to seek medical attention immediately if they experienced symptoms similar to chikungunya. These efforts successfully stopped the disease from spreading further and the outbreak was declared over in February 2008, with 13 reported cases.

However, in May 2008, chikungunya reared its head again and this turned out to be a substantially bigger wave of infection with cases reported in different parts of the country.

Again, the researchers at EHI worked fervently late into the night to track down the strains of viruses. What set this epidemic apart from the previous transmission was that the chikungunya virus was a variant spread by the mosquito species, *Aedes albopictus*, rather than *Aedes aegypti*. Armed with this newfound information, NEA promptly revised its ground strategy to target *Aedes albopictus*, by shifting its focus from the indoor environment, where *Aedes aegypti* thrive, to outdoor forested areas that are favoured by *Aedes albopitus*. Thermal fogging of outdoor areas was carried out and residual spraying on the external walls of buildings was conducted. Search and destroy efforts focused on eradicating outdoor breeding habitats of these mosquitoes.

These efforts finally brought about the control of the epidemic spanning close to a year. By the time the epidemic was over around mid-2009, close to a thousand cases were reported. The preparedness and response of NEA have often been upheld as a success by experts, some of whom have witnessed the disease infecting 30% to 50% of a population within six months of its onset in other places.



"As a WHO Collaborating Centre, EHI will forge closer linkages between scientific experts from different disciplines, so that we can synthesise innovative solutions to the dengue problem. EHI will also work with other international institutions to promote a regional surveillance network, to facilitate the timely exchange of surveillance information. This will improve the region's collective readiness and response to any arboviral disease outbreak in the future."

— Dr Vivian Balakrishnan, Minister for the Environment and Water Resources at the first ASEAN Dengue Day on 15 June 2011

AN INVALUABLE INSTITUTION

Since its establishment in 2002, EHI has been performing an important role in guarding Singapore's public health. But beyond that, this young institution has also made significant contributions to the world in various ways, from its research breakthroughs to the applications it has developed.

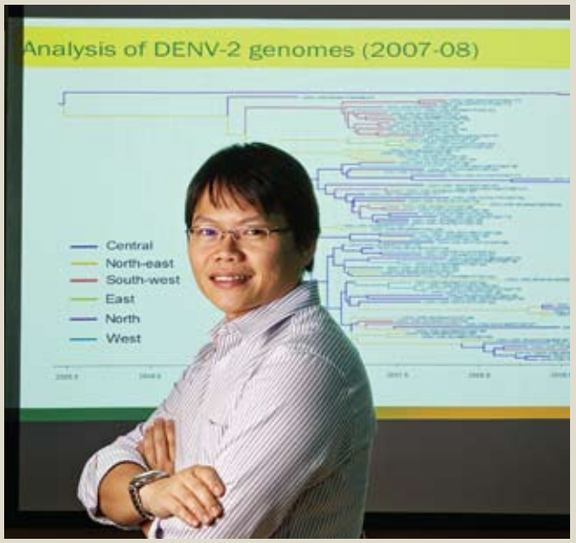
The fields of epidemiology, vector control and environmental hygiene encompass many aspects and evolve rapidly and continuously. The timely exchange of information is thus crucial to ensure that scientists worldwide can work collectively for the good of humankind.

As part of the global scientific community, EHI participates in and hosts regional and international events related to public and environmental health issues. It also partners other research institutes around the world in the quest for knowledge. For example, in 2007, EHI joined forces with the renowned Institute of Pedro Kouri from Cuba to research dengue control. Like Singapore, Cuba is afflicted with dengue outbreaks and has invested heavily in its dengue control programme. Through the collaboration, both institutions benefitted immensely from each other's expertise and experience.

EHI's research excellence is well-noted in the international community as well. Over the years, its researchers have served as advisors to WHO in meetings advocating stronger collaboration and coordination among different sectors for further global efforts against epidemics. In 2011, it was appointed by WHO to be its Collaborating Centre for the Reference and Research of Arbovirus and their Associated Vectors. As a WHO Collaborating Centre, not only can EHI strengthen its research regime, it will also play a leading role in developing a regional dengue surveillance network.

MOLECULAR MICROBIOLOGISTS AGAINST DENGUE

Dr David Lee Kim Sung, Programme Head, Research (Vector-borne Diseases) Section, is one of the molecular microbiologists behind NEA's battle against dengue.



What is it that molecular microbiologists do in EHI to combat dengue?
We sequence the virus, which helps us understand where the virus originates from and how it spreads, so that we can carry out risk assessments. The information is passed to our colleagues in vector control operations, who can then strategise how to pre-empt and manage outbreaks.

How does your work relate to dengue control in Singapore?
Through our work, we can identify which strains are more likely to cause epidemics. So when we find human cases infected by such viruses, we will intervene with our vector control to ensure they don't spread. We also find that virus strains causing outbreaks in different years and different regions can be genetically different. As the virus does not respect country boundaries, it is important for us to be aware of the situation in the region and at the same time share such information regularly with our counterparts in other countries, so that together, the region becomes stronger and more united in tackling this public health menace.

What are the challenges you face in your work?
Sometimes we have limited data available for analysis to come up with meaningful interpretations of the situation. It takes time to obtain samples from the cases, sequence and analyse them. When outbreaks occur, it is urgent to turn out the results as fast as possible, especially if they are associated with an uncommon virus strain. It's always a race against time.

What do you enjoy most about your work?
The work in EHI is very meaningful as I can contribute my knowledge and skills in science in a more practical manner, with my work having an impact on public health policies.



Setting up a mosquito trap in the course of field work.

TRACKING ZOONOSES

Grace Yap, Senior Research Officer, leads the team in EHI on the surveillance of zoonotic diseases, i.e. infectious diseases that spread from animals to humans. The team looks at diseases caused by animals that live amongst us and other zoonotic diseases that may be introduced into Singapore and have the potential to be transmitted by vectors, particularly mosquitoes. Blood samples are taken from different animals for research, including migratory birds. These samples are screened for different pathogens. One of the viruses the team watches out for is the West Nile virus, which is carried by birds and mosquitoes, both of which are found in Singapore.



PORING OVER EVERY PART

Chatting with Lam-Phua Sai Gek about mosquitoes is like stepping into a different world. Most people see nothing but an annoying insect. Not Lam. "This," she points to a photo in a booklet, "is what we call a Gigantic Mosquito. It is very big and its body is very shiny. Its larvae can feed on other mosquito larvae. And this small one here," she points again, "emits a kind of luminescence. It is so beautiful. When I look at the mosquito under the microscope, they are all different. They have different colours, different features, so you can feel their beauty. Not all mosquitoes bite people and not all are vectors."

Lam, who has been documenting mosquitoes since 1974, is a Principal Research Scientist at EHI. She is the taxonomist and chief curator of the institute's impressive collection of over 6,000 mosquitoes representing at least 90 of the 137 species known to be found in Singapore. Taxonomy encompasses the identification, nomenclature and classification of organisms. Such work is fundamental to vector control operations and research as it allows accurate and fast identification of mosquitoes.

Says Lam, "Our work is to identify the mosquitoes found in Singapore – both the adult and the larval stages. The mosquitoes are collected from different parts of Singapore, both urban and forested areas. We preserve the larva, the pupa, the exuviate (The skin shed by the adult form of the mosquito as it emerges from the pupal stage) and the adult. We catalogue all the specimens according to their species and the locations where they were found. We look at the features of the mosquitoes – the head, the thorax, every part – all using our eyes and with the help of the microscope."

This information has been put to various uses, such as developing a taxonomy key for identifying mosquitoes, training materials for ground officers and as a regional reference centre for taxonomic research. The taxonomy key, which guides users through an algorithm to determine the identity of the mosquito, is an invaluable aid in the field.

Apart from taxonomy and curation, Lam's team is also responsible for establishing and maintaining colonies of four species of mosquitoes, *Aedes aegypti* (dengue), *Aedes albopictus* (chikungunya), *Culex quinquefasciatus* (West Nile) and *Anopheles sinensis* (malaria). The mosquitoes are used for research purposes, for example, as subjects in insecticide sensitivity tests.

"We try to collect as many specimens as we can to build a database. Over the years, we have added nine new records to the documentation of mosquito species in Singapore," says Lam. "This documentation is important as it aids risk assessment. For example, the species of mosquitoes that can transmit the West Nile virus are found in Singapore too, but the disease has not occurred yet. We need to be on the alert and we have the record to show the geographic spread of these mosquitoes. So if one day, the West Nile virus shows up here, we know where to target our control."


Toxorhynchites sp.
Toxorhynchites sp. 6.5.87
Toxorhynchites sp. VCRD Lab Mar '87
Toxorhynchites sp. VCRD 2.4.98
Toxorhynchites sp.
Toxorhynchites sp. March '87
Toxorhynchites sp VCRD 2.4.98
Toxorhynchites sp VCRD 2.4.98
Toxorhynchites sp. 04 03 03
Toxorhynchites sp VCRD 2.4.98
Toxorhynchites sp. VCRD 12.8.85
Toxorhynchites sp. CP. LAB 21/3/87
Toxorhynchites sp VCRD 2.4.98

SARS: A CRISIS TO REMEMBER

"When SARS happened, everybody was living in fear. Because nobody knew what SARS was. So there were a lot of people living in fear. We mobilised all our resources. NEA was on the frontline helping to ensure public hygiene standards. The officers went out there, walked every hawker centre, every toilet, working day and night. It was non-stop operations. And the research community was trying to understand what the virus was. But they needed virus samples. At that point in time, the whole of Singapore, not a single laboratory had the facility in place to culture the virus. But our colleagues at EHI volunteered. They felt they had the process competency given their experience working on the dengue virus. So they risked their lives to handle the SARS virus, process it, prepare the samples for the research community. And they succeeded. To me, that is the display of human courage and professional commitment. Because they could have easily said, 'We too, like everybody else, were not set up to handle SARS.' But they did not. They came forward. Yes, they made a mistake post-SARs, but we should never forget the sacrifice they made during the time we were fighting SARS."

— Lim Swee Say, Minister for the Environment (2002 - 2004)

It was a crisis few would forget. In 2003, the deadly SARS (Severe Acute Respiratory Syndrome) virus broke out across the world. Singapore was not spared. Between March and May, the outbreak held sway over the island, severely impacting public health, the economy and the daily lives of residents. By 30 May 2003, when Singapore was finally declared SARS-free by WHO, a total of 238 people were infected and 33 lives taken.

NEA was barely eight months old when the crisis took place. Though it had inherited a wealth of operational systems and expertise from ENV in managing environmental health, things were still coming together on different levels. For instance, its district offices had just been restructured to form five regional offices with new operating boundaries on 1 March, the very same day SARS broke out. Ground operations and networks had yet to be fully organised, and in some areas such as public outreach and engagement, NEA was just beginning to learn the ropes.

Yet, despite these factors, NEA rose up to the call of duty. Together with the rest of the nation, the agency did the most it could to the best of its ability to see Singapore through the crisis.

TRACKING DOWN CONTACTS

When news of an atypical pneumonia case broke out, NEA was one of the first agencies to be notified due to its close working relationship with MOH on environmental health and epidemiological issues. As the infection spread, it became clear that this was no ordinary virus.

Amidst the growing urgency to consolidate and organise the clinical response to SARS, MOH also had to cope with tracking down every single person who had been in contact with SARS patients. NEA was immediately roped in to help with contact-tracing, and a number of NEA officers were assigned to coordinate this right from Environment Building.

Every day, the hospitals would gather the details of contacts to be traced and send them to MOH, who would in turn pass the list to the team of NEA officers tasked to undertake the task. This team would follow up by tracking down some 500 people per day. After the Infectious Disease Act was invoked on 24 March, NEA officers were further tasked to serve home quarantine orders, a duty that was later outsourced to CISCO. The coordination of contact-tracing was transferred to MINDEF in late April after SARS broke out in the community and the ministry's expertise was needed to track down some 2,400 people. Lam Joon Khoi, who was CEO then, remembers, "It was a national effort. NEA was supporting in some areas and taking the lead in others. We went on overdrive then, and commandeered resources from other areas, even the corporate services staff, to reinforce the environmental public health arm. It was not just part of NEA doing the fighting. Fortunately for Singapore, the contact-tracing and quarantine were working quite well. So while we were stretched, we did not reach breaking point."

ENVIRONMENTAL HEALTH – MORE CRUCIAL THAN EVER

During the crisis, environmental health and hygiene were more crucial than ever. While it was clear how deadly the virus was, information on how it spread was still incomplete. Following the first two deaths resulting from SARS on 25 and 26 March, the government took the safe approach and announced that those who died from SARS must be cremated within 24 hours, while Muslim victims had to be double-bagged in their burial.

As the government crematoria and cemetery came under NEA's management, the agency had to ensure that the deceased were handled with care to prevent any transmission. NEA was also responsible for the safe disposal of medical waste from the hospitals, including Tan Tock Seng Hospital, where the war against SARS was being waged. When an infection cluster was traced to Pasir Panjang Wholesale Market on 19 April, NEA worked swiftly with other government agencies to close down the facility. Lee Kheng Seng, who was Head of NEA's South West Regional Office (SWRO) then, recalls, "The SWRO team worked several days and nights to clean up and sanitise the Pasir Panjang Wholesale Centre. It was a multi-agency effort and the crisis brought everyone together. Following this incident, NEA also deployed officers to go round the markets and hawker centres to make sure that hawkers keep to their daily temperature self-checks as a means to prevent similar cases of infection cluster from developing."

SUPPORTING THE CLINICAL RESPONSE

At the same time clinicians were fighting against SARS in the hospitals, it was a race against time to decode the SARS virus. Yet, there were no existing



Temperature screening was a norm during the SARS outbreak in 2003.

diagnostics test, no understanding and no treatment available for the virus. MOH thus formed a consortium of research institutes and laboratories including the Defence Research Medical Institution, Tan Tock Seng Hospital, Genome Institute of Singapore and EHI. The work was distributed, with some institutions to work on diagnostics, the universities to research the pathogenicity of the virus and Genome Institute of Singapore to sequence it. EHI, which was only less than a year old, focused on cultivating the virus to facilitate the research and development.

Another important way in which EHI supported the medical community during the crisis was the processing of samples from suspected SARS cases. Former CEO, Lam Joon Khoi, highlights, "EHI was one of the few centres in Singapore then which could test if the samples were infected with SARS. They had to work round the clock, because there was a huge volume of samples. SARS symptoms were similar to flu and there were many cases of flu symptoms. The amount of effort put in by EHI staff to cope with the volume of work should not be underestimated."

SINGAPORE'S OK!

Former ENV Minister, Lim Swee Say, hit upon the idea of launching a campaign to build public confidence by raising hygiene standards in public areas. He aptly named the campaign, "Singapore's OK" (SOK). Rolled out progressively, the campaign targeted different categories of public areas – food centres, toilets, pre-schools, markets, hostels and construction sites etc. Those who volunteered to take part would be issued a decal if their premises met the criteria, so as to mark their commitment to uphold cleanliness and good hygiene practices. On 6 May, Lim Swee Say unveiled the SOK label for public toilets, the first category to be launched. The campaign received a further boost when Prime Minister Goh Chok Tong launched SOK in the community on 1 June.

Some 600 NEA officers were mobilised to visit every premises covered in the campaign, encouraging the owners to take part. The officers also worked with these premises and NGOs to promote good hygiene habits such as regular temperature checks, keeping public toilets clean and not spitting or littering. Says Dr Pang Fung Yin who was then Head of 3P Partnership in Environmental Health Department, "SARS not only resulted in the loss of lives but also affected the social and economic well-being of Singapore, and caused many people to shun public places. We were concerned with the association between disease transmission and public hygiene and responded promptly by introducing the SOK campaign. The strategy taken was one which focused on shared ownership, leveraging on the voluntary commitment and collective effort of the people, private and public sectors. It was heartening to see many grassroots organisations and owners of premises such as hawker centres, food shops, markets and pre-schools responding positively in championing the common cause of making the environment safe and healthy for all."

A SHARP LEARNING CURVE

Despite its youth, NEA provided critical support in many ways, from taking care of environmental health issues and ensuring high public hygiene standards to contributing through laboratory work and restoring public confidence.

Unfortunately, a few months after the SARS outbreak, a lapse in safety procedures at EHI led to a postgraduate research student attached at the institute to be infected by the SARS virus. Following the incident, an international expert panel appointed by ENV identified a number of shortcomings in the physical set-up and procedures at the laboratory.

The findings led EHI to suspend work that required Biosafety Level 3 facilities till its new premises at Biopolis were ready in 2006. Biosafety levels indicate the level of containment needed in the handling of biological agents, with BioSafety Level 1 as the lowest and Level 4 as the highest. In the new premises, emphasis was placed to the proper design and maintenance of the new high containment facility. EHI also set up its Biosafety management programme and appointed staff dedicated to facility management. In 2006, MOH granted approval to EHI to possess and handle biological agents in its new Biosafety Level 3 facility, which was certified to meet or even exceed MOH's prescribed standards.

Besides this sober lesson in biosafety, NEA had matured in various ways through the crisis as well. Its ground operations had been considerably strengthened and the seeds for greater community involvement in the environment had been planted. Lam Joon Khoi recaps, "The SARS crisis reinforced my view that there is a need to develop strong partnership with our key stakeholders – community leaders, NGOs, industry associations and other public agencies and such during periods where there are no crises to achieve our environmental goals. That same relationship can then be harnessed for swift and effective actions when crises do emerge. Coming out of the SARS crisis, there was greater public awareness on the importance of good environmental health, increased appreciation of NEA's contributions to Singapore's environmental health by the public and a stronger sense of unity among NEA staff as they overcame the difficult situation together." These insights, in a way, prepared NEA for future crises to come.

COMING INTO ITS OWN

May 2005. Cases of dengue infection began hitting more than 100 every week. By August, the weekly figure had soared to more than 400. The rise was relentless. Hospitals and clinics were severely stressed, the public was increasingly concerned and the virus showed no signs of slowing down. By September, it had climbed to more than 700 infections a week.

In a way, this was not out of the blue. Dengue cases had been steadily climbing since 2000, hitting 9,459 cases in 2004. Research has shown that dengue transmission generally peaks every six years, and the last peak had taken place in 1998. But in 2005, the sheer scale of the infection was unexpected, with 14,006 laboratory-confirmed cases and 27 fatalities in all. It was the worst dengue outbreak in Singapore's history.

Yet it was also the event that saw NEA coming into its own in controlling outbreaks. Since the SARS crisis in 2003, NEA had been honing its operational strengths and involving other government agencies, community organisations and volunteers in its programmes. These relationships united the groups together in the fight against dengue.

DEVISING A NATIONAL STRATEGY

Things came to a head in September 2005, when the infection rate spiked. NEA had relied on its routine vector control programmes to cope with the situation. But with the outbreak reaching epidemic proportions, more had to be done.

Singapore's leadership began formulating a coordinated response on a national level. Dr Yaacob Ibrahim, then Minister for the Environment and Water Resources, combined forces with Khaw Boon Wan, then Minister for Health, to chair an inter-ministerial committee overseeing dengue control efforts. On the working level, multiple government agencies formed a task force led by NEA. The mayors of the five community districts united and established a Dengue Watch Committee to work on outreach efforts. Singapore also appointed an expert panel comprising local and international experts to advise the government on prevention and control measures. At the peak of the crisis, Prime Minister Lee Hsien Loong stepped out to front the campaign, giving it much-needed support and endorsement.

AN ALL-OUT BLITZ

It was all or nothing. The task force embarked on a nationwide blitz against mosquitoes and their breeding sites on 18 September. Mobilising more than 6,000 volunteers from government agencies and social organisations, NEA led them in carrying out "carpet combing" of public outdoor areas for six consecutive weekends. The term refers to officers going over the ground with a "fine toothed comb" to seek out every mosquito breeding site. More than 1,000 spots and 8,400 potential breeding sites were destroyed.

The war made its way into homes with the "10-Minute Mozzie Wipe-Out". This massive outreach programme encouraged the public to spend just 10 minutes a day to remove stagnant water from their residences. Some 10,000 volunteers distributed the campaign pamphlets to 880,000 homes. The message was spread to construction sites, shipyards, factories and foreign domestic helpers as well.

On its end, NEA increased its field deployment of officers from 110 to 510. The entire NEA was mobilised too and for two months, officers from other departments took part in the carpet combing exercises. More than 934,000 premises were inspected and 52,000 sites surveyed by NEA in 2005. A public hotline dedicated to dengue was set up by the agency in September, and more than 16,000 calls were received in total. Every call was investigated within 24 hours and tips on breeding sites were followed up.

TARGETING DRAINS AND ROOF GUTTERS

Through the task force, NEA mobilised government agencies and private organisations to take ownership of eradicating breeding sites on their premises. Infrastructural checks, repairs and renovation were carried out to eliminate sources of stagnant water. Cracks were sealed and land was backfilled. The frequency of checks and cleaning of public drains and roof tops was increased. For closed drains and hard-to-reach areas, misting of larvicides was carried out. Points of contact were identified in each organisation so that NEA could liaise with the right person to react swiftly in cluster outbreaks. NEA also helped the agencies evaluate their procedures to cut down mosquito breeding.

Adopting a long-term view, NEA also worked with HDB and BCA to review designs of drains and roof gutters to improve water flow. Roof gutters were banned for new developments from then on, and 10 km of public drains were repaired.

IN THE LAB

Given the speed of infection, time was of the essence both for vector control and clinical treatment. In the aftermath of the epidemic, the researchers at EHI developed a polymerase chain reaction (PCR) based diagnostic assay for dengue which cut short the time needed to diagnose the virus. This meant that doctors could detect cases more quickly and operational response on the ground could be initiated much earlier. Previously, the diagnosis could take a few weeks. The new assay gave an accurate diagnosis as early as the first day of the disease, and even determined the serotype of the virus. However, the assay was less able to detect the virus from the second week of infection. Subsequently, EHI also developed another assay which could detect infection using saliva from the patient, with a 93% sensitivity between four and eight days after the onset of fever.

BUCKING THE TREND

The intensive efforts from September onwards finally led to a slowdown in the outbreak by November 2005. As a result of the campaign, the public had become much more aware of the importance of dengue prevention. NEA had also strengthened its vector control and surveillance programmes considerably.

But it was no time to let up. Dengue was still a significant threat. NEA continued investing more resources in its programmes to control the disease. One significant transformation was the development of a system which integrates information from cases of infection, virus and vector surveillance and the ecology of mosquitoes to plan vector control operations.

In 2007, another outbreak took place. Based on the research on the serotypes responsible for the outbreak, EHI concluded that it would be massive in scale. NEA's operations immediately swung into action, and launched intensive vector control exercises. The impending outbreak was successfully curbed and dengue cases in subsequent years did not reach the same scale as in 2005. The dengue cycle had been successfully bucked.



"It was the most serious dengue outbreak the island has seen, with the highest number of dengue cases and deaths ever recorded. Fighting the outbreak required the mobilisation of the entire nation to get everyone to play his or her part to prevent mosquito breeding and break the chain of dengue infection. This was a tremendous challenge which NEA met head-on. Officers were rostered over a period of about two months including weekends to conduct "search and destroy" operations in partnership with grassroots organisations and volunteers to comb through every single public and private estate and other public areas to remove mosquito breeding grounds. No stones were left unturned, no stagnant water was left untouched."

— Lee Yuen Hee, CEO, NEA (2005 - 2009)











"They used to be called 'teh-gu, teh-gu'. They were seen to be very harsh on people, on street hawkers and so on. But if not for these environmental health officers, our environment will not be that clean. They are the ones who make sure that public health standards are kept. They are not afraid of enforcing the rules because they know they need to do so in order to safeguard the interest of the majority. Over the years, they have also changed in the way they undertake their roles on the ground; now they also educate as they enforce. Our environmental health officers should be recognised for the important role they have and will continue to play in keeping Singapore clean and safe."

– Khoo Seow Poh, Deputy Chief Executive Officer, NEA



WITH DUTY AND PRIDE

They have been called *Kerani Sampah*, *Kerani Jamban* and *Kerani Longkang**. Older generations of Singaporeans nicknamed them "*teh-gu*"**.

These unflattering terms arose during a time when tough action was needed to remedy the deplorable state of environmental health in Singapore. Environmental health officers, first under MOH, followed by ENV, were the frontline in enforcing new stringent laws. Today, these officers in NEA continue the tradition of safeguarding a clean, healthy city. With some 2,000 of them on the beat, NEA environmental health officers are almost synonymous with the agency. Many of them have been in this line for decades, some even from the days of MOH.

Their work is often onerous, unpleasant and even dangerous at times. Says Koh Char Boh, who joined this line in 1972, "I started out with collecting night soil from the *kampongs* (Malay: villages), and also worked at the Lorong Halus dumping ground. Even if we changed our clothes after work, people could still tell we came from the dump when we took the bus home."

Cracking down on illegal hawking, for instance, was fraught with risks in early days. "During my time, there were so many assault cases! Some of us were chased by hawkers with choppers, some were even slashed," recalls Mohamed Jumahin Bin Ibrahim, who started out in 1969. "There would be commotion when we arrived. The public would scold us, 'Give them chance *la*, don't disturb their rice bowl.' But we have to do our duty. When we set off for work, our families would remind us to be careful."

These same people also ensure food hygiene standards, control vector population, manage operations in the cemetery, crematoria and columbaria, and audit waste collection and cleaning services. When public calls come in requesting site visits, they are the first to respond. They know the ground intimately and often put the knowledge to good use, for example, in vector control. "We look high and low, check roof tops, drains," recounts Saliman Bin Mordi, who has been hunting mosquito breeding sites since 1968. "If we can't find the breeding spot, we look for clues. For example, if we see a bird drinking or taking a bath in some hard-to-reach area, or small plants growing on a signboard or such, there must be water nearby, and mosquitoes are bound to be there."

Their ground connections and knowledge are all the more important during crises. Ravindran Nair, who joined ENV in 1979, remembers the dengue crisis in 2005, "We had to train the volunteers from other agencies and the grassroots fast and plan the logistics. For one whole month, it was carpet combing every day."

Over time, the image of the work has evolved. Now, officers educate at the same time they enforce. Mohd Hussain Bin Yusoff joined ENV in 1975 and sees this change first-hand. He explains, "Environmental hygiene issues are very dynamic, and to keep up with them, we need to change public mindsets. We have to reach out to different groups of people including licensees and food handlers, and educate and motivate them to improve their standards."

Yet, despite the fact that their work is so essential to our lives, it is most of the time underappreciated. Unpleasant encounters with members of the public are part and parcel of life. Imran Bin Mohammed Aleh, part of the workforce since 1995, says, "Some residents scold us when we find mosquitoes breeding on their premises. But they don't really understand that when they breed mosquitoes, it will affect their family and neighbours as well."

What fuels these officers is a strong sense of duty and pride, as well as the knowledge that they are making a difference. Akbar Bessir has been in this line since 1972. He says, "Every year, we see improvement in hygiene standards. If you compare the *pasar malams* (Malay: night markets) 10 years ago and now, so much has been achieved in the operations of the stalls and observance of food hygiene. There is also less dis-amenity caused to residents, because of the efforts we put in. It's a lot of satisfaction."

Mohd Zulkefli Abdul Rahman is one of the younger members, having joined in 1995. "To be honest, our work here is very interesting and challenging. Not many people want to work in environmental health, but every time there is a public health crisis, we are the ones who enter the danger zone first, such as during the SARS and dengue outbreaks. That is why I feel proud to be in NEA."

For S Mohan, a veteran who joined in 1978, it is about camaraderie too. He says, "We take pride in our jobs, even if when the work is dirty. All of us played a small role, but we contributed to the bigger picture. This is where my big family is. My strength, my colleagues and my friends are all here."

* In Malay, these mean "Garbage Officer", "Toilet Officer", "Drain Officer".

** This Hokkien term translates to "Ox on the ground", probably to describe the chaos caused when the Special Squad arrives to bust illegal hawkers, akin to an ox running wild.



ON A DAILY BASIS

Daily-rated employees are a significant part of NEA's ground operations force. Many of them have been in this line since ENV times. They play an important role in maintaining the cleanliness of public areas in Singapore and preventing mosquito breeding on a daily basis. To help increase their productivity, enhance their personal development and improve their image, NEA took steps in 2011 to redesign their work scope, introduce new uniforms and increase the pay scale. Besides taking care of sanitation and vector control, the employees will also report on irregularities observed on the ground. This will allow NEA to respond more readily. Supervisory positions have also been added. Daily-rated employees in such positions are in charge of deploying and supervising staff and reporting on irregularities through Personal Digital Assistant devices.

Care For The Departed

NEA's work touches almost every aspect of our lives, even during the times when we lose our loved ones. This is one lesser-known aspect of NEA's work – that it oversees the day-to-day operations and maintenance of government cemeteries, crematoria and columbaria.

A NEW BURIAL SYSTEM

A key challenge faced by NEA when it comes to after-life services is the fact that Singapore is land-scarce. The practice of burials is unsustainable given our land constraints and developmental needs. In 1962, the government built the first public crematorium, located at Mount Vernon. Over time, cremation had become the preferred choice for the majority of the population.

At the same time, Singapore has to cater to those who wish to opt for land burials, such as those whose faiths require the deceased to be buried, including followers of Islam, Judaism, Zoroastrianism (Parsi), the Ahmadiyya faith and Bahá'í faith. These needs have to be met within the country's only cemetery in use, the 318-ha Choa Chu Kang Cemetery.

Choa Chu Kang Cemetery began operations in 1947. By the late 1990s, it was estimated that the cemetery could only last till 2013 based on existing land burial methods. A way to prolong the lifespan of the cemetery had to be found.

In 1998, ENV introduced a policy to limit all burials to 15 years, after which the graves will be exhumed. This policy was drafted after ENV studied similar legislation in places such as Hong Kong and Turkey. For those whose religions forbid cremation, the remains will be re-interred in smaller plots. Otherwise, the remains will be cremated and stored in a columbarium. The exhumed plots will then be used for fresh burials. This policy extended the cemetery's lifespan by about 50 years.



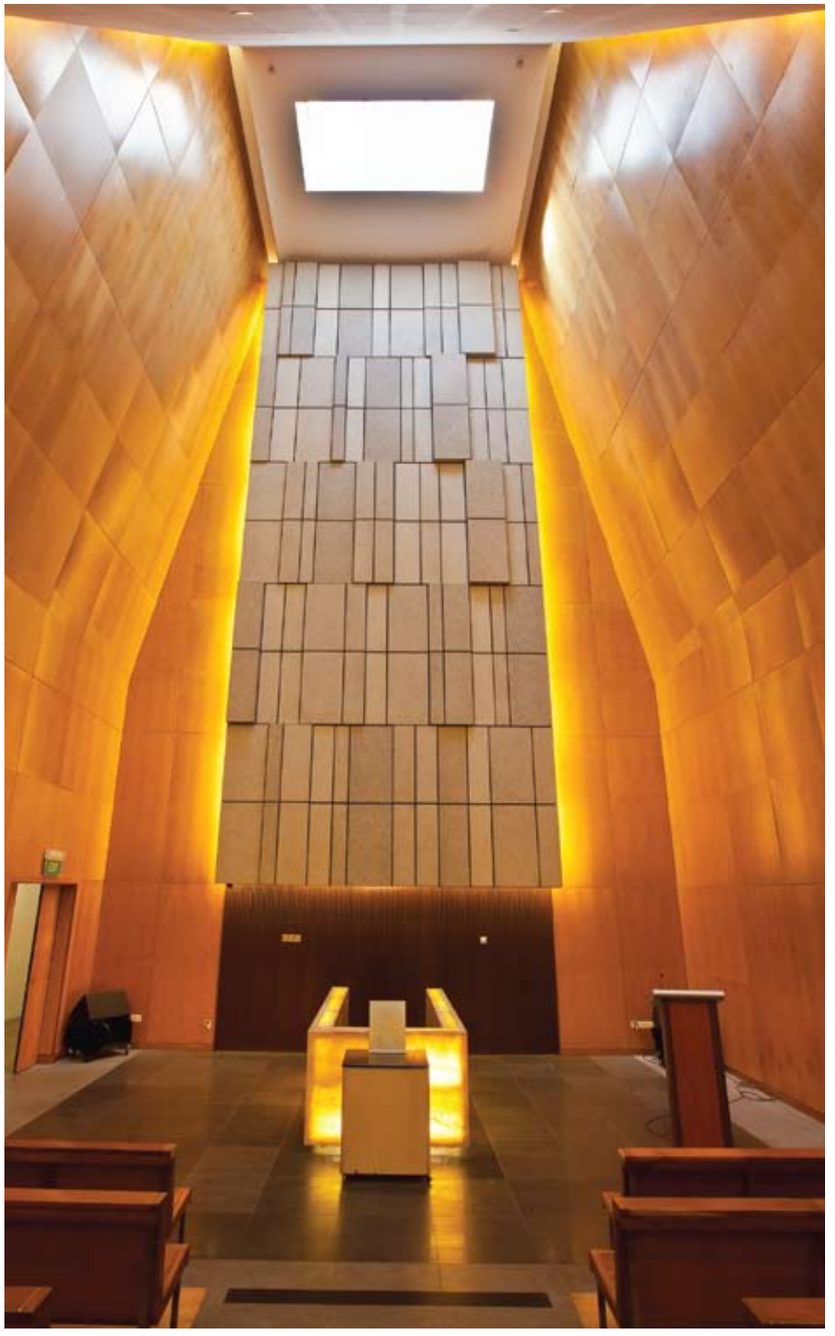


Next, the government began looking into new methods of burial. After NEA took over the management of these facilities in 2002, it continued researching how other land-scarce countries handled burials. Eventually, NEA decided to partner HDB, tapping on the board's technical expertise to develop an engineered crypt system. The system, called New Burial System, comprises prefabricated rectangular concrete boxes without bottoms, laid in a grid on the ground. Each box is fitted with a cover. When a burial takes place, the cover is lifted and the coffin or remains are lowered into the crypt, which is then backfilled with earth and covered.

There are several advantages to the new system. Manual digging of graves is no longer required. The plot density is increased significantly. In traditional land burials, graves must be spaced at least 45 cm apart for the soil to be stable. The distance is now reduced to 15 cm as the grave is separated by a thin concrete wall. Burial and exhumation processes have also become easier with the new system.

Before the system was built, consultations were held with key stakeholders including religious groups and authorities such as Majlis Ugama Islam Singapura. The system received approval and endorsement from the stakeholders and the New Burial System was implemented in 2007 in Choa Chu Kang Cemetery.

As of May 2012, a total of 23,301 crypts were constructed for burials of the recently deceased and re-interment of exhumed remains. The new burial system clinched the Institution of Engineers Singapore Prestigious Engineering Achievement Award in 2009 as it showed that engineering solutions can also be applied to traditional fields such as burials. More importantly, the system, together with the 15-year burial policy, has extended the lifespan of the cemetery to beyond 2130.



SUBTLE CONSIDERATIONS

Constructed in 2004, the planning and design of the Mandai Crematorium and Columbarium Complex actually began way back in 1994 under ENV.

Singapore being such a diverse society, the planners took care to consider specific religious and cultural needs in the design, from the use and flow of space to the type of facilities. "We consulted various religious bodies and funeral directors as we needed to understand the sequence and rituals relating to specific funeral services," explains Chow Mei Ling, who was on the team implementing the system. "For some religions such as Hinduism, believers have a custom of collecting by hand the cremated remains of their loved ones. So we built a private space for the family members to collect the ashes."

The attention to details can clearly be seen in the complex's design, from the controlled ambient temperature and staggered schedule of cremation sessions to the speed at which the coffin is conveyed to the furnace. Quiet rooms provide space for users to grieve in private. The facility's layout is intuitive with clear directional signage. Subtle features such as lighting design, landscaping and colour schemes help ease the next-of-kin's state of bereavement.

"Mandai Crematorium and Columbarium Complex was designed to cater to the needs of different religious groups, while at the same time staying non-sectarian so that it is all-encompassing. We make our considerations for religious and cultural beliefs very subtle and in the background, so that anyone who uses the facilities does not feel uncomfortable."

— Chow Mei Ling, Senior Assistant Director, NEA

Technological solutions applied at the complex are cost-effective, durable and designed to meet Singapore's long-term needs. The density of niches in the columbarium is also higher than in other columbaria to cater for future use. Says Andrew Lau, Head, Facilities, Planning and Development Section, NEA, "Even though the land use is quite intensive, you don't feel the space crunch when you are at the facility because we really took care to consider how people will feel when they visit their loved ones here. We see to it that the facility caters to the deceased as well as the living. Here, the focus is on the living's memory of their loved ones."




and the
Weather tomorrow is...



"Within a city, the weather still plays a crucial role. It affects our daily life, it affects the vitality and business prospects of a city. It can disrupt transport, whether you're flying in on a plane, or in a car driving through a flash flood. Over the last two decades, the focus of the Meteorological Service Singapore has widened. Now it's not just about temperature and wind direction and rain but indeed multi-hazard monitoring and early warning services for trans-boundary smoke haze, volcanic ash, earthquakes and recently, even extremely unlikely tsunamis, or radioactive fallout. Today, meteorological services are also widely used by the larger community. It's no longer just scientists or administrators, but indeed, ordinary people, literally the man and the woman in the streets, and people running businesses and schools. Now with rapid advances in IT and in communication technologies, accurate weather forecasts and real-time weather alerts have also become the norm."

— Dr Vivian Balakrishnan, Minister for the Environment and Water Resources





Meteorological Doppler Weather Radar

It's a familiar announcement at the end of the daily news – "And the forecast for tomorrow's weather is ..." In the papers, on the radio and even on the web and mobile phone, these forecasts shape our lives and schedules constantly.

Sunny days during the school break – family picnics at the beach. Overcast skies anticipated for the weekend – postpone that photography outing. Thundery showers in the late afternoon? Remember the umbrella, book your taxi in advance.

Behind these forecasts is a dedicated team of meteorologists, climate scientists and technicians at the Meteorological Service Singapore (MSS). They scan the skies, monitor radar and satellite images, analyse complex weather models and pore over esoteric charts to churn out some 500 forecasts and briefings a day, complete with readings of rainfall, wind, temperature and relative humidity. This information enables a diverse range of users to make better decisions on their activities and improve their safety and comfort.

But weather forecasting is not the only thing MSS does. It also conducts scientific research on our climate, strengthens Singapore's emergency response plans in case of natural or environmental hazards and provides scientific data and risk assessment for uses ranging from governance and policy-making to business operations. It actively collaborates with regional and international climate research institutes and experts in the field to share research findings and weather data. Today, MSS is not just the national meteorological authority of Singapore but also a leading centre in the region.



The forecasts we get every day are the result of intensive efforts by teams of meteorologists and technicians working round the clock every day. At the Main Meteorological Office at Changi, the meteorologist's day shift starts at 7.30 am. By 9 am, the first weather forecasts for the next three and 12 hours are released. Forecasts will continue to be churned out every three hours and as and when they are needed. Warnings will also be sent whenever severe weather is expected to affect the island. At 5 pm and 5 am, the weather outlook for the next three days will be issued to the broadcast and print media. The work progresses through the night and the cycle goes on.

Behind The Forecasts

"MSS' societal mission has been achieved through the professional expertise and dedication of MSS staff, who deliver reliable and timely services 24 hours a day, every day of the year. For many of us, including myself, our meteorological professions have turned out to be lifelong careers. With the growing demand for meteorological services, along with the increasing threat of climate change to our sustainable development, MSS will have many opportunities to contribute, as never before, to the social and economic well-being of Singapore. By expanding our research collaborations, leveraging on technology and deepening staff expertise, I am confident MSS can meet these challenges and opportunities."

— Wong Chin Ling, Director-General, Meteorological Service Singapore



Barometers, anemometers, rain gauges, thermometers, and such – these make up the basic set of instruments for capturing weather information. But weather science requires much more than that. To provide timely and reliable weather forecasts, MSS relies on a system of weather monitoring, analysis and research.

One of the biggest challenges in forecasting the weather is having up-to-date, precise data on atmospheric conditions. Across the island, weather data collection takes place at MSS' network of over 60 automated weather stations and five manned meteorological stations. Up-to-date information on parameters such as rainfall, wind, temperature, cloud cover and height, solar radiation and visibility are collated every day. With that, more precise forecasts can be provided for Singapore.

Next, technology is a key tool in the weathermen's trade. Images from both geostationary and polar-orbiting satellites are used for observation of clouds and weather systems. A wind profiler radar measures wind conditions at different altitudes in the atmosphere, up to a height of about 5 km.

In 2010, MSS added a more powerful tool to its array — the Meteorological Doppler Weather Radar (MDWR) system. A similar system was first commissioned in 1997. This latest system, however, gives much more information. The MDWR uses the Doppler Effect to gather wind velocity information and also enables MSS to detect and track the movement and intensity of precipitation within a radius of 480 km. MSS can thus better estimate accumulated precipitation, and provide stronger support to the authorities in water resource and flood management.

The weather knows no boundary, and Singapore's weather system is part of a larger regional weather pattern. Hence, MSS also compiles weather data from national meteorological services in the region.

Analysis is done after the data is collected. To do so, MSS employs an advanced meteorological data processing system to handle the sheer volume of information generated daily. Complex mathematical models of the atmosphere, known as numerical weather prediction models, are then run using the processed data. The outputs from these models are analysed to produce a forecast.

The forecasts then benefit a wide range of users, from airlines and ships to the military, businesses and the public.

A RICH ARCHIVE

In the archives of MSS, the earliest weather statistics go back as far as 1820. The first full-scale meteorological station with more elaborate equipment and detailed schedule of observations was set up at Mount Faber in 1929. This rich archive of information enables MSS to better understand historical weather trends and how our climate has changed since the 1820s. It is also a valuable resource in research relating to the climate.



The automated weather station at Kim Chuan Road provides continuous real-time measurement of rainfall, wind, temperature, relative humidity and pressure.

"Before meteorological services were transferred to NEA, the major users were from the civil and military aviation sectors. Our earlier efforts focused on developing our aviation support services to provide better forecasts of flight-path weather conditions and warnings of adverse meteorological situations. In the later years, when our aviation and marine meteorological services, were well developed, we were able to divert more resources to enhance our environmental support and public weather services. Joining NEA enabled us to achieve better synergy to expand and strengthen our public, environmental and climate-change related services, such as addressing the public's need for timely update of weather conditions, frequent occurrences of smoke haze in the region, climate change and its impact."

— Woon Shih Lai, Director-General, Meteorological Service Singapore (1991 – 2003)

BON VOYAGE!

In 2010, over 301,700 aeroplanes took off from Changi Airport. Ensuring that every flight takes off in safe conditions is no mean feat. MSS plays a critical role by providing essential weather information to airline operators and the Civil Aviation Authority of Singapore.

Weather conditions are a significant factor in flight planning, shaping decisions such as the computation of fuel requirements, navigation and routes. The information supplied by MSS helps organisations make better decisions.

MSS undertakes weather surveillance over Changi Airport and the airspace controlled by Singapore. The continuous monitoring allows it to prepare detailed en-route weather forecasts for flights as well as issue advisories and warnings of severe weather. MSS also conducts training on meteorology to air traffic personnel.

On top of this, MSS provides round-the-clock broadcasts of regional weather information to aircraft in flight. Airlines which subscribe to MSS' services are supplied with regional weather information, latest satellite and radar images, direction and speed of upper level winds, upper level temperature and weather prognoses. As the Operational Meteorological Databank in the region, a responsibility MSS has taken on since its designation by the International Civil Aviation Organisation in 1995, MSS also facilitates the exchange of operational meteorological data between Asia Pacific and Europe.

PLAIN SAILING

Just as weather affects flight, it, too, impacts travel by sea.

Storm warnings, sea state and wind forecasts, marine weather assessments and advisories. Such are just some of the services MSS provides to shipping companies and port operators to help them improve on safety and efficiency at sea.

MSS issues weather bulletins for shipping twice daily, as well as warnings of rough sea states and strong winds. It also collects and distributes weather reports made by ships at sea and assists vessels in the calibration and installation of onboard weather instruments.







FROM NOWCASTS TO LIGHTNING ALERTS

Need to plan an outdoor event? The first thing to do is probably to check the weather forecast for the day.

Besides meeting the needs of specialised users, MSS also caters to the public, its largest group of users. Over the years, MSS has improved its services and products to serve the public better.

For example, local weather forecasts are provided not just for the next 24 hours, but for the next 12, and even the next three hours. Taking convenience a step further, location-specific, three-hour forecasts (Nowcast) are made readily available via the Internet and smart phone applications. In 2009, the service was further enhanced with the launch of Weather@SG, which allows the public to access weather information and forecasts via their mobile phones. Today, the data is also made available to smart phone users through NEA's mobile application, myENV.

Lightning risk alerts are also an important service offered by MSS. Singapore has one of the highest rates of lightning activities in the world. For more effective monitoring, MSS has in place a system to detect lightning activities in Singapore. The lightning detection system and the Meteorological Doppler Weather Radar are the key tools used to assess such risks within an area. MSS issues lightning risk alerts via SMS to registered users such as the military, schools, golf clubs and construction companies so that they can plan and take precautionary measures while carrying out outdoor activities.

Aside from these, MSS also offers a whole range of relevant services and products for businesses, including weather assessments and data for planning the design of buildings and structures, and for legal and insurance purposes.



TAKE THAT UMBRELLA ALONG!

Remember those mornings when you wake up to torrential rains which seem to last forever? Well, they are usually caused by "Sumatra" squalls. Squalls are organised lines of intense thunderstorms which are often accompanied by gusty surface winds and low clouds. After they pass, light rain usually persists for some time. These squalls generally build up over the Straits of Malacca at night and move over Singapore before dawn and in the early morning.

Throughout the year, Singapore's weather is marked by uniform temperature and pressure, high humidity, abundant rainfall and no distinct wet or dry season. Rainy days happen once every two or three days. Between mid-November and end January, the northeast monsoon brings surges of heavy rains. The southwest monsoon between June and September is characterised by more frequent "Sumatra" squalls.

In between the heavy downpours and storms, light rains generally fill in the gap. Take note though, the term "drizzle" does not really apply to Singapore. Precipitation is classified as drizzle when the diameter of the water drop is less than 0.5 mm. Any bigger and it is termed "rain". In Singapore, we usually experience either rain or showers. Rain generally forms from layer clouds and is steadier in nature. Showers are short-lived, begin and end suddenly, and usually fall in small localised spots instead of over large areas. So the next time you hear someone talking about drizzle in parts of Singapore, it is actually light rain.

MANAGING EMERGENCIES

You may ask, what does the weather have to do with emergency planning? Well, take one example. During the Fukushima Daiichi radiation crisis in 2010, MSS undertook modelling studies to assess if radioactive plumes would hit Singapore, and to gauge the level of threat posed by these plumes.

When it comes to natural hazards and environmental incidents with trans-boundary effects, MSS' robust weather monitoring and research regime provides critical data and expertise in assessing their impact on Singapore.

For instance, MSS uses satellite images and other meteorological information to monitor the development and spread of fires and smoke haze in the region, particularly during the dry season. When necessary, advisories on occurrence of haze are provided to the public and the relevant agencies in Singapore. On the regional front, the Association of Southeast Asian Nations (ASEAN) Specialised Meteorological Centre, hosted by MSS, monitors land and forest fires to assess and predict the occurrence of trans-boundary smoke haze affecting the region. It also provides regular weather and haze outlooks to ASEAN members and advises them when significant trans-boundary haze is likely to occur.

SENSING THE TREMORS

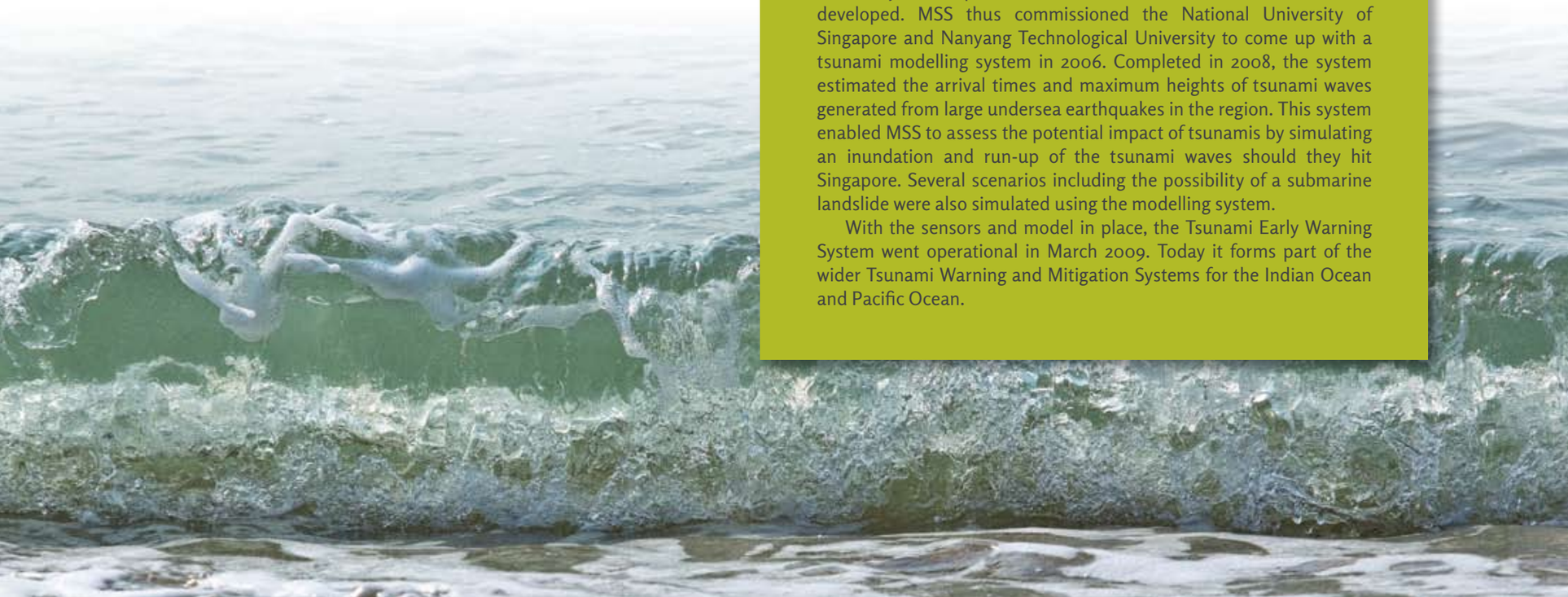
It was one of the most destructive natural disasters in the 20th century. On 26 December 2004, an earthquake of magnitude 9.2 struck off the west coast of Sumatra, triggering a destructive tsunami in the Indian Ocean which took more than 200,000 lives. The sheer scale of the disaster prompted governments worldwide to develop emergency response plans for tsunamis.

Singapore is located about 400 km from the nearest earthquake source in Sumatra. No earthquakes are known to have been recorded here. Studies show that the shallow and narrow Straits of Malacca also make it unlikely for a tsunami to affect us. However, the government decided on a prudent course, and set about putting an emergency response plan in place. The plan required a tsunami early warning system and MSS was tasked to develop it.

Tsunamis are generally caused by earthquakes under the oceans. Hence, having a seismic monitoring network to detect the earthquakes is an important part of a tsunami warning system. MSS has had in place a network of sensors to detect seismic waves since 1996. The sensors were linked to a central data processing system at the meteorological office in Changi. This system was upgraded in 2006 so that Singapore could exchange real-time earthquake data with seismic networks in other countries. This improved the speed and accuracy of seismic detection. Next, MSS installed three new seismic sensors and upgraded existing components to enhance the system's reliability. MSS also worked with the Maritime and Port Authority of Singapore (MPA) to share real-time sea level data. Such data was important to confirm if a tsunami had formed after an earthquake.

Finally, a computer simulation model for tsunamis had to be developed. MSS thus commissioned the National University of Singapore and Nanyang Technological University to come up with a tsunami modelling system in 2006. Completed in 2008, the system estimated the arrival times and maximum heights of tsunami waves generated from large undersea earthquakes in the region. This system enabled MSS to assess the potential impact of tsunamis by simulating an inundation and run-up of the tsunami waves should they hit Singapore. Several scenarios including the possibility of a submarine landslide were also simulated using the modelling system.

With the sensors and model in place, the Tsunami Early Warning System went operational in March 2009. Today it forms part of the wider Tsunami Warning and Mitigation Systems for the Indian Ocean and Pacific Ocean.



"First of all, I need to see if officers are making proper entries to the programme and that our observations are correct. Then, we have to send timely weather data to meteorological centres ashore. Based on data received from the ships and other sources, the forecaster prepares weather charts, which are available for the ships within two to three hours. Most importantly, more data from the ships gives possibility for more accurate predictions, so that we have better weather charts and better forecasts for our voyage. This way, we can expect more accurate surface analysis and weather forecasts for the next 24 hours, 48 hours and more and this can help us to plan our voyage more efficiently. This is very important for captains and officers and can make us more aware of what is around, so that we can make faster decisions. Thanks to correct weather forecasts we can avoid storms, typhoons, improve safety of our ship and save time and fuel."

– Captain Andrzej Gomulski, Master of the vessel NYK Olympus, on the VOS scheme



FRIENDSHIP ACROSS THE HIGH SEAS

Ong Chin Hong holds the one and only job of its kind in Singapore.

He gets on board ships, whether docked in our ports or anchored out at sea. There, he meticulously checks and maintains the ship's weather instruments, from barometer to anemometer, making sure that they are performing accurately. Where necessary, he also shows crew members how to use and log in weather observation data into a standardised system.

Singapore's only Port Meteorological Officer, Ong provides these services to ships taking part in the Voluntary Observing Ships (VOS) scheme. The VOS scheme is an international programme launched by the World Meteorological Organization (WMO). Under the scheme, ships are recruited to observe weather conditions while on the high seas, encode the observations and transmit the data every six hours to the national meteorological services agencies in the WMO network. Singapore supports the scheme too. VOS ships benefit from the goodwill services and support given by the member countries as well.

Says Ong, "Once a ship is recruited, it is my duty to take care of it. I see to the ships recruited in other countries as well. It's like one whole big family. When ships come in, and they need our services, they will send us an email, or the port meteorological officers from the respective countries will send us emails requesting visits. We'll then see to their needs, such as comparing the ships' instruments with ours for calibration and providing weather charts."

This scheme is of particular importance to the provision of meteorological services worldwide because of the data collected. The oceans play a significant role in our global climate system, and data on weather conditions at sea can greatly enhance the development of climate research. At the same time, the real-time data also supports more accurate weather forecasting, climate analysis and provides validation for satellite observations. However, meteorological data over the oceans is sparse, so these ships constitute an important source.

Every participating country in the scheme has at least one Port Meteorological Officer to tend to these ships. These officers also help to recruit ships for the scheme. "For new ships, we will go on board to see what equipment may be lacking. Most of the ships have the necessary meteorological instruments to conduct observation, like dry and wet bulb thermometers, anemometers, sensors for measuring sea surface temperatures and barographs. At the same time I brief them on how to use the electronic log book. We also provide reference books and materials to refresh the seamen's knowledge of meteorology," Ong explains.

For Ong, who understudied the work for a few years before he officially became a Port Meteorological Officer in 2011, the work he does is meaningful and interesting. "In a way, I'm honoured to represent Singapore in this scheme. On land we have stations to collect weather observation data. But two-thirds of the world is covered by oceans, and apart from satellite remote sensing, the only way to get observations from the oceans is through these ships."


DELL

THE SCIENCE OF WEATHER FORECASTING

If you think that predicting the weather is merely a question of looking at the skies, you are pretty far off the mark.

Weather forecasting starts with collecting data – loads of it. Looking at the skies is the first step. Hazman Bin Mohani, Technical Officer, supervises one of the teams tracking weather. These technicians are specially trained to conduct weather observation. "We observe the clouds and check the visibility around the aerodrome using physical markers such as landmark buildings and structures such as the Swissotel," describes Hazman. "Certain clouds are associated with precipitation. In the day, the blue sky acts as the background for us to survey the clouds. At night, we use light sources in the sky, such as stars and the moon, to estimate the cloud cover. When city lights are reflected on the clouds, we know that these are low-lying. To conduct observation at night, one must be a good observer by day. Generally, one needs about three years to be good at reading the skies."

A plethora of weather instruments is maintained by these teams as well. The data collected, including temperatures, dew point and barometric pressure, is submitted hourly. Reports are produced whenever there is a change in weather conditions, such as the start or end of a thunderstorm.

This ground data is only part of what a weather forecaster has to analyse. Goh Wee Poh, Executive Meteorological Officer, elaborates, "To do a weather forecast, we need to analyse a lot of data. For shorter-term forecasts for up to several hours ahead, observation data from the meteorological stations, upper-air sounding data, satellite and radar images are used." For longer-term forecasts beyond twelve hours, an important tool is the numerical weather prediction model, which is a computer model to simulate the future state of the atmosphere. These models provide useful guidance in predicting large-scale weather features, such as the Northeast Monsoon surges, up to several days ahead.

Despite the latest technology, predicting the weather is never a breeze. Firstly, weather dynamics is a complex science which has yet to be fully understood. Secondly, a good forecast requires reliable information about the current state of the atmosphere while there are large observation data gaps over oceans and remote land areas. Thirdly, while the numerical weather prediction models are generally useful in predicting large-scale weather features, they are less skillful in predicting smaller-scale systems.

On top of that, Singapore's weather is often characterised by small-scale thunderstorms. Rain-bearing clouds can form quickly, bringing showers to an area. They can also dissipate as fast as they develop, at times within 10 minutes. Hence, it is highly challenging to detect weather trends or predict in advance when the storms are building up. Says Goh, "Every day is different. We don't have a day where weather conditions are exactly the same. But throughout the year, the temperature variation is small and showers commonly occur, so to many people, it seems that our weather is the same all the time."



Technical officer looking out for physical markers such as landmark buildings and structures to determine visibility.

The constant scrutiny of weather has an unexpected toll on those working behind the scenes. Hazman for example, recalls, "When there are thunderstorms, we have to immediately report on it. So in my first few years here, even when I was off-duty and asleep at home, I would wake up at the first sounds of thunder or rain!" The same goes for Goh, who used to have the added anxiety of putting out inaccurate forecasts. "Especially for the first one or two years, I would keep monitoring my forecast to see whether it was accurate or not! It was hard to get a good sleep during those years and sometimes my friends would ask me why the forecasts were not accurate. But the thing is, people tend to remember the few times when the forecasts were not accurate, and not when they were accurate."

With advances in weather prediction and computer technology, the accuracy of weather forecasts has gradually improved over the years. In 2011, the accuracy of Nowcasts (three-hour forecasts) was around 89%. The occasional inaccurate forecast is something that forecasters have to contend with. "Predicting the weather accurately is a challenge," says Goh. "We strive to produce accurate forecasts, based on the data available and our best assessment of the weather situation. By giving accurate forecasts, we can impact the lives of many people in a positive way."

A Leading Centre For Research

"I am delighted to have signed this agreement between the UK Met Office and the Singapore National Environment Agency. We have a mutual interest in continually improving the scientific understanding and modelling of climate in order to inform decision making around our changing climate. This partnership will enable us to advance climate research, modelling and prediction in Singapore, the wider Southeast Asian region and across the globe."

— John Hirst, Chief Executive, UK Met Office



The effects of climate change are increasingly felt throughout the world. In order to adapt to these changes, governments are seeking answers from science to develop the appropriate response. MSS is establishing the Centre for Climate Research Singapore (CCRS) to respond to the need for better understanding and improved prediction of tropical weather and climate systems affecting Singapore and the wider Southeast Asia region.

The tropical weather and climate systems in the region are poorly understood and less well-researched than mid-latitude weather systems. The broad areas of research in CCRS include improved understanding of tropical weather and climate systems (e.g. El Niño and La Niña, the Asian Monsoon), assessment of local climate change and variability (e.g. droughts and intense rainfall) and improved weather prediction techniques.

CCRS will build up its capability by developing research partnerships with leading climate centres. In May 2011, MSS inked an MOU with the UK Met Office for a multi-year programme involving the joint development and implementation of climate models, exchange of scientists and regional research. The models will be used to generate robust long-term projections of Singapore's climate for various time scales up to 2100. Government agencies will use these projections to develop measures to strengthen the resilience of Singapore's infrastructure to withstand climate change.

Recognising that climate science demands a wide range of inter-disciplinary skills, the Climate Science Experts Network was launched in January 2012 as a platform to engage local and international researchers. For a start, MSS is collaborating with the Earth Observatory of Singapore in Nanyang Technological University and the Centre for Environmental Sensing and Monitoring in National University of Singapore on relevant climate projects.

Past long-term trends of temperature and rainfall are key indicators of changes in Singapore's climate. In 2011, MSS conducted a study of heavy rainfall events in the last 30 years and found that the intensity and frequency of such events have increased across Singapore during the period. The trend is in line with observations of more extreme weather in many parts of the world.

In the area of weather research, MSS has implemented the United States National Center for Atmospheric Research (NCAR) Weather and Research Forecasting model to produce forecasts for up to three days ahead. This model is widely used by many meteorological services in the world to produce operational forecasts for their region and for studies. Research is ongoing to improve the model to better predict weather conditions in Singapore and the region.









FORGING INTERNATIONAL TIES

The World Meteorological Organization (WMO) is the United Nations system's authoritative voice on weather, climate and water. It comprises membership from 189 countries around the world. Singapore became a member of WMO in 1966. A key programme of WMO is the World Weather Watch Programme which coordinates the daily exchange of meteorological observations from about 11,000 land, sea and upper-air stations around the world, including Singapore's. The data is essential to the operations of all national meteorological services. In addition to data exchange, MSS participates in other WMO programmes, conferences, seminars and workshops to keep abreast of international developments in meteorology and to enhance its capabilities in weather and climate prediction. The platforms provided by WMO also enable MSS to develop close working relationships with its international counterparts.

At the regional level, MSS is host to the ASEAN Specialised Meteorological Centre (ASMC), which was established in January 1993. ASMC was set up as a regional collaboration programme among the national meteorological services of ASEAN member countries to enhance regional capacity in weather and climate prediction. Over the years, ASMC has also played a leading role in developing a number of technical capacity-building projects involving the participation of ASEAN national meteorological services.

TRANSLATING SCIENCE INTO POLICY

The global climate system is extremely complex, and our current understanding of the physical processes is rudimentary at best. As it is impossible to perform climate experiments in the laboratory, we are reliant on model simulations of the climate system to provide basic understanding of atmospheric processes. Early climate models were only able to represent key features of the climate system such as incoming solar radiation, rainfall and carbon dioxide concentration. Over the years, such models have become more sophisticated, in part due to greater computing power and better understanding of the atmosphere. The current generation of models also includes cloud processes, a broad range of atmospheric constituents such as sulphates, aerosols and land use etc.

Generally, global climate models are able to represent large-scale features of the global climate fairly well but are less able to do so on a regional and local scale. Such information, however, is needed by decision makers to develop strategies to combat the impact of climate change. The CCRS is set to develop new competencies in climate modelling and research. "Currently, global climate models run by established centres worldwide simulate the climate of the whole globe. These models are based on the best understanding of the science, using mathematical equations," shares John Low, Director, Climate Science Department. "We need to downscale these models to get very fine-scale information of Singapore's climate. These results will be more useful to us. Research on tropical weather is still somewhat lacking in the world, so our centre will develop niche research in this area."

This scientific research is undertaken to further understanding of the climate of Singapore and the region and how it will impact our lives in the future. But most importantly, it will support the translation of climate science into important public policies.

Low elucidates, "Of particular interest now is how Singapore will be affected by climate change in the long term, in 50 years' time, in 100 years' time. And how the weather is going to be like for this region. We want to help develop climate resilience for the country by providing the government with the best science available to develop the policies for Singapore to adapt to climate change. And based on these long-term projections, the government can take decisions on future infrastructural development and economic policies, for example. We recognise that climate science is evolving and new research is being conducted every day around the world, and new information on ice-caps melting, sea levels rising or aerosols will affect our projections. There is a need for MSS to have this capability to look at all this research and assess how it is going to affect our region."



"Of particular interest now is how Singapore will be affected by climate change in the long term, in 50 years' time, in 100 years' time. And how the weather is going to be like for this region. We want to help develop climate resilience for the country by providing the government with the best science available to develop the policies for Singapore to adapt to climate change."

— John Low, Director, Climate Science Department, NEA

Opposite: Weather balloons carrying sensors are released twice a day at the Upper Air Observatory to collect wind, temperature and humidity data of the atmosphere.



The Meteorological Service Singapore is the national authority on weather and climate. It provides forecast and information services to the public, government agencies and industry.

a novel Approach




30
art
Enhance Our Urban Environment
Encourage Community Ownership and Participation
sion
ively and Liveable Singapore
lore Resource Efficient,
mpetitive & Resilient Economy

A Novel Approach

People-Private-Public – Talk about NEA's outreach efforts, or that of other government agencies, and you are bound to come up with these three "P"s. This efficient and concise term, 3P, was popularised by the very minister behind the formation of NEA, Lim Swee Say.

The term did not arise out of nowhere. In the late 1990s, the world was excited about moving into a new millennium and the 21st century. Singapore, likewise, was caught up in the enthusiasm. The government launched an exercise, aptly titled "Singapore21", to envision what Singapore could be in the next decade. Public consultations were held and task forces were set up to work on this new vision. In a parliament session on 5 June 1997, Goh Chok Tong, Prime Minister (1990 - 2004), shared his vision for Singapore21, "More than a house, Singapore must be a home ... Singapore is about people. It is only in Singapore that we, as Singaporeans, have the freedom to build the kind of environment we want to live in, work and bring up our children. That is what being an independent, sovereign nation means. In no other country will we have the same opportunity to shape a country we can be proud of. I hope that Singaporeans from all walks of life will rise to the challenges and make a difference, for Singapore and for your children."

The idea that people and partnership should take a central role in Singapore's future was reiterated in the report released by the Singapore21 committee, captured in recommendations such as this, "Active citizenship must be enlightened by commitment to the values and principles that underpin Singapore society. It paints a powerful vision of a people sector, energised by active, involved citizens, which share in the task of nation-building with the private sector and the public sector."

The notion that nation-building should involve the people, private and public sectors, struck Lim Swee Say, who was on one of the Singapore21 task forces. He recalls, "The 3P approach came from the Singapore21 exercise. One of the key takeaways is that, as the way forward for Singapore, we should adopt the 3P philosophy to engage the people, private and public sectors. That makes a lot of sense, that kind of co-ownership. So I thought, why don't we do that for the environment? If we want people to litter less, we can fine them more. If we want people to consume less water, we can charge them more. So we can use a pricing mechanism to produce short-term results. But if you really think about long-term sustainability, I think nothing is better than adopting a 3P approach."

Lim set upon pushing the approach in his new appointment at ENV straightaway. In his parliamentary speeches, he began introducing the term "3P partnership", compressing the concept of partnering people, private and public sectors into a handy phrase that would soon go viral. In office, he began urging his deputies to adopt the 3P strategy.

In many ways, it was a novel approach. Until then, ENV's primary strategy was regulation and enforcement. Its public outreach efforts were usually didactic in tone, aiming to teach and educate rather than engage. The approach was undeniably top-down.

Putting 3P partnership to action was a tough act. ENV was used to its regulatory role. When NEA formed in 2002, a major mindset shift had to be made. Lam Joon Khoi, appointed CEO of the new statutory board, remembers, "One challenge for NEA then was to nurture the public to take greater ownership of the environment. We envisioned Singapore to have a clean and healthy environment loved and cared for by everyone. We need the public to feel that they have a stake and responsibility in ensuring that our environment is kept clean. Hence, our strategy was to foster a strong collaborative partnership between NEA and the community, creating a shared ownership of the initiatives. We were mindful to balance between the enforcement and environmental promotion roles of NEA. Our focus was on the latter."

"One challenge for NEA that my team and I recognised then was to nurture the public to take greater ownership of the environment. We envisioned Singapore to have a clean and healthy environment loved and cared for by everyone. We need the public to feel that they have a stake and responsibility in ensuring that our environment is kept clean. Hence, our strategy was to foster a strong collaborative partnership between NEA and the community, creating a shared ownership of the initiatives. We were mindful to balance between the enforcement and environmental promotion roles of NEA. Our focus was on the latter."

— *Lam Joon Khoi, CEO, NEA (2002 - 2005)*



With the clear direction from the ministry, Lam began to incorporate 3P partnership in NEA's operations. Singapore was divided into six district offices under ENV for regional management of environmental health. In 2003, Lam restructured them to form regional offices aligned with the boundaries of the five Community Development Councils (CDC). The CDCs were in charge of community bonding and building social cohesiveness on the ground. This strategic move paved the way for NEA's ground operations to be integrated with grassroots initiatives.

Outreach activities were revamped to engage and empower the different sectors. The new CEO saw to it that more resources were allocated to promoting 3P efforts, and engagement of target groups such as schools was stepped up. The 3P Partnership Fund of $750,000 per year was launched in 2003 to provide seed funding for those who wish to initiate environmental projects.

When Dr Yaacob Ibrahim joined the ministry in 2004, he brought the 3P strategy to a higher level with his extensive experience working with the community during his previous post at the Ministry of Community Development and Sports (present-day Ministry of Community Development, Youth and Sports). He says, "Bringing people on our side was, I thought, very important. The message is, 'This is your environment. You have to take care of it together with us and let's work together.' So we engaged the NGOs (non-governmental organisations). And it was something that we were a bit uneasy about in the early days. But we soon realised that if we didn't get the people and NGOs on our side, we were going to have a big problem in bringing the whole nation together to talk about sustainable development."







Operationally, 3P partnership efforts received an even bigger boost under NEA's second CEO, Lee Yuen Hee, who joined in 2005. More resources were provided for such efforts on the ground. The 3P partnership department in NEA was raised to the level of a division to accord its work the rightful significance. "At that time, the 3P partnership approach was already recognised and embraced as the key to NEA achieving its mission. The 3P partnership force was further beefed up and provided with the resources needed to carry out its work," shares Lee. "Another important component was outreach to the young in kindergartens, schools, polytechnics and tertiary institutions. It is only through inculcating our people with the right environmental values from young that we can hope to build and sustain the pro-environmental mindset."

Partnership with the different sectors became more focused during the term of NEA's third CEO, Andrew Tan, who joined in 2009. Programmes were devised to specially target specific groups. For instance, creative initiatives such as music and fashion competitions with an environmental focus were organised to draw young Singaporeans. The flagship environmental event for youths, Youth for the Environment Day, was launched with a bang in 2011.

After a decade of focused efforts, the 3P partnership strategy has successfully become a key thrust in NEA's mission. That said, balancing the duty to regulate and the mission to partner the community is still a fine act. As Dr Vivian Balakrishnan, who took up office as Minister for the Environment and Water Resources in 2011, puts it, "The challenge is that we need to maintain our standards, and have the public and NEA working collectively. We cannot afford to have an adversarial relationship between NEA officers on the ground and the public. What we need is a partnership."

But clearly, within NEA itself, a mindset shift had taken place. For an agency which started out with strict enforcement as its key approach, NEA now sees partnership as important as its regulatory role. Its strong network on the ground with grassroots organisations, schools, NGOs and private corporations testify to how far the 3P partnership strategy has come. As for whether this can eventually bring about Singapore's long-term sustainability, it now depends on how Singapore evolves as a community. Andrew Tan summarises, "We cannot be everywhere all the time. The more philosophical question is, do we want to be everywhere all the time? We want to move towards a society where people take on greater responsibility and we can work together with the community than work against the community. It depends on how Singapore evolves. We want to move towards a more promotional, facilitator mode of engagement rather than enforcement. But we can only do that if everyone plays their part. And the public still wants the authorities to take irresponsible members to task. It is a balance we have to strike. We are conscious of these two roles. For NEA officers, working in NEA, we are in this job to make a difference. We are here to safeguard the environment and we also have a role to educate Singaporeans and get them to be more responsible citizens."


MAKE YOUR
COMMITMENT
TO THE

In The Heartlands

Going down to the district level to manage public health and environmental issues is a strategy that has been used since colonial times. What marks the difference today is the level of participation from the ground. NEA's regional offices not only see to the effective management of public health and the environment, they also partner the communities around them to come up with ideas and projects of their own. Every community has different needs, and these needs can differ vastly from one community to the other. Working with the very people who live there ensures that the environmental initiatives are relevant for the community.

At housing estates in Pasir Ris and Punggol, some 250 volunteers from the grassroots formed a strike force against mosquitoes. Banding together with NEA officers, the strike force covered more than 3,000 residences in August 2011 to drive home the dengue prevention message. The participation from the ground was vital as active dengue clusters were found at Pasir Ris Street 12.

Over at the five community development districts, NEA officers sat down with community leaders and facilitators from grassroots organisations and the CDCs to work out their respective environmental plans and programmes.

These volunteers and community leaders make the all-important difference in NEA's outreach efforts. Carefully nurtured over the years, the relationships between NEA and the grassroots were further cemented during national crises such as SARS in 2003 and Singapore's worst dengue epidemic in 2005, when hundreds of volunteers rallied with NEA to combat the outbreaks.

When NEA's district offices were realigned to the same boundaries as the CDCs, the move paved the way for the agency to show the grassroots that they could work towards common goals. Public and environmental health projects were shown to be effective in producing a higher standard of living for residents and nurturing community bonds. The mayors of the five CDCs supported the move wholeheartedly and took the lead to integrate the environment as a core component in their plans for the community districts.

In addition, NEA trains and guides grassroots volunteers, equipping them with the skills and knowledge to run the programmes on their own. Recycling days, environmental and public health workshops and seminars, and energy conservation campaigns are just some of the usual activities run by these volunteers. The agency supports the grassroots by providing exhibits, educational kits and collaterals. It also supplies intelligence on locations of dengue clusters when outbreaks take place.









Clockwise from top left: Exciting and creative initiatives such as the Eco Music Challenge galvanise Singapore's youths to act for the environment; The Envirobot spreads environmental messages through interactive means; Volunteers from the community go house to house to promote activities such as recycling; Operation MACE injects an environmental lesson in gaming, as participants take down mosquitoes in this first-person-shoot-out game.

A PLAN FOR EVERY DISTRICT

There is the overarching mission to encourage green lifestyles, and there are specific strategies when one gets to the ground. Today, the environment is an essential feature in the CDCs' plans for their districts. But instead of a one-size-fits-all approach, every district has a long-term plan tailored to its specific needs. These plans, developed jointly by NEA and the CDCs, are the result of consultations with grassroots organisations, schools, corporate partners, town councils, voluntary welfare organisations and members of the community.

"Clean City, Enviro-Friendly Lifestyle, Green Community"

The Central district of Singapore is the most diverse in terms of population profile. As such, NEA and Central Singapore CDC use common goals of residents to build unity and bonds within the community. This gives rise to the tagline for the 10-year Central Singapore Environmental Sustainability Plan launched in 2009 – "Clean City, Enviro-Friendly Lifestyle, Green Community". "Clean City" aims to achieve the cultivation of good social and hygiene habits, while "Enviro-Friendly Lifestyle" seeks to educate residents to adopt green practices such as saving energy and practising the 3Rs (Reduce, Reuse, Recycle). "Green Community" provides avenues and platforms for all stakeholders to contribute and create a green movement in the district.

"Actions by the Community for the Environment"

Community ownership is underscored when it comes to North East district's green plan, aptly entitled "ACE Plan" (Actions by the Community for the Environment). The North East ACE Plan aims to raise public awareness and garner support from the community to sustain a clean and green environment in the North East district. This district's five-year work plan, which came into effect in 2009, takes a topical approach. "Clean Environment", "Resource Conservation", "Energy Efficiency", "Encouraging More Community Efforts" and "Creating a Sustainable Green Future" make up the five core objectives of the plan.

"Green Living @ North West"

Promoting community bonding and social cohesion, cultivating a green lifestyle, increasing environmental ownership and achieving environmental sustainability outline the objectives of North West district's 10-year ECO Plan, Green Living @ North West. The plan, launched in 2009, is supported by five pillars comprising "Clean Every Day", "LEAD Everyone Against Dengue", "Fight Climate Change", "3Rs" and "Clean Toilets". North West CDC targets to build the largest green network in the district by 2014 with 45 community green clubs and 2,000 volunteers to reach out to 90,000 households under Recycle @ North West, launched in 2011.

"Go Green @ South East"

South East district's GreenPlan is an all-out call for the community to "Go Green". This five-year plan launched in 2010 identifies the key groups in its district – green schools, green homes, green grassroots and green organisations, and their roles in protecting the environment in areas such as dengue management, energy-saving and resource conservation. What's interesting about the plan is its "T-model". The horizontal bar of the "T" represents the broad-based approach to reach out and involve as many people as possible to do a small part for the environment. The vertical bar symbolises the strategy to engage interest groups and organisations for more in-depth projects.

"Tomorrow Starts Today"

Over at the South West district, the environment and community are inseparable when it comes to long-term planning. Which is why its 10-year plan for the district is called ECo Plan, i.e. Environment & Community Plan. Released in 2009, the plan emphasises building sustainable and engaged communities in a cleaner, greener and healthier home for its residents. Five key thrusts are formulated to develop the district into a cool, green, clean, caring and pro-active community. Targets for the district include reducing energy consumption by 10%, instilling a higher level of social graciousness in the area of adopting good public health and hygiene practices, planting a million native plants and encouraging more residents to volunteer.











CHAT WITH A MAYOR

Dr Amy Khor, whose passion for the environment is well-known, has contributed to a better, more sustainable Singapore through her former posting in MEWR. She shares her thoughts as the Mayor of South West District, a post she has held since 2004.

Having been actively involved in policy-making and issues related to the environment for so many years, could you share with us your perspectives on how the scene has changed over the years?

Four decades ago, way before sustainability became a catchphrase, we were already pursuing it as one of our goals. It was out of sheer necessity that we had to pursue water and environmental sustainability. Through judicious land planning, investment in environmental infrastructure, strict legislation on pollution control and close partnership with the people, private and public sectors, we have built for ourselves a clean and green city that is in harmony with good economic growth.

In 2008, the Inter-Ministerial Committee on Sustainable Development was set up to develop a national framework and key strategies for Singapore's sustainable development for the next two decades. We widely consulted key stakeholders and the public. We received over 1,300 suggestions and more than 700 people participated to contribute their ideas and aspirations to a sustainable Singapore. I am heartened that corporate and community organisations and civic-minded individuals are increasingly taking a keen interest to be part of this co-creation process.

In recent years, we see more organisations contributing towards the environment as part of their corporate social responsibility. More volunteers are stepping forward every year to champion various environmental causes. In South West district alone, we see a steady increase in the engagement of participants and volunteers in the environmental programmes.

With your years of working with NEA, what are your views on the agency's work over the last decade?

As the national agency devoted to environmental issues, NEA has played a crucial role for Singapore's development, and I am proud of it. One of the key changes I witnessed was NEA's active efforts to engage a greater number and wider spread of partners to attain a higher level of buy-in for environmental causes. As a matter of fact, NEA had foresight and was progressive when it established decentralised regional offices to be closer to the ground and work with the community.

When NEA was formed in 2002, it was heir to three decades of hard work undertaken by ENV (present-day MEWR) since 1972. Over the last decade, NEA has focused on shaping lifestyles and habits by engaging the 3P partners more fully and empowering them.

However, the work for the environment is never done. NEA's work is not easy as the challenges faced by the agency run from everyday issues like ensuring clean public toilets and littering to more complex ones like climate change and trans-boundary haze. In a changing social, economic and geopolitical landscape, the target is always moving, and NEA must stay ahead of these trends to remain relevant. It has done commendably well in this regard.



As the Mayor of South West District, can you share with us your thoughts on the district's green plan?

Through the South West CDC's 10-year ECo (Environment & Community) Plan, we aspire to build sustainable and engaged communities, and create a healthier, cleaner and greener home. We also hope that the community will take ownership of the environment, thus building a sense of belonging.

Hence, it is imperative that the community be involved in the co-creation of our plan, right from the stage of conceptualisation. We had a committee formed, made up of very passionate people, from both the private and public sectors, to spearhead the plan's creation, coupled with comprehensive consultation with the community.

These goals then provided the direction for our environmental programmes. For instance, our youth volunteers are trained to raise awareness and inculcate the values of energy efficiency, waste recycling and water conservation in our residents through house-to-house visits.

Additionally, local green plans and programmes must dovetail with national efforts to create a sustainable Singapore. It is in this spirit that we mooted the 10-year plan in 2008 and launched our climate change strategy in 2011 to support the meeting of Singapore's 2020 target to reduce carbon emissions.

We recognise that we can do more for the environment and the community when we synergise and pool our resources together. I am glad that NEA (South West Regional Office) has a dedicated 3P partnership team. Their expertise in environmental issues and programmes has been integral in the success of our efforts. They act as an enabler, as they provide training for volunteers and deploy them in various programmes to engage the community.

THE COMMUNITY GOES GREEN!



"Whenever residents in Canberra see anything that may cause pollution or lead to breeding of mosquitoes, they can call and I will follow up. We patrol the different precincts and note anything that needs improvement. For example, when NEA rolled out recycling bins in our area in 2006, many residents thought they were big waste bins. So we gathered volunteers to stand by the bins, educated residents on recycling and encouraged them to share the information with their families and friends. I also run a emailing list compiling daily updates on dengue fever and clusters from NEA's website. I use the network to pass on new environmental messages as well. If everyone does just a little bit for the environment, it can make a difference. One small group of people can influence people around them and the next generation to change the environment."

Peter Lo, retiree, has been recruiting and training volunteers for environmental causes in Canberra since 2003. Together with the volunteers, he drives initiatives such as recycling projects and patrols to battle mosquito breeding.



"Our first collaborative effort with NEA in raising environmental awareness in Mountbatten was an event in 2007 encouraging everyone to adopt the '3Rs' in their lifestyles. Its success encouraged me greatly. Since then we have worked with NEA to create awareness on environmental issues during our activities. I hope to persuade more residents to take part in our green initiatives by organising events that are creative. During our Mountbatten Commemorates Earth Hour event in 2012, we used minimal lighting and electricity, and the half-lit stage came alive through creative events, such as youths performing beat-boxing and a cappella. It is important that we should all make a concerted effort to save our environment, for our generation and for generations to come. When we see residents engaged through our outlets, it is certainly a wonderful affirmation of the good work we have done for the community."

Amatul Jameel Suhani (A J Suhani Sujari), teacher, is an active grassroots leader raising awareness on environmental issues among her community in Mountbatten.



"I really want to see our children grow up in a clean, healthy environment. The SARS and dengue outbreaks have shown us that we should take care of the environment and our hygiene. It is not easy to convince people to practise the right habits, such as getting rid of stagnant water to prevent mosquito breeding. We conduct house visits often to make sure that the environmental messages get through, and we distribute brochures from NEA to residents. NEA officers work very closely with us, especially in terms of projects for the environment. Making sure that our residents benefit from the projects and helping them understand what we are trying to do so that one day they will appreciate it and practise responsible habits to protect themselves and their loved ones – this is what drives me. After all these years, we can see things changing. People are becoming more aware now, especially when it comes to dengue prevention."

Raja Mohan, who works in a charitable home, has been advocating good hygiene and cleanliness in his community at Hougang since the SARS outbreak in 2003. Besides organising environment events with his fellow grassroots volunteers, he also shares the message through talks and presentations on cleanliness.



"We come up with interactive ways to spread the environmental message, such as games and skits. For example, we enact situations where a litterbug is caught in the act and how passers-by react. This gets the attention of the residents, who learn about taking initiative to stop littering. NEA works very closely with us to drive and visualise our ideas. The past four years of initiatives have definitely led to some changes. When I set off for work at 7 am, I would go round the back of my building. It used to be chocked with litter. But now the ground is much cleaner. Personally, when I see that the estate is clean, and when residents tell me that our events are really educational, it is rewarding. Every word and compliment from residents encourage and motivate me to do more."

Alice Tavamani, an executive in the aerospace industry, strives passionately with her fellow grassroots leaders to enhance the environment at Boon Lay, from dengue prevention and stopping littering to energy conservation.

Opposite: NEA officers get down to the ground to encourage households to conserve energy, one light bulb at a time.


CLEAN
GREEN

CLEAN AND GREEN SINGAPORE

A look at how NEA's flagship environmental event has grown over the years

From a week-long event to a campaign which lasts a year, Clean and Green Singapore has undergone a radical transformation since it began in 1990 under ENV as Clean and Green Week. This campaign has always been a significant event on the national agenda – the fact that it is always launched by either the Prime Minister or the Deputy Prime Ministers underscores its importance. Held during either October or November every year, the campaign started out as an attempt to instil awareness and appreciation of a clean and green environment among Singaporeans. When NEA took over Clean and Green Week in 2002, it immediately took steps to refresh the campaign. More partners were invited to co-create the events and more activities were launched. And the question arose – why slate a week for Singapore to be clean and green? Why not a campaign that called for a clean and green Singapore through the year? Clean and Green Singapore was thus born. Today, the campaign covers a diverse range of topics, from protecting the environment and climate change to public health and hygiene. Over the years, the involvement of the 3P sectors has also grown steadily. The result – a Clean and Green Singapore that is better, bigger and way more vibrant than ever!

2002 The first Clean and Green Week organised by NEA, its focus was on resource conservation. A noteworthy highlight of the week was an exhibition co-organised with Central Singapore CDC, on different ways to reduce, reuse and recycle. This is a key step in the strengthening of partnerships between NEA and the CDCs.

2003 The SARS crisis just blew over. But the lessons were not forgotten. Hence, the theme "We Care, We're Ok" for the year's Clean and Green Week was adopted from NEA's Singapore's OK campaign for higher standards of public health and hygiene. This was the first time such issues took central stage in the week-long event.

2004 NEA jointly organised the event with a CDC, South West CDC, for the first time. It was an important milestone in NEA's efforts to nurture community ownership of the environment. Singapore's OK continued to be the theme for the event to encourage more social responsibility and graciousness. With the launch of "Adopt-a-school" programme, the carnival for schools held during the week was to become more exciting from 2004 onwards.

2005 Nurturing environmental ownership continued to be a primary thrust during the event. The Schools' Carnival during the week proved to be more vibrant than ever as schools took to designing games, competitions and quizzes revolving around environmental topics.

2006 This was the last year in which the event was named Clean and Green Week. From 2007 onwards, it would be known as Clean and Green Singapore. Instead of a week-long event to create more awareness, NEA decided to turn it into a year-long commitment to keeping Singapore clean and green. All the other various campaigns under NEA would fall under the bigger umbrella of Clean and Green Singapore. This would inspire Singaporeans to be environmentally friendly not just for a week, but for each day of the week.

2007 Clean and Green Singapore campaign was launched, marking a new era for NEA's campaigns for environmental sustainability. The planning of environmental events became more cohesive and systematic now that they were seen as part of a larger campaign. Changing the mindsets of Singaporeans on a daily basis also became more prominent in this newly-refreshed event, with the theme "Adopting an Environmentally-Friendly Lifestyle".

2008 The campaign aimed at encouraging Singaporeans to be more conscious of the environment on a daily basis. NEA continued to co-organise the event with external partners, including South East CDC, PUB and NParks. The agency also took care to make the carnival a low-carbon event.

2009 40 years of Singapore's clean and green journey were celebrated at the year's campaign. Singapore's progress and development over the past four decades were traced in an exhibition. Efforts to inspire and empower the community continued to be stepped up through the diverse events organised by NEA and its partners.

2010 Sustainability became a buzzword with the launch of the Sustainable Development Blueprint in 2009. Clean and Green Singapore supported the message of the blueprint. Events were planned and designed to bring forth the importance of personal responsibility in creating a sustainable Singapore.

2011 Community pride and ownership of the environment remained a key focus. NEA also launched two new flagship events – The community-led "Keep Singapore Clean" and "Youth for the Environment Day" to mobilise Singapore's youths to be more involved in environmental causes.



"Through all our initiatives over the years, we have progressively transformed Singapore into a City of Gardens and of Water. Looking ahead, we must learn to develop sustainably as well. The government is coming up with programmes, but these programmes need support from the people sector and the private sector to make them work. I am confident that we can achieve our goals by working together so let us put our efforts together, do it cohesively and ensure a clean and green Singapore for many generations to come."

— Lee Hsien Loong, Prime Minister, at the launch of Clean and Green Singapore 2010, 30 October 2009





NEA 3P officers are the drivers and movers in engaging the community to act for the environment. They work closely with the local communities to create and plan long-term initiatives, and set targets for a better environment. They encourage and support residents, schools, grassroots organisations and corporations to come up with exciting projects of their own. As facilitators, they help different groups of passionate advocates for the environment to connect and network with each other for greater synergy in their initiatives. 3P officers are also posted to every community district, where they have a strong presence on the ground, nurturing generations of Singaporeans to care for and cherish the environment they live in.





Youths For The Environment

Recognising the pivotal role of the youths in shaping a better environment, NEA takes exceptional care to reach out to youths in ways that are relevant and fresh, so as to nurture a strong sense of environmental ownership.



"Let's keep it clean, green, beautiful, Like Heaven on Earth, We'll get by with the simple things in life, We'll give it what it's worth ..." so croons Haramain Osman in a self-composed ode to the environment at the Eco Music Challenge 2011 organised by NEA.

Haramain Osman is one of the many youths inspired by NEA's initiatives to put his creativity to use for Planet Earth. Music challenges, fashion shows of designs that use recyclable materials, climate change photography competitions, creating sculptures from urban waste – these are just some of the exciting things concocted by NEA to galvanise youths for the environmental cause.

Recognising the pivotal role of youths in shaping a better environment, NEA takes exceptional care to reach out to youths in ways that are relevant and fresh, so as to nurture a strong sense of environmental ownership.

Not only that, principals and teachers are joining the cause as well. Today, the environmental movement is picking up at schools around Singapore. Every school has a teacher who doubles as the Environmental Education Advisor, trained by NEA, to plan and coordinate environmental programmes and incorporate environmental issues in the students' learning process. Environment or green clubs are also springing up in schools, birthing grounds for future environmental champions. This network took the agency years of work to build up, by working with principals and teachers on a school-by-school basis, and consultations with the Ministry of Education.

The firm foundation with the schools made it possible for NEA's flagship environmental platform for youths, Youth for the Environment Day (YED), to be created. Launched in 2011, YED falls on Earth Day, 22 April, every year. Young Singaporeans can champion environmental ownership and renew their commitment to care for their environment on that day, and take part in a wide range of programmes such as recycling drives, energy conservation, clean-up of their surroundings and workshops on environmental issues.




UTH FOR THE
ENVIRONMENT DAY
Launch of
YOUTH FOR THE
ENVIRONMENT DAY
Guest-of-Honour: Dr Yaacob Ibrahim, Minister for the Environment and Water Resources
2011
ECOFRIEND
AWARDS
CEREMONY

GROOMING STEWARDS OF THE ENVIRONMENT

Step into any school today and chances are that you will meet budding environmental champions busy with recycling drives, dengue prevention programmes or designing experiments on environmental solutions.

The green movement among the young is making waves, starting from the schools. Principals and teachers are now more and more involved in integrating environmental education with student life. Koh-Leong Swee Ling, the teacher co-ordinating environmental education in Dunman High School, explains the importance, "The situation in this world today is such that sustainability is a real issue. We do need to teach our children so that they adopt a green lifestyle and they know the concern about resources. To prepare them to achieve sustainability in the long run, they need to be educated on the environment. They need to have the passion to find out more, to innovate and to find solutions. It is vital for our students to have environmental education."

These views are echoed by many educators. Incorporating environmental issues into the schools' curriculum and activities also help to achieve the bigger mission of student development. Chong Chee Wee, the teacher in charge of the green club at Anchor Green Primary School, shares, "We want to build character in our pupils, and through the platform of environmental education, they can learn to be gracious citizens."

The surge in such awareness came about thanks to NEA's 3P partnership with the schools. The agency promotes environmental education, provides resources and training for educators, tailors events from workshops to competitions and ties schools up with private companies for environmental initiatives. "NEA provides many avenues for students and teachers, such as the Environmental Champion workshops and Environmental Education Advisor networking sessions, to provide training and learning opportunities to further develop environmental advocates. Through the networking sessions and provision of funding and resources, we are able to better align our programmes with NEA's agenda," shares Annie Lim, Senior Teacher for Environmental Education at Commonwealth Secondary School. "Our long-standing partnership with NEA has been invaluable and instrumental in developing the green agenda within the school and community. Being involved in the maturing and growth of NEA's programmes has propelled the school's progress. We are recognised as the centre for environmental education among schools in the west zone and in 2009, we were the first school to receive the President's Award for the Environment. We anticipate closer collaboration as we work together in nurturing the next generation of environmental stewards, advocates and guardians."

With the resources and support from NEA, the schools go on to create a myriad of activities, such as school fairs, green competitions, workshops and classroom learning sessions covering topics from dengue prevention to energy conservation, eco-camps and hands-on projects.

What is most encouraging is the fact that schools are beginning to see a mindset change taking place among the students. For example, at Dunman High School, students used to complain about having to use recycling bins. In recent years, students have switched to complaining about improper use of recycling bins. Anchor Green Primary School no longer sees the need to remind its students that Wednesday is recycling day, as they have internalised the message. Anglo-Chinese School (Independent) teacher, See Boon Tiam, says, "The viewing rate of exhibitions on the environment has gone up, now they create quite a buzz among the students. Our green club started with only 14 members and now it's over 100 strong. There are also more and more student-initiated projects."

Summing it up, Lim Bin, teacher-in-charge of Woodgrove Secondary School's green club, says, "We really appreciate the opportunities and trust given by NEA. The platform it provides for the students is not something you can find in textbooks."

"Sometimes when you teach the students, they may or may not internalise everything they are taught, such as taking care of the environment. But perhaps years later, they will recall and connect what they were taught when they were young to their present, and make a difference."

– Lim Bin, teacher, Woodgrove Secondary School






END
COASTER

Under NEA's Corporate And School Partnership (CASP) programme, companies can partner schools on different levels, from sponsorship and mentoring students to internships and even implementing customised larger-scale programmes.



Launch of Environment Champions Industry Module for students in 2010, a collaborative effort between NEA and Panasonic.

MORE THAN BUSINESS

Since the Industrial Revolution in the 19th century, the relationship between businesses and the environment has never quite been amicable. Thankfully, this is changing as industries and corporations mature and begin to take on social responsibility in caring for the environment. In Singapore, NEA plays a central role to facilitate the private sector in their corporate social responsibility initiatives.

Under ENV and later on NEA, working relationships with the private sector have always existed. What marks the change today is a growing partnership. "In the past, engagement with the private sector was operationally driven. We consult them in terms of policy changes, how to roll out and smoothen the implementation," says Tan Wee Hock, Director of the 3P Network Division in NEA. "Now, we leverage quite strongly on the private sector to educate the young and the community. The private sector has resources to help certain groups. They can provide manpower to mentor youths and funds to sponsor programmes. It is quite important, because some of the companies can develop environmental programmes and go out and do a lot of good work. They also have expertise to interest the young in areas such as developing clean energy. They can develop flagship programmes to further spread the message. This is what we mean by engaging the private sector now."

To cater to different levels of involvement from the private sector, NEA has developed a range of programmes. Getting the private sector to raise funds for environmental causes and NGOs is an important component of NEA's strategy. A key fund-raising event is the annual Semakau Run, organised at Semakau Landfill by NEA to encourage corporate giving. Since its launch in 2007, the Semakau Run has raised more than $1.8 million. The funds benefited environmental NGOs and social service charities with pledges made by participating companies.

Today, after a decade of facilitating private sector partnership, NEA has small and medium enterprises and large corporations taking part in its 3P programmes. More importantly, greater synergy for the environment is created as NEA helps to bridge the private and people sectors.

"Traditionally, we have been focusing on safeguarding, ensuring that we continue to enjoy a high standard of environment. If you look at the history of NEA, besides the infrastructural development, the enforcement role is actually quite central. Enforcement definitely has an impact. First of all, if most people abide by the rules, and some of them don't, and they are not punished, it defeats the purpose. After a while, it erodes the whole social contract. Therefore enforcement against the minority who are offenders and bringing them to task is critical. Secondly, when you enforce, it still does have an impact of converting offenders. For example, during Corrective Work Order, they will realise the impact caused by their actions, for the simple fact that they have made the life of the sweeper more difficult and it's not trivial. But when we move towards green issues, that's where enforcement may not be the best model. To encourage people to recycle, if we just enforce, we may get results. It will work up to a certain extent, but do we have the resources to sustain it all the time if we make it mandatory? We cannot be everywhere all the time, it's not feasible. That's where we think, at the end of the day, the people must take ownership. Basically a lot of our work is changing behaviour. We want to create social pressure. Just like queuing. If you jump queues, people will stare at you. So going forward, we have to leverage more on public consultations and discussions. Ultimately, if you look at our programmes, we used to be driving towards a clean and green Singapore. Now we want to move towards a clean and green Singaporean. If you are a clean and green Singaporean, you will always do the right thing, it doesn't matter where you are. This is the set of values you will exhibit."

— Tan Wee Hock, Director, 3P Network Division, NEA



A CARNIVAL OF PARTNERSHIPS

Under NEA's Corporate And School Partnership (CASP) programme, companies partner schools to further the environmental cause. A cornerstone of NEA's outreach to schools and the private sector, the programme was previously known as "Adopt-a-school" when it was launched in 2004. Through the CASP programme, companies can ramp up on their corporate social responsibility initiatives by supporting environmentally minded youths. Schools and students also benefit through the funds, advice and mentorship provided by the companies. The students are also exposed to real situations and scenarios first-hand, such as using clean energy in industrial settings.

The partnerships culminate in the annual two-day Schools' Carnival. This national platform exhibits the joint projects by students and the companies. Converting waste heat into electricity and using hydrogen cells to power vehicles, coming up with new ways to use old things, designing environmental education kits and organising eco-camps – students embark on a journey of learning and discovery and put in their best shot for the environment.





BECAUSE THEY ARE OUR FUTURE

It is a power plant, yet it has received accolades for its environmental initiatives, among which are its school outreach programmes. Each year, Senoko Energy allocates resources for environmental projects with students. For example, through its National Weather Study Project which involves more than 150 schools, students are educated on climate change and encouraged to launch research initiatives.

Senoko Energy is in this for a good reason. "We are conscious of the impact we have on the environment and the community. It is our business to be a good neighbour and that means care for the environment and community engagement are a priority on our agenda," explains Kwong Kok Chan, General Manager.

On its own, Senoko Energy has been consistently adopting best practices in environmental protection in its operations. But building internal capacity alone is not enough.

Says Kwong, "We wanted to engage the schools as well as the community and even NGOs. When NEA approached us in 2004, asking us if we were interested to adopt schools for environmental initiatives, it was the chance we were waiting for. And immediately, we said, 'Yes.' We would adopt all the schools in our neighbourhood."

Following the breakthrough, Senoko Energy went on to build strong collaborative networks with the schools. Today, it shares expert knowledge on clean energy, mentors and funds student projects, influences the schools' curriculum for environmental education and organises internships.

The resources pumped into Senoko's outreach programmes are considerable. Just the National Weather Study Project alone called for millions in funding. Senoko's staff also volunteer regularly at different schools. But to the company, the returns are immeasurable. "We have benefitted immensely," asserts Kwong. "We have seen some really imaginative projects by the students and we have also learnt new things along the way. But the main objective is also how we can add value to the learning journey of the students, because they are our future. Through NEA, we have worked out a very meaningful programme to build bridges in the community, make the first step into the schools and make a difference."

IT'S FOR A CAUSE

For Wah & Hua, it all started with the line they are in. "We have been collecting tonnes of rubbish every day," Marcus Koo, Business Development Manager, says. "And we have been thinking, how do we push this further – cutting down the waste volume at source."

Starting at source means engaging with the community, so as to spread the message of minimising waste. "It was fortunate that NEA shared with us the opportunity to work with schools," recalls Koo. "Without NEA, we wouldn't have this opportunity. We've got to give credit to NEA that in these few years, rather than just enforcement, they are also building bridges, and linking people with the private sector."

Being a small enterprise, Wah & Hua employs an innovative approach to educate students. "We give talks. We do presentations and teach students about recycling. But more importantly, we engage the students creatively," Koo highlights. "I bring the students to our facility. We collect industrial



waste, so there are a lot of materials that can be re-used. We get the students to conceptualise art projects and pick their materials from the waste. This gives waste a new lease of life. And for all our projects, we don't spend a single cent!"

And happily, the students get to unleash their creativity too, as they experiment with using waste, learn about the value of resources and become more conscious of the waste management issues faced by Singapore. Wah & Hua also shares its knowledge of waste management, recycling tips and industrial knowledge with the schools.

"It's for a cause," summarises Koo. "Doing things for the environment is not about profit. A lot of people think that we need to do very big projects to help the environment. But each of us just needs to change a little habit, like recycling, and the environment can improve a lot. So you never know who you are going to benefit. Hopefully we can spread the message, and more people will join in."

WE LIVE IN AN INTER-DEPENDENT WORLD

Even before corporate social responsibility became a global business trend, real estate developer, City Developments Limited (CDL), had started on this track more than 15 years ago when it adopted environmental protection as a key pillar in its business strategy during the mid-1990s.

Says Esther An, General Manager of Corporate Affairs and Head of CSR (Corporate Social Responsibility), "We are all living in an inter-dependent world. One cannot draw a clear line between a business, the community and the environment. To achieve business excellence, we have to do well not only financially, but also in the community and the environmental aspects."

Outreach is a key element in CDL's environmental efforts to achieve buy-in from stakeholders and change behaviour. Since the 1990s, it has been actively working with ENV. Apart from proactively engaging its supply chain to track and improve environmental performance at work sites, CDL has also initiated environmental projects such as Project: Eco-Office in 2002 and One Degree Up Campaign in 2007 etc. Today, CDL's efforts range from engaging its tenants and homebuyers and NGOs to collaborating with schools under NEA's CASP programme. "Apart from funding their projects, we work closely with teachers and students on eco-projects that benefit the students and the environment," An recounts.

Over time, NEA and CDL have nurtured a strong working relationship. An puts it this way, "To us, NEA is not just a government agency; it's a partner. When we talk to our tenants on our green initiatives for example, with NEA's support, there's a stamp of trust and confidence that it is a public-private engagement. And we have worked together on many projects such as NEA's 10% Energy Challenge and Semakau Run. In the past, when we started out on Clean and Green Week, there wasn't so much public-private sector collaboration. But now, NEA is working to engage not only the corporate sector, but also the grassroots, the community centres, the academics and the schools. I'd say that NEA has done a very good job in encouraging and supporting the corporate sector and the community to drive and take on environmental initiatives."

The Word Is "Ownership"

The key to change is "ownership", when we see the environment as ours to safeguard and cherish. When we see it as a norm that every individual takes responsibility for ensuring long-term sustainability. NEA's 3P partnership efforts, networks and communication efforts have come a long way in nurturing this change in Singaporeans.

Enforcement and regulation can only go so far in achieving long-term environmental sustainability. The government cannot be everywhere to make sure we follow rules, and it would take too much resources. The question of keeping public areas clean is a pertinent example. Every year, the government spends millions on keeping our streets, roads and public areas free from litter. In 2006, the bill was a hefty $36 million. By 2010, the figure had risen to $50 million. The amount can be cut down considerably if we do not litter in the first place.

To push the transformation further, the agency has mounted schemes to facilitate a sense of ownership in the people, private and public sectors.

The 3P Partnership Fund uses co-funding to enable any organisation or company registered in Singapore to carry out environmental projects which engage schools, the private sector or the community-at-large. These projects are conceptualised and undertaken by the applicants.

The other strategy adopted by NEA is to nurture "change-makers". Through the programmes it runs, NEA identifies individuals who are passionate about the environment, and further equips them with skills and knowledge to do more. These individuals go on to spark change in their community, be it the workplace, home or school.

The Environment Champion programme, inaugurated in 2004, is a case in point. The programme consists of modules covering a wide range of topics. In 2005, the programme was extended from schools to private and public organisations. For the corporate environment champions, the modules impart know-how on planning and implementing environmental programmes, such as minimising wastage of paper, electricity and ensuring good hygiene in the work place. The champions also learn to conduct basic self-audits to gauge the performance of their pro-environment initiatives.

Forging ahead, NEA's Youth Environment Envoy programme was launched in 2005 to complement the Environment Champion movement. Youths who are keen to become leaders of change can apply to be trained as Youth Environment Envoys. To empower these youths, NEA holds workshops to enhance their public speaking, networking, facilitation and project management skills to prepare them for leadership. These young envoys will then move on to start projects of their own to contribute towards Singapore's environmental sustainability.

In the venture to support environmental ownership, NEA collaborates with NGOs which advocate the same causes. The support comes in different forms, such as funding, partnership and using NEA's communication platforms to promote certain initiatives. For example, the agency supports the Singapore Environment Council in its effort to champion a greener workplace and schools through its various awards and audit programmes. NEA also partners the Restroom Association of Singapore to encourage a clean toilet culture among toilet owners and users by endorsing the association's programmes.

The work to effect change is a never-ending task. Since its inception, the 3P Partnership Fund has grown to $1.5 million. Schools, community groups and NGOs have since created and launched diverse programmes as a result of the funding. Environment Champions now advocate in private and public organisations and in their community. Annual surveys on 3P Partnership programmes have shown positive reception towards NEA's initiatives and support. These are important steps towards a vision of Singapore in which each and every one of us claims ownership of our environment.



THIS IS MY PASSION

Pop by any green event in Ang Mo Kio or Hougang, and chances are you would run into this unassuming young man, working behind the scenes with volunteers.

Ong Chee Siang is a familiar face around his neighbourhood. He chairs the local youth club, champions environmental causes, mentors students, brainstorms and drives community projects with volunteers and helps out regularly at charitable organisations. This is a far cry from the wayward boy who once aspired to be the best gangster in Singapore. Two years in a boys' home changed his perspectives.

Giving back to the community was at the top of his mind when he left the boys' home in 1998, and volunteer work became a daily affair for him. Championing environmental causes entered the picture later, when a dengue outbreak took place in 2004. "The incident was so impactful that I told myself, I must do something for the environment. NEA was running dengue prevention programmes then, and I decided to take part," Ong recalls.

His interest in environmental issues grew as he became more involved in programmes run by NEA, and took part in the Youth Environment Envoy (YEE) programme launched in 2005. Soon, his interest infected the other members of the youth group he chaired, Ci Yuan Community Club (CC) Youth Club. "I thought to myself, why don't I focus on the environment, and use it to bond the community?" Ong shares.

Led by Ong, Ci Yuan CC Youth Club went on to push for a greener lifestyle in their community. A project in 2007 saw the club manning a station in Hougang to collect recyclables from residents, who welcomed the move. This convinced Chee Siang to persuade the public waste collector of the area to provide recycling bins for every two blocks in his neighbourhood. Seminars, workshops and activities relating to the environment were organised regularly as well.

The club was game to take on large-scale projects too, such as working with NEA in 2009 to organise the biggest youth-initiated environment project in Singapore – the National Youth EnvirOlympics. For a group of youths who had little experience in organising large-scale events, it was a challenging feat. "We had meetings till midnight with NEA staff in the Environment Building!" laughs Chee Siang. "The staff guided us step by step, and showed us how to manage the project. I was very motivated when I saw them staying up with us and putting in so much effort."

Launching project after project for the environment is only part of what Chee Siang does, who devotes a large part of his time working with youths as well. To reach out to them effectively, he forms interest groups within the youth club, each focusing on a different aspect of environmental sustainability such as recycling, community engagement and biodiversity. On his days off from work, he shuttles between several schools, mentoring students and helping them with their projects on the environment. He explains, "The youths I'm mentoring now, I hope they will become advocates to change the mentality and practices of Singaporeans. At the end of the day, the ground-up approach is the best way to change things."

Reflecting on his years of environmental advocacy, Chee Siang, who was awarded the EcoFriend award in 2008, says, "The YEE programme by NEA had a great impact on me. This belief in youths was very important, especially for me, who had come out of a boys' home. They didn't ask, why would I want a gangster on my committee? The people in 3P Partnership, they have touched and helped me. And because they have shaped me, now I can help shape the younger generation. This is my passion, to remember the people who moved me, and to move others."



"I find it really interesting when we go out to educate the public. Most people find it difficult to accept why teenagers will take up such activities. Sometimes, the public will avoid us because they think we are selling things, and they will go, no, no, no and walk off."
– Andy Wee, Litter-Free Ambassador

GOING ALL OUT AGAINST LITTER

Many teenagers would prefer spending their spare afternoons hanging out, learning the latest hip-hop moves or chasing the latest pop trends. But these youths from the Institute of Technical Education (ITE) College East would rather patrol the bus interchange and MRT stations near their school. Their mission: spread the litter-free message. If they spy a litterbug, they will politely ask the culprit to pick up his or her trash and bin it. Those who bin their trash get a cute token as encouragement.

These students are part of the Litter-Free Ambassador programme rolled out in 2010 by NEA. For years, NEA has been working with schools to run litter-free activities and nurturing students to keep the environment clean. Grooming student ambassadors to get out of their schools and into the community to combat littering marks a huge step forward. It is in fact, empowering students to take ownership of a clean environment and stand their ground against litterbugs.

At ITE College East, some 50 students enrolled in the programme in 2011. They are passionate about the environment, and want to do their bit in making Singapore cleaner. "I want to help the community and to set an example of not littering," says Jeffrey Din, one of the Litter-Free Ambassadors. "We go out in teams of 10 to 15 people, and patrol public areas. We spread the message, bin your litter, so that our environment can be clean and green."

Telling litter-bugs what to do, however, is not exactly a pleasant task. But these students are all up for it. Andy Wee enthuses, "I find it really interesting when we go out to educate the public. Most people find it difficult to accept why teenagers will take up such activities. Sometimes, the public will avoid us because they think we are selling things, and they will go, no, no, no and walk off." His fellow ambassador, Keith Kwa, is upbeat about the patrols, "Even if the litterbugs scold us, we just smile! At least I know what I'm doing is right. If they refuse to pick it, we pick up their garbage and throw it for them."

Still, to many youths, the idea comes across as odd. After all, much of our society is not used to taking a strong, public stance against littering. Keith shares, "Some of our friends would ask, 'Why are you doing this?' But after a while, some of them would join me, and slowly they get the hang of it."

Things are changing, slowly but surely. "We also patrol in the school," says Gay Yee Ying. "So we can see a small difference. Sometimes we see a student walking past litter, and stopping to pick it up and throw it into a bin." Keith agrees, "We know a small group has changed. We do see that around the school, and it is a form of motivation, when I see someone picking litter and binning it!"

Changing societal norms take time, and Singapore still has some way to go before littering becomes a true social no-no. These young litter-free ambassadors, however, are optimistic about their mission. Andy sums it up philosophically, "What we say, they do hear. But we don't really know if they take it to heart. But I think it's a matter of habit. If you keep doing something, other people will follow you. It will take time, it will be slow. Now, we are a minority trying to influence the majority, so it will take time for everyone to get the message."

RECOGNISING CHANGE-MAKERS

More and more, individuals from different walks of life are stepping up to contribute to our environment. These people put in time and effort, without any personal gain, to help bring about a clean and green Singapore. They touch the lives of many others, influencing people around them to care more for the environment.

Every bit of contribution makes a difference. Collectively, the efforts of such change-makers contribute to our long-term sustainability. To recognise these individuals, NEA launched the EcoFriend Award in 2007.

The award is given to those who have actively played a role in protecting, promoting and enhancing our environment. Recipients may come from the public and private sectors, NGOs, grassroots organisations and educational institutions. With the award, NEA hopes that EcoFriends are motivated to carry on their good work, and that more people will be inspired to emulate them.






the Environment
is ours



NEA exerts itself to strengthen the execution of MEWR's policies on the ground. It has to ensure that operationally, there are resources that can support and enhance the ministry's strategies.

The environment draws no borders.

Rapid urbanisation around the world, growing energy consumption, globalisation and the cumulative impact of over two centuries of industrialisation. These bring about consequences that are faced by the entire world, such as climate change, diminishing resources, trans-boundary pollution and epidemics. Given the scale of these issues, and how interconnected the world is today, the way every country manages its environment has the potential to impact other regions as well.

On top of that, what seems to be an environmental issue on the surface actually cuts across many other spheres of interest. Take, for instance, energy conservation. On a nation-wide scale, energy efficiency encompasses many aspects, from infrastructure and work processes to consumer products and mindsets of Singaporeans. Singapore also needs to take note of the diminishing supply of resources worldwide for energy production, explore new technologies developed elsewhere and adapt them to its use.

Singapore is in a particularly vulnerable position with its small land area, high population density and status as a travel hub. Yet it has to uphold its environmental performance and ensure that the expectations of its people are met continually.

Tackling such cross-cutting issues requires a whole-of-government approach. Environmental management thus has to be a key thrust in governance, and strategised in alignment with national development.

NEA exerts itself to strengthen the execution of MEWR's policies on the ground. It has to ensure that operationally, there are resources that can support and enhance the ministry's strategies.

NEA addresses this on various fronts. First it actively identifies emerging environmental challenges and invests in building capacity to meet them, both within NEA and the environment industry. Next, it collaborates with other government agencies to develop holistic strategies. Lastly, it uses various means to promote Singapore's expertise in environmental management on the international stage. This enables Singapore to contribute its best practices and knowledge to the global discourse on the environment.




Maybank
HSBC
2.7 M

Collaboration Is The New Normal

Close collaboration among government agencies to tackle environmental issues is more crucial than ever before. NEA has always been working with other government agencies on issues ranging from land-use planning and emergency responses to environmental incidents to vector surveillance and food safety. However, this is usually at an operational level to smoothen out processes and implementation. What is required today goes further than that. Agencies need to come together to jointly develop and take part in realising a national vision for sustainability, from tackling climate change and promoting energy efficiency to pollution control.

BANDING TOGETHER AGAINST CLIMATE CHANGE

Battling climate change requires a concerted and co-ordinated global effort due to the scale of its potential impact, such as rise in sea levels, drastic changes in weather patterns, increased spread of vector-borne diseases and damage to ecological systems. In the scientific community, it is widely held that the build-up of greenhouse gases in the atmosphere, predominantly carbon dioxide, is the main cause for global warming which results in climate change. In this connection, large amounts of greenhouse gases are produced by human activities such as burning of fossil fuels to produce energy.

While climate change is a global challenge, Singapore remains committed to doing our part to tackle this issue. Singapore ratified the United Nations Framework Convention on Climate Change in 1997. In 2006, Singapore acceded to the Kyoto Protocol. The Inter-Ministerial Committee on Climate Change (IMCCC) was established in 2007 to coordinate inter-agency efforts on climate change. In 2010, Singapore pledged to reduce greenhouse gas emissions by 7% to 11% below business-as-usual levels in 2020. If a legally-binding agreement is reached, in which all countries in the world implement their commitments in good faith, Singapore has further committed to reduce its emissions by 16% below business-as-usual levels in 2020.

Under IMCCC, there are three Working Groups. The International Negotiations Working Group will work out Singapore's strategy in the international climate change negotiations under the UN Framework Convention on Climate Change. The Long Term Emissions and Mitigation Working Group will study how Singapore can stabilise its long-term emissions. The Resilience Working Group will study Singapore's vulnerability to the adverse effects of climate change and develop long-term plans to ensure that Singapore is able to cope with the phenomenon.

NEA is a member of all the three Working Groups and supports the whole-of-government approach in addressing climate change. For example, the climate change vulnerability study commissioned by NEA in 2007 was the first nationwide study to look into the potential long-term impact of climate change on Singapore. The two-year vulnerability study investigated the effects of sea level rise and changes in meteorological patterns such as temperature and rainfall. The findings support the work of the Resilience Working Group. NEA will commission future studies to update the findings as climate science and models improve over time.

"It is inevitable that as our knowledge base grows, new questions will arise and old assumptions will be overturned. We must have a constant sense of inquiry – sharing timely and relevant insights across disciplines, striving to prepare early against uncertainties."

— Desmond Kuek, Permanent Secretary, MEWR, at the launch of the Climate Science Experts Network on 3 February 2012

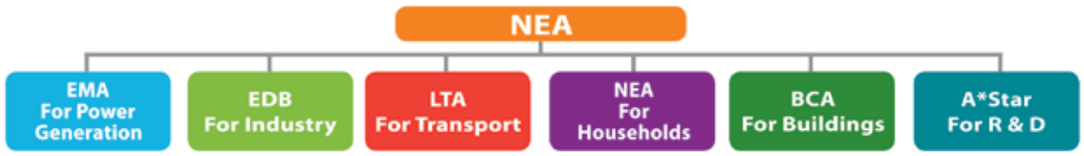








From left: Raising awareness of climate change during events such as this training workshop in 2007; In 2008, NEA jointly organised an exhibition on climate change with the Singapore Science Centre and the Shell Companies in Singapore; The multi-agency Energy Efficiency National Partnership was launched in 2010.



Energy Efficiency Programme Office (E2PO)

SAVING ON WATTS

Like many developed countries, Singapore relies on the burning of fossil fuels to meet our energy needs. As our population grows and economy develops, our energy demand will increase as well. In the bid to fight climate change, cutting down on energy use is critical.

This was why the Energy Efficiency Programme Office (E2PO) was formed in 2007 to increase energy efficiency in Singapore so as to reduce greenhouse emissions. Co-led by NEA and the Energy Market Authority (EMA), E2PO comprises several other government agencies as well. The programme office promotes and facilitates the adoption of energy efficiency in Singapore. This is done through four strategic thrusts — stimulating demand for energy efficiency, developing human and institutional capabilities in this area, promoting emerging energy-efficient technologies and innovation, and profiling and highlighting energy efficiency internationally through platforms such as the Singapore International Energy Week.

In Singapore, the key sectors contributing to energy consumption can be broadly classified as power generation, transport, industry, households and buildings. NEA actively promotes energy efficiency in households, industrial facilities as well as public sector buildings. In true 3P partnership spirit, NEA reaches out to government agencies, private companies, schools and grassroots to spread the energy-saving message to households through the 10% Energy Challenge. Launched in 2008, the challenge encourages Singaporeans to cut their energy use by 10% or more through the adoption of energy-saving habits.

Another key programme is the Energy Efficiency National Partnership (EENP), launched in April 2010 with 49 founding members. Jointly spearheaded by NEA, EMA and EDB, the programme reaches out to industry stakeholders, urging them to adopt an energy management system and measures to improve energy efficiency. To help the industry develop such practices, an EENP learning network has been developed, providing opportunities for the industry to learn about energy efficiency ideas, technologies, practices, standards and case studies through platforms such as conferences, workshops and seminars.





Top: The inaugural 10% Energy Challenge exhibition at Suntec Convention Centre, 2008.
Bottom: The Energy SAVE roadshow at Tampines, jointly organised by NEA, HDB and EMA in 2009.

In April 2012, a new Energy Conservation bill was passed in parliament. The resulting Energy Conservation Act will consolidate all energy efficiency-related legislation currently found in different acts. From 2013 onwards, the new act requires all large energy users in the industry and transport sectors consuming more than 15 gigawatt-hours of energy equivalent per year to appoint an energy manager. These companies also have to monitor and report energy use and greenhouse gas emissions, as well as submit energy efficiency improvement plans.

LABELLING FOR EFFICIENCY

It may be the cheapest refrigerator in the store, but if it comes with only two ticks on the energy label, it can easily cost you $65 more in electricity bills every year. That is, if you compare it to a refrigerator with four ticks on the label.

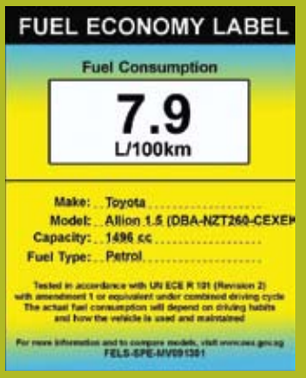
Since 1 January 2008, all household air-conditioners and refrigerators sold in Singapore are required by NEA to have a label indicating their energy efficiency. By 2009, clothes dryers were added to the list too. These three household appliances are among the most energy-consuming appliances in a home.

Energy usage is usually a hidden cost when it comes to purchasing electrical appliances. The more inefficient an appliance, the more costly it actually is in the long run. By employing such a labelling scheme, called the Mandatory Energy Labelling Scheme, NEA makes energy conservation relevant to the average consumer. The message – reduce energy use, reduce utility bills. With the labels on these big-ticket items, consumers can now easily decide if the more costly but energy-efficient product beats the cheap inefficient appliance. This move also gives greater impetus to manufacturers to design better, more efficient goods.

To help consumers avoid high energy consumption and energy costs brought about by inefficient appliances, NEA introduced the Minimum Energy Performance Standards (MEPS) in September 2011 for household air-conditioners and refrigerators. MEPS removes models that fall short of minimum specified energy performance levels from the market. At the same time, consumers benefit from cost savings while suppliers are encouraged to import more energy-efficient appliances as technology improves.







A NEW WAY TO LIVE AND WORK

Just imagine. A home in an estate where energy-efficient features are integrated with the infrastructure. Or a workplace in an eco-business park with clean technology. This may very well be the norm in the near future.

Punggol Eco-Town and CleanTech Park are two flagship projects in Singapore which encapsulate the notion of sustainable development. Both projects are made possible thanks to close inter-agency collaboration and collective thinking.

Solar panels to power common area lighting and lifts, roof-top greenery, dedicated recycling chutes and a system to collect rainwater. These eco-friendly features can be found in Punggol Eco-Town, at Treelodge@Punggol which was developed in 2010. The town also serves as a test-bed for new environmental technologies such as integrated energy systems with a lower carbon footprint.

This project is driven by HDB, with the aid of working committees comprising representatives from URA, NParks and EDB, just to name a few.

NEA enters the picture when it comes to urban solutions and engaging residents to play a role in environmental protection. The agency contributes its experience and technical knowledge, and coordinates solutions for issues such as 3P partnership efforts, recycling and waste management. For instance, in the new public waste collection contracts for the eco-town, NEA requires the contractors to also collect recyclables from dedicated chutes.

On the industrial side, Singapore's first eco-business park, CleanTech Park, is another story in point. Companies that embrace environmental sustainability and "green" corporate social responsibility are welcome to set up here. Besides, the park is also dedicated to nurturing the growth of research and development relating to clean technology. In addition, this project contributes to Singapore's goal to be a "living laboratory" where companies can develop, test-bed and commercialise green and urban solutions.

The 50-ha park, which master plan was unveiled in 2010, is home to a host of eco-friendly features – efficient building design, a fuel cell plant for renewable energy, solar panel installations, as well as eco-concrete for roads and recycled construction materials for its buildings. Like Punggol Eco-Town, CleanTech Park is the result of several agencies working together. The project is jointly driven by JTC and EDB. Agencies such as NEA are consulted for their respective expertise.

A key thrust in CleanTech Park is to go beyond the provision of a sustainable shell. Tenants are also required to set up sustainable laboratories and offices, and incorporate green work practices on an ongoing basis. With NEA's support, JTC engaged the International Institute for Sustainable Laboratories, a non-profit organisation, to draw up a guidebook for tenants on good design practices. In the guidebook, specific strategies to achieve energy efficiency, as well as safe, healthy and eco-friendly workspaces, are laid out. These can help tenants to save on their energy consumption by as much as 50%. The development of the guide was funded largely by NEA under its Design for Efficiency Scheme.

Between eco-friendly homes and clean technology parks, sustainable development is now shown to be achievable, thanks to the combined efforts of public and private organisations in Singapore.







Top: Treelodge@Punggol is HDB's first eco-precinct.
Bottom from left: Energy-efficient lighting is used in Punggol Eco-Town; Chutes dedicated to recyclable waste are found in certain HDB estates; Artist's impression of CleanTech Park.
Opposite: A vision of what CleanTech Park will be like in the future.





The 50-ha business park (below), which master plan was unveiled in 2010, is home to a host of eco-friendly features – efficient building design, a fuel cell plant for renewable energy, solar panel installations, as well as eco-concrete for roads and recycled construction materials for its buildings.



Building Capacity

"The establishment of the SEI (Singapore Environment Institute) *is significant on two counts. First, it is a major undertaking by Singapore to help transform our environmental workforce into a knowledge driven and skill intensive one. Second, it will also add to our overall efforts in positioning Singapore as a leading hub in environment management, knowledge sharing and technology transfer, thereby contributing to capacity building in the developing world."*

— Lim Swee Say, Minister for the Environment (2002 - 2004), at the opening of Singapore Environment Institute on 18 February 2003



SHARING SKILLS, KNOWLEDGE AND EXPERTISE

What was groundbreaking when the Singapore Environment Institute (SEI) was established under NEA in 2003 was this – it was no longer solely about meeting technical competencies internally. Instead, the full-fledged institute was to play a pivotal role in shaping Singapore's environmental future, by building capacities both within NEA and in the industry. Its work was to help develop a knowledge-driven and skill-intensive workforce, enhance the country's readiness to meet future environmental challenges and contribute to Singapore's position as a leading environmental hub in the region.

The new institute had to identify existing gaps in environment training and education in Singapore and explore ways to plug these gaps by tapping on NEA's wealth of expertise and experience. SEI also began forming strategic partnerships with other institutions and professional bodies to develop programmes which required a different set of competencies from those found in NEA. Within its first year of establishment, SEI had rolled out more than 150 courses, covering both technical operational issues and higher-level topics such as urban environmental management. Today, SEI continues to live up to its mission by setting trends in environmental education, nurturing and incubating the environment industry and integrating different fields and spheres of knowledge through strategic partnerships.

SETTING TRENDS FOR ENVIRONMENTAL EDUCATION

SEI scans the horizons constantly to make sure that it continues to stay ahead. It identifies new and emerging areas where skills and knowledge may be lacking and spearheads new initiatives.

This is reflected in the evolution of SEI's programmes over the years. For instance, in 2003, the institute launched a professional speaker series where experts from the industry were invited to impart their insights and knowledge. In 2004, it collaborated with the meteorological service arm of NEA to introduce weather-related courses on topics including smoke haze and lightning risks. To bring environmental education to life, the institute launched the Programme for Environmental Experiential Learning (PEEL) in 2007. PEEL consists of a series of learning trails showcasing environmental facilities in Singapore and providing behind-the-scenes information on how Singapore manages and maintains a clean environment.

As energy concerns became more paramount, SEI collaborated with the Institute of Energy Economics, Japan, in 2007 and 2009 to co-organise a workshop on renewable energy in Tokyo so that local environmental professionals could learn from Japan's experience in this area. In 2010, SEI expanded the workshop to cover such topics with a new partner, the Energy Conservation Center, Japan.

Incubating capacity in the environmental sector is crucial if Singapore is to be equipped with the necessary resources to adapt and cope with emerging environmental challenges. Through the Singapore Environment Institute and a bold industrial development agenda, NEA contributes to the growth of capacity in this sector, enhancing its pool of competencies, knowledge, skills and trained manpower.

With the recognition that environmental issues are more and more cross-cutting in nature, SEI launched a new platform in 2011, where global thought leaders in the fields of design, technology and environmental innovation are invited to give talks on solutions and ideas for environmental sustainability.

INCUBATING GROWTH

To fulfill its role, SEI nurtures and incubates a skilled workforce for the industry through its training programmes. The wide range of programmes it has launched shows the extent to which SEI has shaped the industry.

In 2003, SEI collaborated with Work Development Agency and the Singapore Pest Management Association to set up the first Approved Industry Assessment Centre to standardise and formalise the level of skills across the pest management industry and to upgrade the capacities of workers. SEI was also the Assessment Centre for Basic Food Hygiene.

Over the years, SEI has engendered and upgraded diverse competencies in the industry, from recycling and resource recovery to energy efficiency. Working with different partners, SEI has launched numerous certification courses, such as the Singapore Certified Energy







From top: Grassroot leaders taking part in a PEEL trail; PEEL Clean Water Trail in action; Participants visiting a waste management plant during the PEEL Clean Land Trail.

"In the past, we were a third-world country. We could go around and see what other developed countries were doing to manage their environment, and learn from them. Now that we are a first world country, we are facing the same issues as other developed countries – water, environment, waste, energy and so on. The situation nowadays is complex; it is no longer a linear, consequential kind of arrangement. The environment has become a global rather than localised issue. We can no longer have someone else to tell us what to do. In the past 40 years, we have done very well; we have accumulated our processes, our systems and our framework of environmental management. We have enough inner strength. Therefore we need to rely on the knowledge we have, think critically, leverage on sense-making in the environment. We must sense our surroundings, adopt new mindsets and be in tune with the development of Singapore."

— Ong Eng Kian, Director, Singapore Environment Institute, NEA



Manager programme to help engineering professionals build up their capabilities in emerging management and energy engineering. Other initiatives include courses on pest management, food handling and environmental control and even workshops to educate the business community on environmentally sustainable work practices.

INTEGRATING DIFFERENT FIELDS

Synergy is the word when it comes to environmental education. Every institution has its own set of expert knowledge and skills. To have a comprehensive training framework for both NEA and the industry, SEI has been leveraging on the knowledge base of local and international agencies and organisations through training partnerships. This approach also ensures that SEI can integrate different sources of knowledge and contribute to the renewal of the industry's skills.

Since its beginnings, SEI has signed a considerable number of Memorandums of Understanding (MOU) with different groups ranging from local educational institutes, government agencies and international organisations. The Sydney Institute, Institute of Technical Education and the Civil Service College are just some of its partners. A noteworthy MOU on training was inked in 2005 with NParks, PUB, National University of Singapore, Nanyang Technological University, Ngee Ann Polytechnic and Singapore Polytechnic. The programme, known as Singapore Environmental Training Programme (SETP), brings the different agencies together to promote Singapore's efforts through training and technology transfer to the region and beyond.

Apart from such collaboration, SEI also actively promotes the sharing of knowledge regionally and internationally through workshops, seminars and international programmes organised in partnership with various international organisations and agencies such as United Nations Development Programme (UNDP), Asian Development Bank (ADB), Japan International Cooperation Agency (JICA), Korea International Cooperation Agency (KOICA), Hanns Seidel Foundation (HSF), Deutsche Gesellschaft für Internationale Zusammenarbeit GmbH (GIZ) and so on. SEI has hosted many official visits of foreign dignitaries as well. Through these visits, the institute exchanges points of view and information on environmental management with officials from other parts of the world.













Clockwise from top: Signing of SETP in 2005; Signing of MOU between ITE and NEA in 2010; NEA CEO, Andrew Tan (right) with Paul Bennett (centre), in a 2011 lecture on design thinking for the environment, moderated by Professor Kristin Wood from Singapore University of Technology and Design (left); Visionary environmental thinker, Michael Braungart, at a lecture organised by SEI and Centre for Liveable Cities in 2010; Energy study workshop in Japan organised by SEI for industry executives in 2010; Learning about innovative environmental solutions from Kitakyushu in 2012.

More than ever, SEI is focused on partnering environmental experts, academia and governments to develop programmes that impart insights on critical issues. Distinguished speakers and renowned global thought leaders are also invited regularly to share their view on pertinent topics. These programmes are set up to explore relationships between discrete fields such as technology, design thinking and the environment, so as to catalyse innovation.

RISING UP TO NEW CHALLENGES

As environmental issues become progressively complex, environmental education and inquiry need to evolve so as to keep pace with these issues. In the case of SEI, the additional challenge is to ensure that NEA and the industry in Singapore have the necessary competencies to excel in this new playing field.

"When it comes to addressing future needs and challenges, we have to first assess if we have the competencies to meet them. If not, we have to start building the capacity now. We are here basically to fill in gaps that the industry is not able to address now, and get things started. The necessary expertise may not reside with NEA, so one of our key strategic thrusts is to partner centres of excellence outside of NEA. This way, we can shorten our learning curve by adapting their solutions to our context," shares Ong Eng Kian, Director of SEI.

This approach has been consistently employed by SEI. For example, when the government began promoting the use of compressed natural gas (CNG) vehicles in the early 2000s, SEI assessed that the local industry was in its infancy and lacked the competencies to support the operation and maintenance of CNG vehicles. The institute invited members of the industry to undertake a trip to Australia, where the use of CNG was well-established, to study the operations and management of CNG refilling stations and equipment. One of the industry partners who went on the trip eventually set up the first CNG station in Singapore. Another example is SEI's joint seminar on waste management and recycling with Kitakyushu City, one of the most advanced cities in Japan in terms of pollution control, waste management and recycling. Representatives from both public and private sectors took part in the seminar held in 2012 and benefitted from Kitakyushu's sharing.

"Moving forward, it is no longer basic technical training that we are talking about. We want to promote thought leadership, and we have to look at issues that are nexus in nature, such as sustainable development and climate change," says Annie Tan, Deputy Director of the Environmental Learning Centre under SEI.

More than ever, SEI is focused on partnering environmental experts, academia and governments to develop programmes that impart insights on critical issues. Distinguished speakers and renowned global thought leaders are also invited regularly to share their views on pertinent topics. These programmes explore relationships between discrete fields such as technology, design thinking and the environment, so as to catalyse innovation.

SEI has also launched intensive efforts to document NEA's knowledge, practices, capabilities and expertise since 2011. This includes capturing the experiences and perspectives of pioneer environmental engineers and officers who have contributed to Singapore's environmental management. The undertaking allows SEI to distill essential knowledge and lessons learnt from Singapore's environmental journey and build a rich resource base that future NEA officers may benefit from. With these in place, SEI can also shape the direction of environmental inquiry, nurture thought leadership and find better solutions for the future.

Says Tan of her work, "Knowledge is growing at a fast speed. Issues are evolving at a fast pace. We have to keep at the forefront of issues. It's no longer enough to be relevant. We can become obsolete by being relevant only. So we have to be forward-looking, resourceful and responsive to constantly transform and meet the new demands. It is exciting, it's challenging. This is the environment we are in."



"The challenge is how we are going to chart our future. We need to scan the horizon and be constantly on the lookout, reach out to different partners, stakeholders, knowledge partners and institutions. There is no single outfit that addresses all the issues. Traditionally, we have been looking at operational issues. Now we are moving into broader contexts, going upstream to look at innovation and strategic issues. We are moving away from a technical training kind of institution to a knowledge-based institution."

— Annie Tan, Deputy Director, Environmental Learning Centre, SEI



DEVELOPING THE INDUSTRY

Not so long ago, parents would admonish kids who would rather play than study with the dire threat, "Do you want to be a road sweeper one day?"

For years, the environment industry has been seen as a low-skill, unsophisticated sector, despite how essential its services are to our daily lives. But this is changing as the industry undergoes a makeover, thanks to increasing global attention on environmental issues, the steady upgrading of the industry over time and the government's efforts in developing and promoting local environmental companies. More are also realising that a thriving environmental industry is a key component to sustainable urban solutions and systems.

Alongside a training regime, NEA also pursues an agenda to further develop the industry in different ways. The agency works closely with industry associations to develop different sectors in a systematic manner, upgrading the professionalism and image of the industry. In 2009, this strategy received a boost when NEA set up a dedicated office to focus on industry development and promotion.

To these ends, NEA engages companies actively for feedback on their needs. It collaborates with multiple government agencies to drive home the importance of the environmental sector. For instance, one of the challenges faced by the waste management industry is the access to industrial land. Whether land is allocated to an industry is generally based on its economic yield. The waste management sector, though critical to a sustainable city, does not present high economic returns. Thus, when the land needs for such companies are juxtaposed against more lucrative sectors, priority is often given to the user with more economic potential. To address this issue, NEA initiated the formation of a committee in 2011 with agencies such as URA, JTC and EDB to evaluate the allocation of land to waste management and recycling companies.

NEA has also implemented steps to liberalise aspects of its functions and encourage the development of expertise outside the government, as in the case of Waste-to-Energy (WTE) incineration plants. At the turn of the 21st century, the government felt that the operation of WTE incineration plants could become more efficient if it was privatised. In 2001, the government called for an open tender for a developer to design, build, own and operate a WTE incineration plant. However, given the high capital outlay and risks of uncertain profit, no developers came forth.

In 2004, NEA made another call for developers to undertake the project, this time with revised conditions and attractive conditions in place to encourage developers to come forth. This time, a number of proposals came in. The tender was eventually awarded to Keppel Seghers Engineering Pte Ltd for its competitive bid. Singapore's first private incineration plant was thus built and operations began in 2009. Today, following the successful divestment of Senoko Incineration Plant in 2008, about 40% of the incineration capacity in Singapore is now managed by the private sector.

NEA also administers a number of funds to develop and upgrade the industry. The Environment Technology Research Programme, for example, provides assistance to organisations in undertaking research and development projects relating to resource recovery from waste. The Design for Efficiency Scheme encourages investors to integrate energy and resource efficiency in the design stage of their developments while the Innovation for Environmental Sustainability Fund supports projects that use technology to address Singapore's environmental challenges.

These efforts taken by NEA have set in motion a long-term drive to vitalise and energise Singapore's environment industry.



"My vision for NEA is transform the entire environment industry, as it is perceived now to be dirty and low-value add, to first-world standards. And our industry can help position Singapore as a hub for environmental solutions in the region and be a model for others. It is a really very new area — developing the clean environment industry. Singapore is a city-state; there is a limit to how companies can grow here. I hope to see more Singapore-based companies being groomed as world champions and compete with other companies, scale greater heights and bringing back returns to Singapore's economy. By having professional and world-standard companies, it will also facilitate the work of NEA in creating a liveable and sustainable environment for Singapore citizens. The other part of the equation is that people must come in to complement it. So I would like citizens to come forward and call the environment their home, and treat the environment as they treat their home. If we can see all these things, we can see more successful environmental management in Singapore, and that will be a success formula to showcase to other parts of the world."

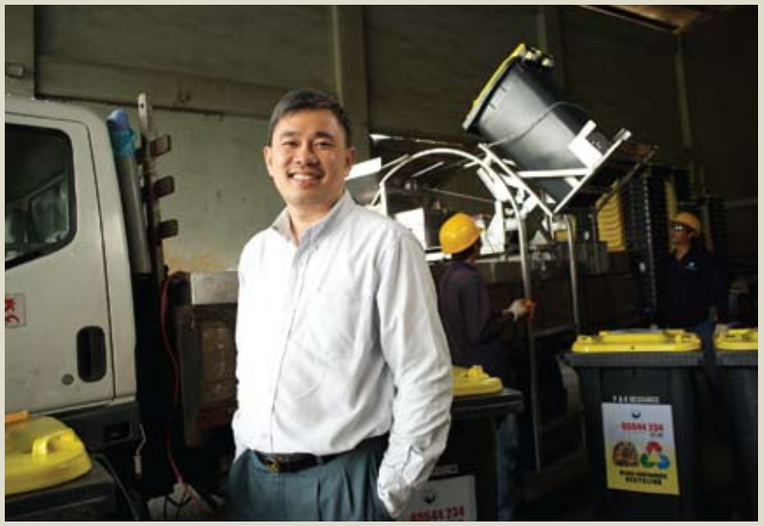
— Dalson Chung, Director, Industry Development & Promotion Office, NEA





"To produce 1 tonne of glass, 1.2 tonnes of raw materials are needed. But a tonne of waste glass yields a tonne of glass. And using recycled glass saves energy and raw materials, while reducing the need for mining, which impacts the environment."

– Patrick Ng, Executive Director, P & R Resource Management



ONE MAN'S WASTE IS ANOTHER MAN'S RESOURCE

You may have seen this 30-second operation near a coffee shop in your neighbourhood – two workers pick up a case of some 200 used glass bottles, dump them at one go into a container, and 15 seconds of noise later, the work is complete.

The humble container, a glass-crushing device, is playing a key role in increasing the rate of glass recycling in Singapore, thanks to the efforts of a local company and Singapore Polytechnic, as well as the support of NEA's Innovation for Environmental Sustainability (IES) Fund.

First, though, why the fuss about glass? Most of us don't think twice about chucking that jam jar or drinks bottle down the chute. But, unlike most waste we generate, such as food, paper and plastics, glass cannot be destroyed during incineration. What goes in as glass comes out as glass. In 2010 alone, 60,700 tonnes of glass were sent to Semakau Landfill, shortening the landfill's lifespan.

Yet the rate of glass recycling in Singapore is still rather low, at only 24% in 2010. The problem lies in the fact that the material is cheap and collection is arduous. Unlike metals, the incentive to recover waste glass on a large scale is not as financially rewarding. Getting households, commercial and industrial premises to put aside used glass for recycling is a challenge. The most immediate hurdle? Waste glass collection is a noisy, back-breaking and time-consuming process that is potentially unprofitable if inefficient.

But to the intrepid businessmen at P & R Resources, waste glass is a resource. Having been in the line of glass manufacturing support for many years, they are convinced that glass recycling makes good business and environmental sense. Teaming up with Singapore Polytechnic, the company convinced NEA to fund its research to tackle the problem of collecting waste glass, starting with commercial premises. Says Patrick Ng, Executive Director of P & R Resources, "As a commercial entity, we don't have strong research and development skills. As a statutory board, NEA is able to fund the research and development. And Singapore Polytechnic provided the skills and know-how." Supported with a grant from the Innovation for Environmental Sustainability Fund, the team began researching in 2009.

The students and educators at Singapore Polytechnic soon identified key problems with the process. "The workers had to lift a heavy load of about 70 kg above their heads and dump it into a container. It's not only laborious, it is also very noisy. After collection, at the plant, the glass would be lifted by a forklift and dropped to crush the glass, so as to facilitate the transport to Malaysia, where the glass is processed. So we thought of crushing it on-site. This also means that fuel can be saved as more glass can be contained. We also thought that if the glass could be stored vertically in containers, it would be more efficient," shares Dr Ng Khee Yang, the project's key consultant in Singapore Polytechnic.

Crushing waste glass on-site proved to be a challenge. The crushed glass had to be of the right size. Too fine, and it would be unusable, and too big, it would have to be crushed again. Noise and safety had to be considered too – the amount of noise caused must not be a public disturbance, and the mechanism must not release glass shards or dust into the surroundings.

The polytechnic students eventually came up with the winning design, which used several rotating hammers with joints to crush the bottles as they were loaded into a container. The joints allowed whole cases of bottles to be dumped in without jamming up the device. "The crushing mechanism is very important because it is the bottleneck of the operation," says Dr Ng. "So once we were able to clear this crushing mechanism, we went on to the problem of lifting the case to the vehicle." Various solutions were proposed for lifting, including a vacuum and forklift system, before a commercially feasible procedure was developed. After three years of research and development, the team finally came up with a safe system that shaved the operation to a mere 30 seconds to collect and crush 75 kg of glass.

A sound procedure alone is never enough. To their dismay, P & R often found food waste, plastic bags and such in the containers meant for waste glass. "This created a lot of problems for us, especially since we are not a general waste collector!" Patrick Ng points out, "Here NEA plays a key role. We approached NEA to educate the food and beverage outlets. At first, not a lot of the outlets were receptive to recycling. But NEA is more persuasive as a government agency, and it helped us solve these teething problems and encourage more businesses to recycle their glass waste." With an efficient system in place and NEA's support, P & R Resources is the biggest player in Singapore's glass recycling today.



"We are close to the region, we have a very good legislative framework in Singapore and doing business here is easy. So we want to attract these companies to position here and partner our companies to step out to other countries. NEA plays a role in attracting some of these big companies. When we have work visits to other countries, NEA comes along. It's always helpful when you have government agencies with you, because you are seen as one Singapore."

– Guah Eng Hock, Chairman, Waste Management and Recycling Association of Singapore

A PARTNERSHIP TO MANAGE WASTE

"In whatever we do, there is the presence of MEWR and NEA. There is a lot of collaboration," says Guah Eng Hock, Chairman of the Waste Management and Recycling Association of Singapore (WMRAS). As far as industry associations go, WMRAS is a new kid on the block, having been set up only in 2001. And that, too, came about at the encouragement of former Minister for the Environment Lim Swee Say.

The waste management and recycling industry began flourishing in earnest after the government began to liberalise the sector in the late 1990s. This growth is reflected in WMRAS' expanding membership over the years, from a mere 20 or so companies to over 100 in 2012.

Since its establishment, WMRAS has matured together with NEA, forging close ties as they work jointly on several fronts. Once a year, a dialogue between WMRAS and the CEO of NEA is held, during which the public and private sectors share thoughts and perspectives. This closeness is crucial to the industry, as Guah explains, "NEA is the agency for public waste collection contracts. It plays a very big role in policy-setting – in terms of allocation of land for waste management companies, in terms of tender policy, whether they go for the cheapest or best-sourcing. These are issues that affect our members and the industry. We constantly engage NEA in these issues, which are brought to the table and openly discussed, and it has led to revisions in tender policies for the betterment of the industry."

Ironing out operational matters is not the only motivation for this close partnership. Both WMRAS and NEA share similar objectives – to develop and grow the industry. "Basically, we want to increase the competencies and skills of the workers," shares Guah. "There is a whole Workforce Skills Qualifications training road map for the industry, covering several levels such as safety, operational familiarity with equipment and identifying hazardous materials. The development of the courses is led by the Workforce Development Agency, with inputs from WMRAS and NEA."

What interest both parties more, however, is putting Singapore's waste management and recycling solutions on the map. The reason for this is simple. "If we look at the Singapore market, it is still a very limited market. So if we want the industry to grow, we need to sell our services and technology overseas. For example, there are companies in Europe with well-tested and developed models which cannot be imported wholesale into Asia. They need good local partners who understand the culture of the region," Guah elucidates. "We are close to the region, we have a very good legislative framework in Singapore and doing business here is easy. So we want to attract these companies to position here and partner our companies to step out to other countries. NEA plays a role in attracting some of these big companies. When we have work visits to other countries, NEA comes along. It's always helpful when you have government agencies with you because you are seen as one Singapore."

To put Singapore's urban solutions in the limelight, WMRAS has made several moves. Recognising the huge demand for waste management capabilities in Asia, WMRAS successfully courted the International Solid Waste Association (ISWA), a leading non-profit organisation promoting sustainable waste management, to open in Singapore a regional training centre for the industry. The MOU to effect this centre, the first of its kind outside Europe, was inked in 2010 between NEA, WMRAS and ISWA. Says Guah, "ISWA has a huge knowledge base. And our waste management industry is still at a very nascent stage, so we stand to benefit from this. Singapore can also help ISWA to reach out to Asia. And NEA has very good relationships with its counterparts in Southeast Asia, so it can interest more people in the centre's activities."

This synergy between NEA and WMRAS is further demonstrated in WMRAS' tradeshow for the industry, WasteMET Asia, which aims to be the leading show for waste management and recycling technologies. The show inaugurates in 2012, at the same time and venue NEA launches its flagship environmental event, CleanEnviro Summit Singapore.

Opposite: Material recycling facility operated by Veolia



Beyond Our Shores

Since Singapore's independence, it has transformed from a polluted, insanitary place to one of the most liveable cities in the world. The lessons and expertise we have built up give us a robust foundation to face future challenges and to contribute towards common ground globally to address the world's environmental issues.

Across the world, demand for environmental solutions is also growing as more countries realise the importance of a clean environment and invest in environmental management technology and systems. In 2004, the global environment market was estimated at US$548 billion and projected to grow to US$800 billion by 2015, according to a report commissioned by the United Kingdom government.[1] Asia is expected to lead the growth in terms of needs and expenditure.

Singapore can play a role in sharing our environmental knowledge and solutions with the rest of the world. In the past four decades, Singapore has built up a reputation for having a sustainable environment and nurtured networks with multiple governments to share environmental practices. Now is the time for Singapore to ride on the foundation that it has built and reach out to the world.

REACHING OUT

As a small city-state, reaching out to the world is a long-established strategy in Singapore's national development, from cultivating diplomatic relations to building our economy. Going forward, this strategy is a key thrust in our environmental management sector.

Since the formation of NEA, the agency has actively taken part in international environmental events such as workshops, conferences, trade shows and seminars. At such events, Singapore's track record in environmental management is presented to participants from other countries. Close collaborative relationships are also cultivated with international organisations such as WHO and the World Bank. NEA also established a business unit in 2010 to provide consultancy services on environmental management to other governments.



Opening of ISWA/WMRAS World Congress in 2008, officiated by Dr Yaacob Ibrahim.

"For many young Singaporeans, I think the environment has a certain appeal. They are now more aware, because of all the global issues centred on the environment. We want to give them a chance to do something here, and beyond Singapore, meaning what Singapore can do as a reference for other cities in the region. We are not saying that our solutions are the best, or necessarily at the forefront. But we have overcome our own challenges, and we are together with other cities, having to confront a common set of challenges going forward, so let's find ways to deal with that. If we give ourselves some reputation to uphold, it raises the bar and we have to work harder towards these goals. We also give Singaporeans at large a greater sense of pride in the country and a greater sense of ownership."

— Andrew Tan, CEO, NEA

1 UK Centre for Economic and Environmental Development. 2006. *Emerging Markets In the Environmental Industries Sector November 2006*. United Kingdom: Department of Trade and Industry.



More importantly, NEA has also started the bid to export Singapore's environmental solutions to the region and the world by promoting our local industry. In 2010, NEA formed the Clean Environment Industry Committee with EDB and other agencies. The objective was to transform Singapore into a vibrant global environment hub which will offer innovative cost-effective solutions to meet global and local needs.

Overseas mission trips are regularly organised by NEA to identify opportunities where Singapore-based environmental companies can be showcased. As the facilitator, NEA matches the needs of other countries with the solutions offered by Singapore-based companies. This work, combined with NEA's efforts to develop the local industry, has led to a growing number of Singapore-based companies gaining a foothold in overseas markets. One such example is a $1.7 billion project for solid waste management that Keppel Seghers secured in Qatar, given its track record in constructing Singapore's first private WTE incineration plant.



Prime Minister of the Kingdom of Thailand, Samak Sundaravej (centre), on a visit to Tiong Bahru Market in 2008.

BRINGING THE WORLD TO SINGAPORE

When the world beats a path to Singapore to look for solutions, we know that Singapore has arrived as an environmental hub.

NEA has been sharing knowledge and solutions with other countries via courses and seminars. NEA collaborates with foreign embassies, governments and international organisations to roll out training programmes that are relevant to the needs of these international participants. These programmes provide in-depth presentations on Singapore's environmental strategies to demonstrate linkages between nation-building, economic development and environmental sustainability, and also impart technical knowledge and expertise. Officials and participants come from a long list of countries, including Afghanistan, Bangladesh, India, the Republic of Korea, Thailand, Malaysia and Iran.

NEA has hosted and facilitated numerous regional and international events related to the environment since its formation. Forums, conventions, regional working group discussions and trade fairs are just some of them. The environment event to watch out for is the CleanEnviro Summit Singapore, held in July 2012. This event is the result of careful research to identify existing gaps in regional environmental shows, so as to come up with a distinctive and relevant event.

CleanEnviro Summit Singapore is poised to play a key role in promoting Singapore's standing as an environmental hub. The event is diverse in its offerings. Besides profiling the capabilities of local clean environment companies, the event also consists of a high level summit for leaders in the clean environment sector. These leaders come from different parts of the world, including Germany, China, the Middle East and the United Nations. A closed door roundtable for regulators and a trade show curated by WMRAS, WasteMET Asia, also form part of the event.

EXPORTING SINGAPORE'S DNA

From Qatar and Brunei to China and India, Singapore has been busy sharing its experience in environmental management with other countries on a much larger scale than ever. More and more, we are stepping out, and exporting our DNA in sustainable development to other countries. Singapore government agencies are actively providing insights and consultancy, working together with other governments on environmental projects, and even bringing our local companies overseas to showcase their competencies.

For example, sharing our experience in managing solid waste with senior government officials from Kerala, India, in 2010, and conducting a feasibility study on the cleaning up of Cooum River in Chennai during the same year. Or, supporting the successful bid of a home-grown company to design and build a waste management facility in Brunei.

The project that takes the cake however, is an exciting eco-city in Tianjin, China. Located on wasteland which had seen years of chemical pollution, the site comprises abandoned salt pans, a deserted beach and a huge waste water pond. Transforming this into a liveable, environmentally friendly and resource-conserving city is a visionary, ambitious undertaking by the Chinese and Singapore governments. When completed in the 2020s, the city will serve as a model for sustainable development. Multiple government agencies from Singapore are involved, such as the Ministry of National Development, Ministry of Trade and Industry and MEWR. Every agency chips in to provide its competencies, from urban master planning to public housing development and water management.

When it comes to environmental management, NEA is the primary consultant. The agency collaborates closely with the Chinese government, sharing its experience in diverse areas such as pollution control, solid waste management and development planning. The advice helps the city planners to put in place infrastructure and policies to meet key environmental performance indicators. Some of these indicators are ambient air quality, noise pollution control, carbon emission, domestic waste generation and overall recycling rate. NEA also provides inputs on guidelines for green construction and co-chairs a task force with the eco-city's Environment Bureau to explore the implemention of a pneumatic waste conveyance system in the city.

Our local environment industry is also active in imprinting its mark on the eco-city. For example, Alpha Biofuels signed a Memorandum of Understanding (MOU) with the Sino-Singapore Tianjin Eco-City Investment and Development Co., Ltd in 2011 to implement an organic waste gasification and biodiesel-from-used-cooking oil system. Another company, LHT Holdings, also signed an MOU to establish a wood waste recycling plant to manufacture eco-wood products. This will help developers to achieve green building standards through the use of recycled materials.

Through these exciting projects, Singapore is able to learn from other countries, build its capacities further, and contribute to a better environment elsewhere in the world.










a Future
we share

Transforming To Be Future-Ready

"I see NEA operating in a national as well as a broader, global context. And that's because of what Singapore is. We are a city situated in this wider landscape and whatever happens out there will impact Singapore. So there are always two dimensions to NEA's work. The first is the national dimension. And that is the core part of the work that we do, which is how to protect Singapore's environment. This task is something we have been doing fairly well since Singapore's independence. Much of the infrastructure and the policies have been put in place over the years. But as we go forward, there are many things we have to re-examine. In terms of whether the infrastructure is adequate and whether the policies are relevant to today's context. And I think developments in recent years have given greater impetus to the re-examination of many of these fundamentals. When you look at big issues like climate change, there's always a concern whether you are operating in the context of uncertainty. So what was perhaps done based on rule of thumb, common sense and the level of knowledge that was required in the early years of our development is no longer sufficient."
— Andrew Tan, CEO, NEA

A decade after its formation, NEA is now gearing up to meet the future — a future where environmental management is expected to be much more complex, more dynamic and fast-paced, where environmental issues are anticipated to be larger in scale, impact and reach.

To get to the starting line, NEA has to re-invigorate its organisational structure, culture and operations as well as its competencies and knowledge level. And these would just be the first steps.

Change is not easy. But over the decade, the agency has built up a firm foundation of organisation excellence, strong operational regimes, a wealth of expertise and a culture of passion and professional dedication. Its efforts have been well-recognised, as the numerous awards it has garnered under its belt since 2002 show – The Quality Management Standard, Singapore Quality Class, People Developer, Public Service Achievement Award, Singapore Service Class and Singapore Innovation Class, just to list some. This continual drive to be better is the cornerstone of NEA's metamorphosis today.



"Change is the only constant. NEA has to gear up to support change, because you can't use an existing structure to support something that is new. New skills, new roles, new resources have to be acquired so that we become relevant, responsive and resourceful. We are gearing ourselves to get to the starting line, so that we can execute our blueprint for the future. Such as connecting all our environmental sensors for air, water, weather, so that the public can benefit from early alerts. With the data, we can also run different simulation models for better situational awareness. The information can also be used by different groups, individuals, research institutes and businesses."

— S Satish Appoo, Director, Strategic Development and Transformation Office and Joint Operations & Planning, NEA



PEOPLE EXCELLENCE

An organisation is only as good as its people and every member of the staff in NEA plays a part in the agency's mandate to protect and sustain the environment. Forward-looking and excellent human resource management and development thus hold the key to the agency's transformation.

Work in NEA is diverse, challenging, and meaningful. NEA employees possess a strong sense of purpose because they know the impact of their work on the daily lives of Singaporeans. The agency strives to attract talent with a passion for the environment. Staff who join NEA with limited knowledge of environmental protection inevitably nurture their love for the environment over time. Progressive human resource policies and schemes such as learning and career development opportunities, varied work-life harmony initiatives and age-friendly employment practices are in place to ensure a conducive work environment.

Future-readiness calls for having the right people and work culture. Passion for the environment, commitment to the agency's vision and mission, and work ethics that resonate with NEA's – these are what will make the difference. In the next decade, the agency will focus on raising the skill levels and competencies of its workforce to better tackle the strategic challenges ahead. It will also focus on creating a first-class, sustainable workforce and a workplace where passionate environment advocates strive together in a dynamic, challenging field to serve Singapore.

WALKING THE TALK ON SUSTAINABILITY

Walk the talk, lead by example – this is all the more paramount as NEA gets down to executing its blueprint for long-term sustainability.

NEA actively encourages eco-friendly workplace practices and puts in place resource-efficient measures. For example, it installed systems to monitor and review its water and electricity consumption patterns and fixed flow regulators on the taps in all its premises including hawker centres. Since 2005, all its offices have been awarded the Eco-Office Label by the Singapore Environment Council for its environmentally friendly workplace measures. In 2010, all of NEA's premises were awarded the Water Efficient Building Label. In 2011, NEA completed a purpose-built meteorological service facility along Kim Chuan Road. The agency took care to equip the facility with green features such as solar panels and a rain harvesting system to irrigate the building's greenery.

Creating and maintaining a sustainable workplace needs continual perseverance, and NEA has stepped up efforts to consolidate and spearhead all sustainability initiatives since 2010.

OPERATIONAL EXCELLENCE

In a world where environmental issues and crises are getting more complex, becoming future-ready hinges on NEA's capacity for situational awareness and its ability to respond fast and effectively during environmental incidents and emergencies.

To intensify its ability to monitor the environment and respond to potential incidents, NEA has launched several exciting projects. These include piloting state-of-the-art sensors for more precise, continuous monitoring, constructing a new operations command centre to coordinate NEA's ground operations in all parts of Singapore and building up NEA's capacities in a myriad of fields such as nuclear science, food hygiene and climate science. These projects, when completed, will bring about a faster, nimbler, future-ready agency with 24/7 standby readiness, with a dedicated team of officers who will help monitor the environment.







A VISION FOR TECHNOLOGY

As environmental issues grow more complex, technology will take on an increasingly crucial role in providing solutions. Recognising that technological applications take time to incubate, NEA has embarked on envisioning exercises. During these exercises, it projects future challenges and defines the competencies required to address them. Next, the agency assesses if it has such competencies in place. Where gaps exist, NEA puts steps in place to bridge them, including identifying possible technologies that can strengthen NEA's future operational capabilities. Once these technologies are identified, the agency incubates them through various means.

For example, NEA invests heavily in budding environmental technology and funds relevant research and development projects. It partners other educational and research institutes to explore solutions and encourages the industry to practise innovation. Since 2009, NEA has also been focusing on addressing its long-term technological needs and coordinating more exciting projects for the future.





NEA's Contact Centre, one of the busiest of such services in the government, is an important channel for the public. It operates round the clock, handling more than 600 calls a day, up from an average of 400 calls in 2002. During environmental and public health incidents such as dengue outbreaks, the calls can spike to more than 800 a day.



NEA's smartphone application, myENV, was launched in 2011.

PROVIDING CHANNELS

All you need is one inconsiderate person to provide a safe breeding spot for *Aedes aegypti*, and dengue can break out in the neighbourhood.

The action of one has the potential to impact many others in today's environment. In NEA's bid to build a more environmentally sustainable future for Singaporeans, the agency has to meet the challenge of reaching out to every single individual. Communication is central to its efforts to heighten awareness of impending environmental problems and to increase the sense of urgency and personal responsibility for the environment.

To send the message out, NEA utilises diverse channels and methods. It partners the Ministry of Manpower to inform foreign workers on dengue prevention steps, and recruits foreign workers to be ambassadors in their community to battle littering. It reaches out to domestic helpers as well to educate them on dengue prevention.

Web, social media and even phone applications are put to good use for engaging the technologically savvy as NEA goes beyond traditional print media. Educational materials, guidelines, information kits for the general public and industries covering topics varying from recycling and waste minimisation to food hygiene, and weather information are just a mouse's click away on NEA's website. To make information on the environment even more accessible, NEA launched a smartphone application in 2011. The application allows users to stay updated on the latest public health and environmental information, such as dengue clusters, track records of food caterers and even lists of licensed pest control operators.

Well-known figures are also roped in for the cause. In 2010, Edmund Chen, a former Mediacorp artiste, was appointed as NEA's Cleanliness Ambassador to influence the community to exercise their social responsibility in keeping Singapore clean.

NEA provides feedback channels on its website and myENV application as well so that members of the public can reach NEA easily at all times to give feedback, raise concerns and contribute their ideas on caring for the environment. The agency also responds to communication sent via letters, emails and faxes, and to relevant issues raised in newspaper forums.

The agency's Contact Centre, one of the busiest of such services in the government, is an important channel for the public. It operates round the clock, handling more than 600 calls a day, up from an average of 400 calls in 2002. During environmental and public health incidents such as dengue outbreaks, the calls can spike to more than 800 a day.

The nature of the calls varies widely, from giving feedback to appeals and enquiries on issues ranging from mosquitoes and rats to noise and smells. It is not uncommon for the centre to receive calls which are unrelated to NEA's scope of work, and even calls from lonely folks who just want to chat. And remarkably, the most common call is for NEA to deal with perceived offences. Officers in this line have to be highly trained to handle the calls efficiently and professionally. Every call is logged and tracked so that NEA can monitor the quality of the response.





AN ENGAGED, SOCIALLY RESPONSIBLE WORKFORCE

As NEA engages the community to be more environmentally and socially conscious, it walks the talk by embracing corporate social responsibility through fostering an engaged, socially responsible workforce. Through initiatives such as the annual Semakau Run and Vertical Challenge, NEA encourages staff volunteerism and raises funds for environmental non-governmental organisations and its adopted charities, the Singapore Children's Society and the Children's Cancer Foundation.





A Blueprint For Sustainability

"The diversity in our roles reflects the importance of NEA in many areas that may not seem apparent at first glance, and which may even be taken for granted at most times. But it's a very important job, to make sure that every morning when you step out of the door, you see that the place is clean. That you feel safe coming back home because you are not going to be bitten by mosquitoes, and that your waste is collected on time and every time. And when you enjoy your weekend, the sky is blue, you can go to the beach, and you see other fellow Singaporeans also similarly being as responsible as you are in terms of keeping the environment clean. That would be the ideal vision."
— Andrew Tan, CEO, NEA

Transformation is necessary for NEA to keep abreast of evolving times. It is also crucial for the agency to begin executing its blueprint to realise its vision for our environmental future.

Coming up with plans for sustainability has always been a key pillar in the story of Singapore's environment. This became even more central when the Singapore Green Plan was inaugurated by ENV (present-day MEWR) in 1992, followed by the Singapore Green Plan 2012 released in 2002 and revised in 2006. Important targets for Singapore's environmental performance were set in the plans, which also served as a guide to balance environmental and developmental needs.

Environmental sustainability took centre stage when the Sustainable Development Blueprint was released in 2009. A result of close collaboration among several ministries and agencies, the blueprint charted the way for long-term sustainable development.

NEA's road map supports the broader objectives of the Sustainable Development Blueprint. It fills in the interim and long-term targets and how NEA will achieve them. Above all, it rests on a vision of our future — one of a beautiful, clean and green environment with the best air quality, clean water and clean land, a Singapore with high standards of public health, and a society that makes the most and best use of its resources.





"Achieving sustainable development enables us to grow our economy whilst safeguarding our living and natural environment. Today, Singaporeans enjoy a high quality environment in a vibrant and prosperous city state. This was achieved as a result of strong political commitment supported by good planning, constant innovation and effective implementation."

— Tan Yong Soon, Permanent Secretary, MEWR (2004 - 2010) at the launch of Nanyang Environment & Water Research Institute Centres, 5 October 2009



CLEAN, HEALTHY AND CONDUCIVE SURROUNDINGS

Singapore is one of the most urbanised environments in the world today. But this has not stopped NEA from setting even higher standards for our air, water and land in time to come. More stringent standards based on international benchmarks will be implemented for our ambient air quality and coastal and inland waters. When it comes to public health and hygiene, NEA has progressive targets to bring about a litter-free, smoke-free environment, with more effective vector control and food hygiene surveillance. Noise pollution will be more tightly monitored and controlled so as to protect public health and to create more comfortable living surroundings.

CHERISHING OUR RESOURCES

Earth's resources are finite. For long-term sustainability, we need to move towards a society where every resource is cherished and used prudently. As Singapore grows more affluent, nurturing mindsets to conserve resources will likewise grow more challenging.

As such, NEA will be making more moves to reduce the waste we generate, encourage resource recovery and promote recycling. More schemes will also be put in place to encourage and drive energy efficiency in Singapore.

BETTER PUBLIC SERVICE

Need to know if the weather is going to be fine for your outdoor event this weekend? Well, meteorological services are to receive a boost to provide more timely, reliable and accessible information in the near future. Singapore is also poised to become a climate research hub in the region.

NEA is gearing up to be even more service-oriented in the future. Hawker centres, an icon of Singapore, will be more exciting places in time to come. On top of being places of affordable, clean food, hawker centres of the future will also be even better places for community bonding, with facilities in place for the aged, young and disabled. Eco-friendly features will also be integrated with these centres.

A HUB FOR THE WORLD

Singapore has already taken firm steps to establish itself as an environment hub. Going forward, more attention will be given to build Singapore as a regional and international environmental centre. NEA will nurture and groom more local companies to develop cost-effective environmental solutions for future challenges. It will also continue to build capacities within Singapore, so that we may continue to share our environmental solutions with other parts of the world.



Our Environmental Future

"We have been able to overcome our limitations and create something we can be proud of. The limitations of land, the limitations of water, the limitations of many things — we can overcome it, we have overcome it. We can continue to overcome our limitations even as society and issues get more complex, and we have proof of it. We have nothing. But out of nothing, we have created something we can be proud of. And that is the story of Singapore."

— Dr Yaacob Ibrahim, Minister for the Environment and Water Resources (2004 - 2011)

Singapore has always been an anomaly. With its natural limitations, it is a marvel how far it has come.

Its story consistently tells of the will to overcome all odds to survive and excel. From an entrepôt to global financial centre, from backwaters to metropolis, from a dirty, polluted island to a clean, liveable place – these transformations demonstrate time and again, that as long as we do not leave anything to chance, Singapore can do it.

In many aspects, the quality of our environment has become better since NEA's formation in 2002. Our ambient air quality has improved, guidelines for recreational water quality are now stricter and standards for pollution control have grown more stringent. Surveillance of public health and hygiene has become even more rigorous. NEA's scientists have conducted significant research in various areas, bringing considerable benefits to public and environmental health. The agency has also played a leading role in galvanising our local environment industry and contributed to the rise of Singapore as a regional environment hub.

At the same time, NEA has transformed radically. Today, it no longer relies on enforcement alone. Over the last decade, it has successfully nurtured an inclusive culture where ordinary Singaporeans from different spheres work together with NEA to bring about a clean, green Singapore.

Resting on its laurels has never really been the agency's approach. Safeguarding a clean and healthy environment, nurturing an environmentally conscious society where every individual cherishes Singapore's beautiful, liveable surroundings – this continues to be NEA's core mission. The agency has also renewed and transformed itself to keep up with times and meet impending challenges effectively.

From the leaders of the agency to officers in field work, every single person in NEA has made a difference. Their passion and commitment towards their work form the bedrock of the quality of environment we enjoy. But their work alone is not enough. Singapore's environmental sustainability hinges upon each and every Singaporean now. This vision of an even more beautiful, clean, green city is possible, if we stand together to cherish and safeguard our environment, for ourselves and for future generations to come.





"In general, keeping the environment safe for human beings is NEA's primary responsibility. It involves first making sure we have the talent pool for the diverse areas of work. We need to make sure that these people are backed up by science, the laboratories, access to the latest information so that they can do their jobs well. This is also an evolving field, whether it is food safety or infections, or the quality of air or water. You need to have access to the latest technology and latest scientific information. NEA also needs to continue its role in public education and in engaging the public so that people become part of the solution. So it is all about creating a network of highly motivated staff, publishing real-time accurate information and exploiting the best technology available to protect our fellow Singaporeans from current and future environmental challenges."

— Dr Vivian Balakrishnan, Minister for the Environment and Water Resources



A 10-Year Journey for Environmental Sustainability

2002

- NEA is established on 1 July.
- Toilet Upgrading Programme is launched.
- "Singapore, Litter-Free" campaign is launched.
- NEA and the Singapore Environment Council (SEC) introduce voluntary energy labelling for household air-conditioners and refrigerators.
- Energy Audit Scheme is launched.

2003

- District offices are restructured on 1 March to form five regional offices aligned to the same boundaries as CDCs.
- Singapore's OK campaign is launched to restore public confidence during the SARS crisis.
- NEA's Call Centre begins operating 24/7. The hotline is changed to 1800-CALL NEA.
- Situation Rooms set up for greater situational awareness and coordination of ground response.
- NEA launches Nowcast, the weather forecast for the next three hours.
- Singapore Dengue Consortium forms under NEA's lead.
- NEA launches CNG Taxi Monitor Project with Borneo Motors Singapore and CityCab, Comfort Transportation and TIBS Taxis.
- NEA and SEC introduce voluntary fuel economy labelling for passenger cars.
- Singapore Environment Institute is established.
- $750,000 3P Partnership Fund is launched to provide seed funding for those who wish to initiate environmental projects.

2004

- Adopt-a-School scheme, later renamed Corporate And School Partnership programme, is launched by NEA to reach out to schools.
- NEA's key 3P outreach programmes, Youth Environment Envoy and Environment Champion programmes, are launched.
- Recycling Day is launched.
- HDB's 92 wet markets and hawker centres are consolidated under NEA's purview.
- One-Stop Service Centre at the Environment Building opens.

2005

- Use of ultra-low sulphur diesel is made mandatory.
- Energy Smart Building Labelling Scheme is launched by NEA and the Energy Sustainability Unit of the National University of Singapore.
- $10 million Energy Efficiency Improvement Assistance Scheme is launched to encourage companies to carry out detailed energy appraisals.
- NEA launches intensive exercises to eradicate vector population during the worst dengue outbreak in Singapore's history, when cases spike at 700 a week. The trend finally slows down by November 2005.

2006

- Smoking is restricted to designated smoking corners at nearly 7,300 non air-conditioned food shops and 121 hawker centres from 1 July.
- NEA commissions a littering behaviour study.
- New diesel vehicles to comply with Euro IV emission standards with effect from 1 October.
- NEA collaborates with Muaro Jambi Regency to build the regency's capacity to deal with land and forest fires.
- Intensive Source Reduction Exercise strategy is implemented to reduce population of *Aedes* mosquitoes. Dengue trend is successfully bucked as cases fall to a four-year low of 3,126 cases, a stark contrast to 14,209 cases in 2005.

2007

- Centre for Radiation Protection and Nuclear Science (CRPNS) is formed.
- Permissible noise limits for construction sites within 150 m of residential premises are tightened for night-time, Sundays and public holidays.
- Publication on hawker centres, "Singapore Hawker Centres - People, Places, Food", is launched.
- New Burial System is implemented.
- Public Sector Taking the Lead in Environmental Sustainability Initiative is introduced to demonstrate the environmental and economic benefits of efficient energy and resource use.
- EcoFriend Award is launched.
- First Semakau Run is held at Semakau Landfill with more than 40 corporations participating to raise funds for environmental NGOs.
- NEA commissions study to understand Singapore's vulnerabilities to climate change.
- Smoking ban is extended to entertainment outlets.
- Mandatory Energy Labelling Scheme (MELS) is launched.
- Singapore Packaging Agreement is signed by NEA and 32 organisations.
- Bring Your Own Bag Day is launched to encourage the public to use less plastic bags.
- Clean and Green Week is renamed "Clean and Green Singapore" and becomes a year-long event from this year onwards.

2008

- New water quality guidelines for recreational beaches are launched.
- NEA introduces new standards for piped drinking water quality standards.
- $22 million Grant for Energy-Efficient Technologies (GREET) is established to encourage companies to invest in energy-efficient technologies.
- NEA introduces Design for Efficiency Scheme to encourage investors in new facilities to integrate energy and resource efficiency measures early in the design stage.
- NEA and the Institution of Engineers, Singapore, co-administer the Singapore Certified Energy Manager (SCEM) training programme to develop local expertise in professional energy management. A $6 million SCEM training grant is introduced.
- $500,000 grant to promote Clean Development Mechanism projects in Singapore is launched.
- NEA launches 10% Energy Challenge campaign.
- Chikungunya outbreak. NEA successfully curbs the spread.

2009

- Fuel Economy Labelling Scheme (FELS) is implemented for passenger cars and light goods vehicles.
- Sociological study to study littering behaviour is commissioned. The study results in the 2011 publication "Towards a Cleaner Singapore: Sociological Study on Littering in Singapore".
- Smoking ban is extended to cover children's playgrounds, markets and exercise areas.
- $8 million 3R (Reduce, Reuse and Recycle) Fund is launched to encourage waste minimisation and recycling projects.
- $15 million Environment Technology Research Programme is launched to develop capabilities in waste management.
- NEA and Microsoft launch Singapore's first mobile weather service, Weather@SG.
- NEA publishes a commemorative book, "Habitats in Harmony: The Story of Semakau Landfill", to celebrate Semakau Landfill's 10th anniversary.
- All new motor vehicles imported for use in Singapore are required to comply with revised noise standards from 1 October onwards.
- Clean and Green Singapore Facebook page is launched.
- NEAsg Twitter is launched.

2010

- Energy Efficiency National Partnership (EENP) is launched with 49 founding partners.
- Oil spill in the Singapore Straits. NEA leads the clean-up of affected coastal areas and steps up environmental monitoring.
- Eco Music Challenge is inaugurated.

2011

- NEA plays a key role in Singapore's measures to safeguard the country from radioactive contamination following the Fukushima Daiichi nuclear crisis.
- NEA's first mobile phone application, myENV, is launched for smart phone users.
- 3P SYNC Magazine is launched to recognise the efforts and contributions of NEA's 3P partners, as well as to keep them informed about environmental initiatives in Singapore.
- First Clean and Green Singapore Music Album created by local youths is launched.
- Youth for the Environment Day is launched.
- NEA signs MOU with UK Met Office and repositions its meteorological services to include climate science research.
- EHI is designated a WHO Collaborating Centre for the Reference and Research of Arbovirus and their Associated Vectors.

2012

- Inaugural technical magazine "ENVision" is launched.
- Time-stamping for catered food is mandated.
- A new Energy Conservation Bill is passed in parliament.
- Sites for seven new hawker centres are confirmed.
- New Upper Air Observatory opens. First book on Singapore's weather, "The Weather and Climate of Singapore", is launched.
- NEA begins to progressively take over all duties for public cleaning except for housing estates.
- Android version of myENV mobile application is rolled out.
- NEA celebrates 10th anniversary.

GLOSSARY

TERMS

3P Partnership: People-Private-Public Partnership. NEA's outreach strategy to involve the people, private and public sectors in its drive for Singapore's long-term environmental sustainability

Aedes aegypti: Species of mosquito that can spread dengue

Aerodrome: An area, including buildings, installations and equipment, intended to be used for the arrival, departure and surface movement of aircraft

Ambient air quality: Condition of the surrounding air

Chlorophyll-a: The most common type of chlorophyll, a green pigment, present in plants and algae

Climate change: Long-term shift in overall weather conditions over time

Columbaria: Buildings with niches for the placement of cremated ashes

Crematoria: Places where the deceased are cremated

Dengue: Tropical disease caused by virus transmitted by the *Aedes* mosquito

Doppler Effect: A shift in the frequency of an electromagnetic or sound wave due to relative movement of the source or the observer

Drizzle: Very small, numerous and uniformly distributed water drops that may appear to float while following air currents

El Niño: A significant increase in sea surface temperature over the eastern and central equatorial Pacific Ocean that occurs at irregular intervals, generally ranging between two to seven years

Endosulfan: A pesticide that has been found to be bioaccumulative and an endocrine disruptor. This pesticide has been included for control under the Stockholm Convention on Persistent Organic Pollutants in 2011.

***Enterococcus*:** Bacteria found in faeces of humans and warm-blooded animals. Presence of *enterococcus* is an indicator of fecal pollution

Entomologist: A person who studies arthropods, including insects, spiders, ticks, and mites

Epidemiologist: A person who studies the distribution, determinants and control of health-related events in a specified population

Genotype: Genetic make-up of an organism which is used to define the biological population it belongs to

Incineration bottom ash: Remains of waste after it is burned at waste incineration plants

Intensive Source Reduction Exercise: Operation to seek out and destroy actual and potential mosquito breeding sites

Ionising radiation: Energy that can produce ions from atoms or molecules in its path

Knock-on effect: Principle whereby a mishap in one facility sets off a chain reaction of problems in neighbouring facilities

Meteorology: The scientific study of the atmosphere

New Burial System: Engineered crypt system where the deceased are buried in prefabricated concrete crypts that are laid out in grids to allow for a more efficient use of land and make the cemetery plots more accessible to visitors

NEWater: High-grade reclaimed water produced from treated used water that is further purified using advanced membrane technologies and ultra-violet disinfection, making it ultra-clean and safe to drink

Noise Exposure Forecast: Measurement of a community's exposure to aircraft noise

Nowcast: Weather forecast for the next three hours

Numerical weather prediction models: Computer programmes that solve mathematical equations which describe the physics of the atmosphere; used extensively by weather services throughout the world to aid weather forecasting

Particulate matter: Generic term for air pollutants varying in size and composition

Pathogen: Micro-organism – virus, bacterium etc – that can cause disease

pH value: Measure of potential hydrogen in a solution. Pure water is neutral, with a value of about 7 at 25 °C. A lower pH value denotes acidity while a higher one denotes alkalinity

Points Demerit System: A systematic approach to suspend or revoke the licences of retail food establishments. Demerit points are given to hygiene offences based on the severity. Licences are suspended for two weeks upon accumulation of 12 or more demerit points within a 12-month period. Upon the second offence, licences are subjected to four weeks' suspension. After the second suspension, the licences are revoked if there is accumulation of another 12 demerit points within a 12-month period.

Precipitation: Any form of water, whether liquid or solid, that originates in the atmosphere and falls to the Earth's surface

Radiological dispersal device: Weapon designed to spread radioactive material over a large area to terrorise or kill

Rain: Steady precipitation which falls in liquid state from layer clouds

Serotype: A variant of a micro-organism, based on the antigens found on its cell surface, that is used to classify its grouping at the sub-species level

Severe Acute Respiratory Syndrome (SARS): An infectious respiratory disease in humans which is caused by the SARS coronavirus

Shower: Precipitation that falls from a convective cloud; these characteristically start and end suddenly, and undergo rapid change of intensity

'Sumatra' Squall: A line of thunderstorms which forms over Sumatra or the Straits of Malacca and approaches Singapore or Peninsula Malaysia from the northwest, west or southwest depending on the wind direction. These systems are often accompanied by torrential rain and strong gusty winds.

Thermoluminescent Dosimeter: Device that measures levels of ionising radiation exposure by measuring the amount of light emitted from a crystal in the device

Tsunami: A series of enormous waves formed in the sea or near the coast, generated by violent undersea disturbances such as earthquake-induced movement of the ocean floor, landslides and underwater volcanic eruptions

Vector: A living organism carrying a pathogen capable of causing an infection in humans

Voluntary Observing Ships (VOS) scheme: A scheme by the World Meteorological Organization (WMO) in which ships plying the various oceans and seas of the world are recruited for performing, taking and transmitting meteorological observations. These observations form part of WMO's global observation network.

West Nile virus: A flavivirus found in both the tropical and temperate regions transmitted mainly by mosquitoes and can cause meningitis or encephalitis in infected humans

World Meteorological Organization: A specialised agency of the United Nations. It is the United Nations system's authoritative voice on the state and behaviour of the Earth's atmosphere, its interaction with the oceans, the climate it produces and the resulting distribution of water resources.

Zoonosis: Any infectious disease that can be transmitted between species (sometimes through a vector) from animals to humans or from humans to animals

ABBREVIATIONS

ASEAN	Association of Southeast Asian Nations
AVA	Agri-Food and Veterinary Authority
BCA	Building and Construction Authority
CCRS	Centre for Climate Research Singapore
CDC	Community Development Council
COPPC	Code of Practice on Pollution Control
CRPNS	Centre for Radiation Protection and Nuclear Science
CSR	Corporate social responsibility
EDB	Economic Development Board
EHI	Environmental Health Institute
EMA	Energy Market Authority
ENV	Ministry of the Environment
HDB	Housing and Development Board
HSA	Health Sciences Authority
HUP	Hawker Centres Upgrading Programme
IBA	Incineration bottom ash
MEWR	Ministry of the Environment and Water Resources
MFA	Ministry of Foreign Affairs
MOH	Ministry of Health
MPA	Maritime and Port Authority of Singapore
MSS	Meteorological Service Singapore
NEA	National Environment Agency
NEF	Noise Exposure Forecast
NGO	Non-governmental organisation
PRCS	Pneumatic Refuse Conveyance System
PSI	Pollutant Standards Index
PUB	Public Utilities Board
SARS	Severe Acute Respiratory Syndrome
SCDF	Singapore Civil Defence Force
SEI	Singapore Environment Institute
TSIP	Tuas South Incineration Plant
URA	Urban Redevelopment Authority
VOS	Voluntary Observing Ships
WDA	Workforce Development Agency
WHO	World Health Organization
WMO	World Meteorological Organization
WTE	Waste to Energy, as applied to incineration plants

PEOPLE

Lim Swee Say
Acting Minister for the Environment (2001 - 2002),
Minister for the Environment (2002 - 2004)

Dr Yaacob Ibrahim
Minister for the Environment and Water Resources (2004 - 2011)

Dr Vivian Balakrishnan
Minister for the Environment and Water Resources (2011 - present)

Professor Simon Tay
Chairman, NEA (2002 - 2008)

Chew Gek Khim
Chairman, NEA (2008 - present)

Lam Joon Khoi
CEO, NEA (2002 - 2005)

Lee Yuen Hee
CEO, NEA (2005 - 2009)

Andrew Tan
CEO, NEA (2009 - present)

ACKNOWLEDGEMENTS

We would like to thank all who have given their time and efforts to make this book possible. Our sincere regrets if we have missed out any person or organisation who has contributed.

BOOK COMMITTEE

Chairman:
Ong Eng Kian, Director, SEI

Members:
Alvin Saw
Andrew Low
Gillian Oh Wee Kee
Ho Cheng Hoon
Janice Tan
Jasmine Low
Koh Min Ee
Koh Su Ching
Lim Bee Lan
Ng Chee Yong

ORGANISATIONS

Anglo-Chinese School (Independent)
Anchor Green Primary School
Ci Yuan Community Club Youth Club
City Developments Ltd
Commonwealth Secondary School
Dunman High School
Hwa Yuen Porridge
ITE College East
JTC Corporation
Menu Pte Ltd
Nestlé Singapore
NYK Olympus
P & R Resource Management Pte Ltd
Senoko Energy
Sin Tong Nam
Tetra Pak
Wah & Hua Pte Ltd
Waste Management and Recycling Association of Singapore
Woodgrove Secondary School

CONTRIBUTORS

Agnes Hoe
Akbar Bessir Ahmad
Albert Lim
Alexandra May Serrenti
Alice Tavamani
Amatul Jameel Suhani (A J Suhani Sujari)
Aminah Yunos
Amy Khor, Dr
Ananda Ram Bhaskar
Andrew Lau
Andrzej Gomulski, Captain
Andy Tan
Andy Wee
Ang Kok Kiat
Annie Lim
Annie Tan
Billy Chew
Bin Chee Kwan
Brandon Low
Calista Lin
Chan Chin Kai
Chanditha Hapuarachchi
Chay Noh Tan
Chia Kai Feng
Chong Chee Wee
Choo Soey Tee
Choo Yew Kian
Chow Chee Kiong
Chow Mei Ling
Choy Kum Cheong
Chua Yew Peng
Clement Chow
Dalson Chung
Daniel Wang
David Lee, Dr
Dennis Lee
Derek Ho
Derk Jan Hartgerink
Devin Otto Kimble
Dulcie Chan
Eddie Foo
Elizabeth Ong Baoxuan
Esther An
Fareena Abdul Rahim
Gay Yee Ying
Geoffrey Kho
Gloria Chin
Goh Wee Poh
Grace Yap
Guah Eng Hock
Gwen Tan
Hasron Ibrahim
Hazman Mohani
Hussein bin Majid
Imran Bin Mohammed Aleh
Indrani C Rajaram
Ivy Ong Bee Luan
Jacques Deschamps
Jeffrey Din
Jelita Teper
Jerine Tang
John Low
Jolyn Lim
Joseph Boey
Joseph Hui
Jothieswaran Poobalasingam
Jumahat Jantan
Jumi Boh
K Panirselvem
Kavitha D/O Govinda Rajoo
Keith Kwa
Khairul Samsudin
Khoo Seow Poh
Koh Char Boh
Koh Kim Hock
Koh-Leong Swee Ling
Kwan Wai Kern
Kwek Chin Hin
Kwok Wai Choong
Kwong Kok Chan
Lam-Phua Sai Gek
Lee Ah Seong
Lee Kheng Seng
Leow Bee Geok
Lesley Choo
Liew Wen Hwee
Lim Bin
Lim Hiong Li
Lim Jing Kai
Lim Kuang Huwak
Linda Kang
Lisa Kok
Loh Ah Tuan
Low Eng Hai
Low Lee Hua
Lu Hong Hong
Lye Hui Hui
Marcus Koo
Michelle Lee
Mohamad Ghazali Bin Ismail
Mohamed Jumahin Bin Ibrahim
Mohd Hussain Yusoff
Mohd Zulkefli Abdul Rahman
Muhammad Khairozi Bin Hassan
Ng Bee Eng
Ng Khee Yang, Dr
Ng Lee Ching, Dr
Ng Meng Hiong
Ng Min Kuan
Noraini Matanawi
Noralha Dollah
Ong Bee Leng
Ong Chee Siang
Ong Chin Hong
Ong Lee Li
Ong Liu Ching
Ong Seng Eng
Ong Siew Lin
Patricia Ee
Patrick Ng
Peter Lo
Phua Tan Tee, Dr
Poon Chiew Tuck
Rachel Ng
Raja Mohan
Rama Chandramogan
Ravindran Nair
Rohaya Saharom
S Mohan
S Satish Appoo
Saliman Bin Mordi
Samson Lim Kah Hui
See Boon Tiam
Selvam Sambandam
Shaik Amina Bee Shaik Mohamed
Sharon Ong
Shelby Ong
Sim Li Ling
Siti Aida MD Yusof
Siti Shahara
Steven Chiang Hock Chuan
Sulaiman Bin Abdul Rahman
Sumaidi Bin Ani
Sunny Tan
Suresh Kulaveerasingham
Tai Jo Fen
Tan Giin Bo
Tan Kee Pin
Tan Shyh Hou
Tan Quee Hong
Tan Wee Hock
Thia Kai Feng
Thng Wei Ling
Trina Ng
Valerio Nannini
Vaneeta Bhojwani
Veronica Choo
Vilma Tay
Wendy Su
Wilson Ang
Wong Chin Ling
Wong Chiu Ying
Wong Kit Yin
Wong Mong Lin
Wong Yeh-Ping
Woon Shih Lai
Yap Shee Pak
Yaw Eng Koon

Special thanks to all the individuals and organisations who are involved in facilitating the photography for the publication.

INDIVIDUALS

Abdul Rahim
Abu Bakar Bin Abdul Hamid
Alancia Winnie Neo
Alfred Tan
Alvin Choy Meng Loong
Amir Hamzah Bin Hamid
Angeline Tang Phui Heng
Au Wing Kuen
B Karnan
Balwinder Singh
Benny Low Zhixiong
Boo Chew Han
Chai Chiang Yap
Chatterji Somnath
Chik Ka Man
Chong Kuek On
Chong Wan Kee
Choo Yew Kin
Christina Lee
Cong Ying
Daren Swee
Elzena Ibrahim
Eugene Chong Wei Ming
Evelyn Siah
Faznie Bin Sulaini
Fiona Karan
Fong Lap Weng
Freddie Chen Wenhui
G Thurainadan
Garyth Lim Jien Keat
Goh Su Xin
Hajjah Nor Ima
Ho Kim Soon
Indrani C Rajaram
Jens Suessmilch
Joel Koh Hon Wei
Jonathan Koh
Jumain Suaip
June Touw
Justin Ng
K. Subramaniam
Karen Wong
Kavitha Govinda Rajoo
Kavitha D/O Rajandran
Khairul Anwar Bin Othman
Kieran Tan
Koh Yew Guan
Kok Foo Lim
Koyil Pillai
Lau Wei Har
Lee Keng Chuen
Leong Kaiyan
Lily Han
Lim Bee Lan
Lim Mei Lan
Lin Jiangxun
Lit Pei Wen
Liu Shiqin
Loo Deliang
Low Kee Chong
Lynda Lim
Margaret Marshall
Mariam Maharobin
Matthew Webster
Mohamed Izhar Bin Abdul Manan
Mohd Ataullah Fazalur Rehman
Mohd Bin Bujang
Mohd Fajar Bin Abd Jalil
Mohd Gafar Bin Ahmad
Mohd Hussain Yusoff
Mohd Norddin Bin Hussin
Mohd Ridzuan B Razali
Mok Jiak Wee
Nazhar Anuar
Ng Beng Tee
Ng Yu Man
Nigel Peh
Ninabi Bin Maeng
Nur Syafawati Adnan
Nurul Sakinah Abdul Bashir
Ong Li Lian
Ong Pei Fen
Ong Siew Lin
Pang Fook Chong
Paula Kesavan
Peter Tan
Phua Kim Tian
Png Ah Bah
Ron Tay Li Rong
Rosli Ismail
Saiman Bin Rasikon
Samson Lim Kah Hui
Sarbini Bin Soori
Shahrum Sulaiman
Shlan Bin Suleiman
Siti Farhana
Tan Aik Keat
Tan Chen Wai
Tan Chia Wen
Tan Geok Chin
Tan Joo Thai
Tan Kah Aik
Tan Mui Hoon
Tang Hui Qi
Tay Koon Seng
Thng Wei Ling
Vincent Choh Shin Tung
Wai Hui Kian
Wan Rosman Akwan
Wong Yeh-Ping
Yap Kar Lin
Ye Yi
Yu Zhen
Zainal Bin Hashim
Zhang Jiannan
Zhang Peisha
Zheng Kai Jie
Zheng Zhihua

ORGANISATIONS

ExxonMobil Asia Pacific Pte Ltd
Greenchem Technology Pte Ltd
Housing & Development Board
ITE College East
JTC Corporation
Khoo Teck Puat Hospital
Marina Bay Sands Pte Ltd, *Garden of Light (2012)* by Hexogon Solution Pte Ltd, *Featured at i Light Marina Bay 2012*
Menu Pte Ltd
NYK Fleet, Columbia Ship Management Ltd
People's Association
Port Authority Singapore
Sembcorp Industries Ltd
TÜV SÜD PSB Pte Ltd
Urban Redevelopment Authority
Veolia Environmental Services

PHOTO CREDITS

Alicia Tan - Page 17 (bottom), 34, 58, 67 (bottom), 134, 162

City Developments Limited - Page 159

JTC Corporation - Page 31, 173

Lua Hwa Beng - Page 8, 10, 11, 35, 43, 48, 51, 75, 84, 97, 129, 167, 169 (top), 183, 184, 196

Ong Chee Siang - Page 161

Richard Koh - Cover, Page 14, 18, 22, 25 (bottom), 27, 29 (background), 33, 39, 40, 45, 46, 49, 53, 55, 56, 57, 59, 61, 69, 70, 73, 77, 79 (right), 81, 83, 85, 92, 93, 95, 96, 98 (top), 100, 102, 103, 106, 107, 108, 109, 115, 117, 118, 119, 120, 122, 123, 125, 128, 130, 131, 133, 135, 139 (top), 147, 151, 164, 172, 179, 181, 182 (bottom), 185, 187, 190, 193, 195 (bottom right), 201, 205, endpaper

Page 62 (left to right): Marcus Ng, James Koh, Loh Kok Sheng, Jeemee Goh

REFERENCES

Chua Lee Hong. 2002. *The Singapore green plan 2012: Beyond clean and green towards environmental sustainability.* Singapore: Ministry of the Environment.

Chua Mui Hoong. 2004. *A defining moment: How Singapore beat SARS.* Singapore: Institute of Policy Studies.

Environmental Protection and Management Act 2002. (c. 94A), Singapore: Attorney General's Chambers.

Environmental Public Health Act 2002. (c. 95), Singapore: Attorney General's Chambers.

Global Leaders of Tomorrow Environment Task Force. 2001. *2001 Environmental sustainability index.* Switzerland: World Economic Forum.

Inter-Ministerial Committee on Sustainable Development. 2009. *A lively and liveable Singapore: Strategies for sustainable growth.* Singapore: Ministry of the Environment and Water Resources and Ministry of National Development.

Jurong Town Corporation. 2000. *The making of Jurong Island.* Singapore: Jurong Town Corporation.

Koh, K W Benjamin et al. 2008. The 2005 dengue epidemic in Singapore: Epidemiology, prevention and control. *Annals, Academy of Medicine.* 37 (7), p. 538

Kong Lily. 2007. *Singapore hawker centres: People, places, food.* Singapore: National Environment Agency.

Lee Hui Chieh. 2008. Virus First Detected Here By GPs' Dengue Network. *The Straits Times.* 24 January.

Lee Kim-Sung et al. 2010. Dengue virus surveillance for early warning, Singapore. *Emerging Infectious Diseases.* 16 (5), p. 847.

Meteorological Service Singapore. 1994 – 2001. *Annual reports.* Singapore: Meteorological Service Singapore.

Ministry of the Environment and Water Resources. 2006. *Singapore green plan 2012.* Singapore: Ministry of the Environment and Water Resources.

Ministry of the Environment. 1972 – 2002. *Annual reports.* Singapore: Ministry of the Environment.

Ministry of the Environment. 1997. *Singapore: My clean & green home.* Singapore: Ministry of the Environment.

Nathan, Dominic. 2001. Singapore: Green paradise or polluted city? *The Straits Times.* 29 January.

National Environment Agency. 2002 – 2011. *Annual reports.* Singapore: National Environment Agency.

National Environment Agency. 2003. *The NEA person code.* Singapore: National Environment Agency.

Ng Fu Chuan Marcus. 2009. *Habitats in harmony: The story of Semakau Landfill.* Singapore: National Environment Agency.

Ng Lee Ching et al. 2009. Entomologic and virologic investigation of chikungunya, Singapore. *Emerging Infectious Diseases.* 15 (8), p. 1243.

Ng Lee Ching. 2011. Challenges in dengue surveillance and control. *Western Pacific Surveillance and Response*, 2 (2), p. 1.

Parliament of Singapore. 1999 – 2011. *Singapore parliament reports.* Singapore: Parliament of Singapore.

Pollution Control Department. 1994 – 2001. *Annual reports.* Singapore: Ministry of the Environment.

Pollution Control Department. 1997. *Code of practice on pollution control.* Singapore: Ministry of the Environment.

Singapore Department of Statistics. 2011. *Yearbook of statistics.* Singapore: Singapore Department of Statistics.

Singapore Environment Institute. 2003 – 2010. *Annual reports.* Singapore: National Environment Agency.

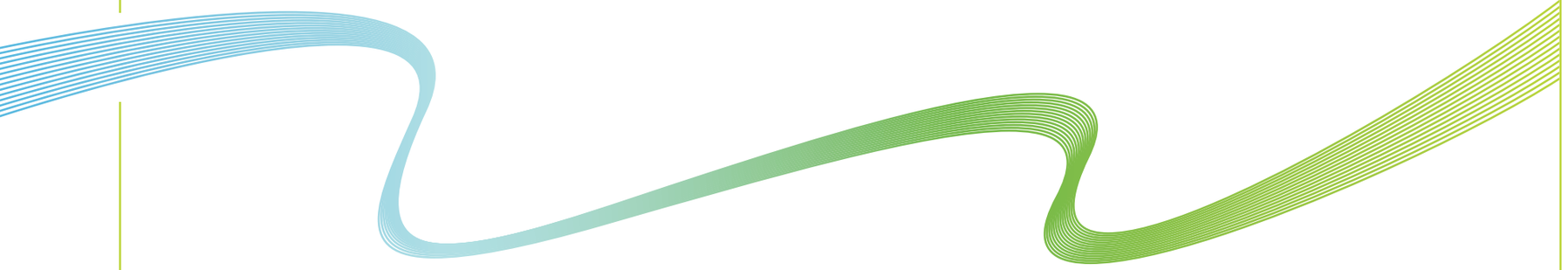
Straughan, Paulin Tay et al. 2011. *Towards a cleaner Singapore: Sociological study on littering in Singapore.* Singapore: National Environment Agency.

Tan Cheong Huat et al. 2011. Entomological investigation and control of a chikungunya cluster in Singapore. *Vector-borne and Zoonotic Disease.* 11 (4), p. 383.

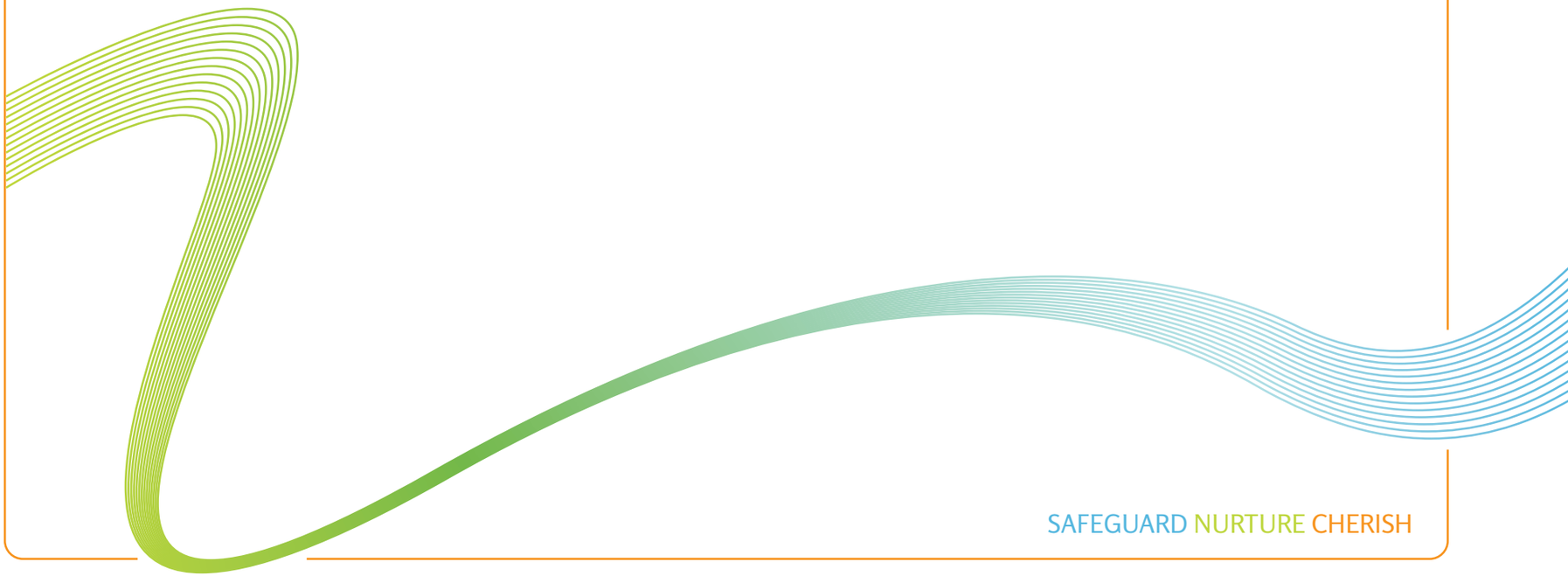
Tan Yong Soon et al. 2009. *Clean, green and blue: Singapore's journey towards environmental and water sustainability.* Singapore: Institute of Southeast Asian Studies.

The Economist Intelligence Unit. 2011. *Asian Green City Index.* Munich: Siemens AG.


SAFEGUARD NURTURE CHERISH



SAFEGUARD NURTURE CHERISH



SAFEGUARD NURTURE CHERISH





National Environment Agency

Safeguard · Nurture · Cherish

ISBN 978-981-07-2528-0